Thesis Project by Group CLS

THESIS 1-3

Thesis Project

Group Members:

- Celles, Aaron Kent M.
- Leviste, Lee Ryan A.
- Santeco, Enrique S.

Section 1. Introduction

1.1 Problem Overview

Large-scale global optimization (LSGO) presents a critical challenge in computational intelligence, where the "curse of dimensionality" leads to an explosive increase in local optima and a high risk of premature convergence. While foundational algorithms like Differential Evolution (DE) are robust, their dependence on crossover and random mutation may restrict guided exploration — reducing particle diversity over time and leading to premature convergence.

This notebook implements and evaluates the proposed **Diversity-Preserving Level-Based Learning Differential Evolution (DP-LLDE)** algorithm, tested against the **CEC 2017 Large-Scale Global Optimization Benchmark** across 29 benchmark functions (F1–F30, excluding deprecated F2) at **10 dimensions**.

1.2 Research Objectives

1. Design and implement DP-LLDE, integrating sequential relative particle mutation and dynamic level-based learning into the Differential Evolution framework.
2. Evaluate DP-LLDE on the CEC 2017 benchmark suite (F1–F30) at 10 dimensions.
3. Examine convergence behavior, solution quality, and the effect of diversity-preserving strategies across unimodal, multimodal, hybrid, and composition functions.

1.3 Algorithm Overview - DP-LLDE

DP-LLDE extends LBLDE (Qiao et al., 2022) with three core contributions:

1. **Sequential Level Assignment** — each individual is assigned a learning source level drawn randomly from {2, …, NL} on the first generation, decaying toward the elite level (level 1) in subsequent generations.
2. **Diversity-Aware Exemplar Selection** — if two exemplars e1, e2 fall below a diversity threshold, one is replaced with the most distant candidate in that level, preventing exemplar collapse.
3. **Normalized Mutation Pressure (Eq. 10):**

$$V_i = e_1 + F_i \times (e_2 - X_i) \times \frac{NL - k}{NL}$$

The factor $(NL - k)/NL$ decays to zero as $k \to NL$, naturally reducing mutation pressure as learning quality improves.

Parameter	Value
Population Size (NP)	100
Number of Levels (NL)	4
Initial μ_CR	0.35
Max FES	10,000 × D
Dimensions (D)	10
Seed	42

Section 2. Benchmark Function Description

2.1 Overview

The **CEC 2017 Large-Scale Global Optimization Benchmark** (Awad et al., 2017) is a standardized suite of 30 test functions designed to evaluate optimization algorithms on complex, high-dimensional search spaces. The benchmark tests an algorithm's ability to avoid false local optima, convergence speed, and solution accuracy.

All functions are defined over the search space $[-100, 100)^D$ with their respective global optima offset by a function-specific bias value.

> **Library used:** `cec2017` by Duncan Tilley — provides the official shift vectors, rotation matrices, and shuffle permutations from the CEC 2017 specification.

2.2 Unimodal Function Overview

Unimodal functions contain a **single global optimum and no local optima**. They are used to test the convergence speed and accuracy of an algorithm in a straightforward landscape.

No.	Function Name	Global Optimum
F1	Shifted and Rotated Bent Cigar	100
F2	Shifted and Rotated Sum of Different Power *(Deprecated)*	200
F3	Shifted and Rotated Zakharov	300

Note: F2 is officially deprecated from the CEC 2017 benchmark but is included here for completeness as it remains implemented in the `cec2017` library.

2.3 Multimodal Function Overview

Multimodal functions contain **multiple local optima**, making them significantly harder to solve. They are used to evaluate an algorithm's capacity to avoid premature convergence and escape false local optima.

No.	Function Name	Global Optimum
F4	Shifted and Rotated Rosenbrock's	400
F5	Shifted and Rotated Rastrigin's	500
F6	Shifted and Rotated Expanded Schaffer's F6	600
F7	Shifted and Rotated Lunacek Bi-Rastrigin	700
F8	Shifted and Rotated Non-Continuous Rastrigin's	800
F9	Shifted and Rotated Levy	900
F10	Shifted and Rotated Schwefel's	1000

2.4 Hybrid Function Overview

Hybrid functions divide the search space into several regions, each governed by a **different basic function**. They assess an algorithm's adaptability to heterogeneous landscapes and mixed search behaviors.

No.	Component Functions	Global Optimum
F11	Zakharov; Rosenbrock; Rastrigin	1100
F12	Elliptic; Schwefel; Bent Cigar	1200

No.	Component Functions	Global Optimum
F13	Bent Cigar; Rosenbrock; Bi-Rastrigin	1300
F14	Elliptic; Ackley; Schaffer F7; Rastrigin	1400
F15	Bent Cigar; HGBat; Rastrigin; Rosenbrock	1500
F16	Schaffer F6; HGBat; Rosenbrock; Schwefel	1600
F17	Katsuura; Ackley; Griewank+Rosenbrock; Schwefel; Rastrigin	1700
F18	Elliptic; Ackley; Rastrigin; HGBat; Discus	1800
F19	Bent Cigar; Rastrigin; Griewank+Rosenbrock; Weierstrass; Schaffer F6	1900
F20	HappyCat; Katsuura; Ackley; Rastrigin; Schwefel; Schaffer F7	2000

2.5 Composite Function Overview

Composition functions **combine multiple shifted and rotated basic functions** into a single search space, creating highly complex and deceptive optimization landscapes. They are the most challenging category in the CEC 2017 suite and significantly test an algorithm's overall robustness and global search capability.

No.	Component Functions	Global Optimum
F21	Rosenbrock; Elliptic; Rastrigin	2100
F22	Rastrigin; Griewank; Schwefel	2200
F23	Rosenbrock; Ackley; Schwefel; Rastrigin	2300
F24	Ackley; Elliptic; Griewank; Rastrigin	2400
F25	Rastrigin; HappyCat; Ackley; Discus; Rosenbrock	2500
F26	Schaffer F6; Schwefel; Griewank; Rosenbrock; Rastrigin	2600
F27	HGBat; Rastrigin; Schwefel; Bent Cigar; Elliptic; Schaffer F6	2700
F28	Ackley; Griewank; Discus; Rosenbrock; HappyCat; Schaffer F6	2800
F29	F15; F16; F17	2900
F30	F15; F18; F19	3000

Section 3. List of Required Libraries

3.1 List of Libraries and Versions

The following libraries and modules were used for optimization, benchmarking, and visualization in this project:

Library / Module	Purpose	Version
`python`	Core programming language	3.10+
`numpy`	Numerical computing and array operations	2.0.2
`matplotlib`	Plotting and saving convergence graphs	3.10.0
`cec2017`	CEC 2017 benchmark functions (simple, hybrid, composition)	Latest (pip)
`cec2017.simple`	Simple benchmark functions F1–F10	—
`cec2017.hybrid`	Hybrid benchmark functions F11–F20	—
`cec2017.composition`	Composition benchmark functions F21–F30	—
`dpllde`	DP-LLDE optimization algorithm (uploaded locally)	Custom
`google.colab.files`	File upload utility for loading `dpllde.py` into the Colab environment	Built-in
`sys`	Clears cached module before re-importing `dpllde.py`	Built-in

Section 4. Environment Setup and Validation

4.1 Installation

Install `cec2017`

```
!pip install cec2017 --quiet

ERROR: Could not find a version that satisfies the requirement cec2017
(from versions: none)
ERROR: No matching distribution found for cec2017
```

4.2 File Validation

Import `simple.py`, `hybrid.py`, and `composition.py` python files for benchmark functions

```
from cec2017 import simple, hybrid, composition
print('cec2017 loaded!')
```

```
cec2017 loaded!
```

Import `sys` control over warning messages for verification that `lblde.py` exists

```
from google.colab import files
import sys

uploaded = files.upload()

if 'lblde.py' in uploaded:
    if 'lblde' in sys.modules:
        del sys.modules['lblde']
    import lblde
    from lblde import LBLDE
    print('dpllde.py uploaded and imported successfully!')
else:
    print('ERROR: Please upload a file named dpllde.py')
```

```
<IPython.core.display.HTML object>
```

```
Saving lblde.py to lblde.py
dpllde.py uploaded and imported successfully!
```

Section 5. Configuration and Helper Functions

5.1 Library Imports

Import libraries `numpy`, `matplotlib`, and `cec2017` for numerical computation, visualization, and CEC 2017 benchmark functions.

```
import numpy as np
import matplotlib.pyplot as plt
from cec2017 import simple, hybrid, composition
```

5.2 Global Parameters

Define global parameters

```
D = 10 # Change this when testing different dimensions

bounds = np.array([[-100.0, 100.0]] * D)
MAX_FES = 10000 * D
```

```
NP, NL, MU_CR = 100, 4, 0.35
SEED = 42
```

Store all results for summary table under `all_results` dict

```
all_results = {}
```

5.3 Function Definitions

Define `make_wrapper` to adapt CEC 2017 functions to accept a 1D input array from LBLDE, returning a large penalty value if the output is invalid (nan/inf).

```
def make_wrapper(fn):
    """Wrap a cec2017 function (expects (1,D) input) to accept (D,)
input."""
    def wrapper(x):
        val = fn(x.reshape(1, -1))
        if hasattr(val, '__len__'):
            val = float(val[0])
        else:
            val = float(val)
        return val if np.isfinite(val) else 1e18
    return wrapper
```

Define `run_and_plot` to execute a single LBLDE optimization run on a given benchmark function, print the results, store them for the summary table, and save the convergence graph.

```
def run_and_plot(fn_name, fn_eval, f_global, fn_label):
    """Run LBLDE on a function and plot convergence graph."""
    print('=' * 80)
    print(f'LBLDE on CEC 2017 {fn_name} — Single Run')
    print('=' * 80)
    print(f'Function       : {fn_label} ({D}D)')
    print(f'Bounds         : [-100, 100]')
    print(f'Global optimum : {f_global}')

    opt = LBLDE(
        objective_func=fn_eval,
        bounds=bounds,
        NP=NP, NL=NL, mu_CR_ini=MU_CR,
        max_fes=MAX_FES,
        seed=SEED
    )
    best_sol, best_fit, history = opt.optimize(verbose=True)

    error = abs(best_fit - f_global)
    print(f'\n{"="*80}')
    print('Results:')
```

```
    print(f'  Best Fitness           : {best_fit:.6e}')
    print(f'  Global Optimum         : {f_global:.6e}')
    print(f'  Error (|best-optimum|) : {error:.6e}')
    print(f'  Best Solution (5 dim)  : {best_sol[:5]}')
    print(f'  D^(0) (init. diversity): {opt.D_0:.4e}')
    print(f'{"="*80}')

    # Store for summary
    all_results[fn_name] = {
        'label': fn_label,
        'best_fit': best_fit,
        'f_global': f_global,
        'error': error
    }

    # Plot
    fes_axis = np.linspace(0, MAX_FES, len(history))
    plt.figure(figsize=(10, 6))
    plt.plot(fes_axis, history, color='steelblue', linewidth=1.5)
    plt.xlabel('Function Evaluations (FEs)', fontsize=12)
    plt.ylabel('Best Fitness', fontsize=12)
    plt.title(f'LBLDE Convergence — CEC 2017 {fn_name}: {fn_label}
({D}D)', fontsize=12)
    plt.grid(True, which='both', linestyle='--', alpha=0.6)
    plt.tight_layout()
    fname = f'convergence_{fn_name}.png'
    plt.savefig(fname, dpi=150)
    plt.show()
    print(f'Graph saved as {fname}\n')

print('All wrappers and helpers defined. Ready to run functions.')
```

```
All wrappers and helpers defined. Ready to run functions.
```

Section 6. CEC2017 Benchmark Functions

6.1 Unimodal Functions (F1-F3)

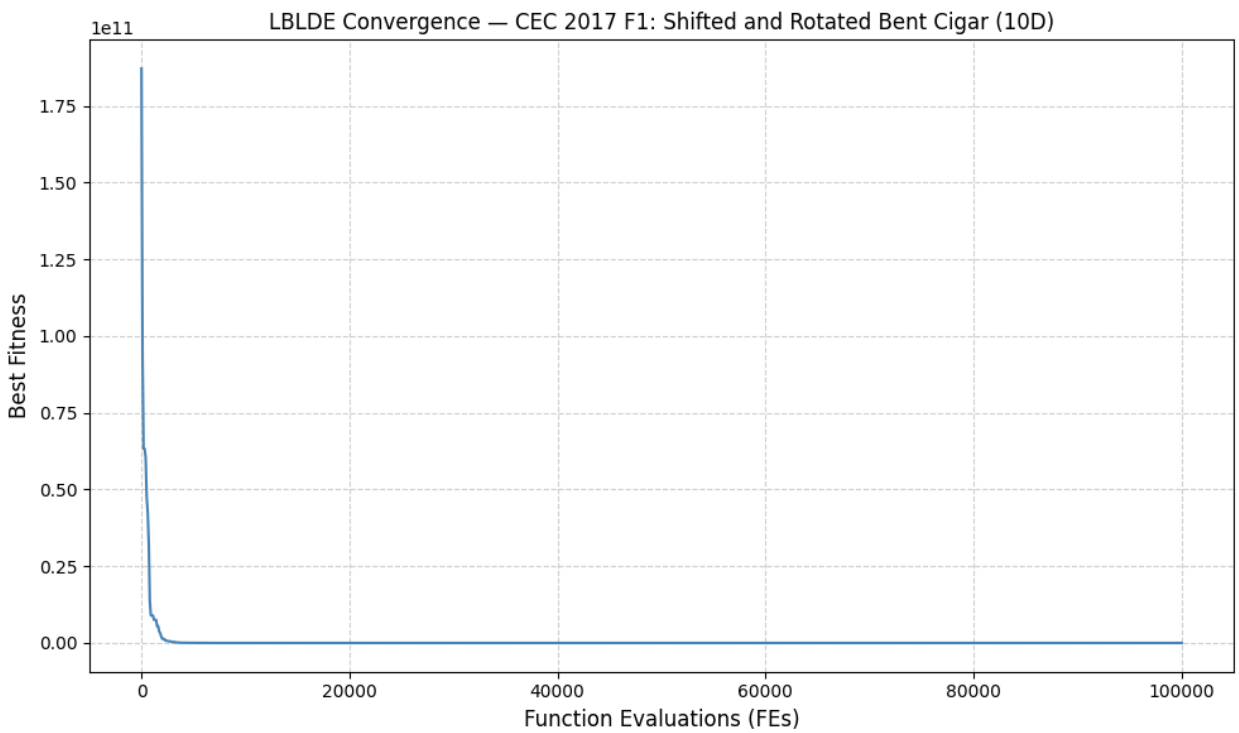
```
# ==============================================================
# F1 — Shifted and Rotated Bent Cigar
# ==============================================================
run_and_plot('F1', make_wrapper(simple.f1), 100.0, 'Shifted and
Rotated Bent Cigar')
```

```
================================================================================
LBLDE on CEC 2017 F1 — Single Run
```

```
==========================================================================
==========
Function       : Shifted and Rotated Bent Cigar (10D)
Bounds         : [-100, 100]
Global optimum : 100.0
Generation 0: Best Fitness = 1.871674e+11
Generation 10: Best Fitness = 8.959256e+09, FES = 1100
Generation 20: Best Fitness = 1.333767e+09, FES = 2100
Generation 30: Best Fitness = 3.339968e+08, FES = 3100
Generation 40: Best Fitness = 8.271954e+07, FES = 4100
Generation 50: Best Fitness = 2.325592e+07, FES = 5100
Generation 60: Best Fitness = 5.531678e+06, FES = 6100
Generation 70: Best Fitness = 1.592522e+06, FES = 7100
Generation 80: Best Fitness = 5.501703e+05, FES = 8100
Generation 90: Best Fitness = 2.206438e+05, FES = 9100
Generation 100: Best Fitness = 8.091999e+04, FES = 10100
Generation 110: Best Fitness = 3.877696e+04, FES = 11100
Generation 120: Best Fitness = 1.181221e+04, FES = 12100
Generation 130: Best Fitness = 3.051162e+03, FES = 13100
Generation 140: Best Fitness = 1.568496e+03, FES = 14100
Generation 150: Best Fitness = 6.579976e+02, FES = 15100
Generation 160: Best Fitness = 2.003476e+02, FES = 16100
Generation 170: Best Fitness = 1.549128e+02, FES = 17100
Generation 180: Best Fitness = 1.365352e+02, FES = 18100
Generation 190: Best Fitness = 1.124532e+02, FES = 19100
Generation 200: Best Fitness = 1.089062e+02, FES = 20100
Generation 210: Best Fitness = 1.033126e+02, FES = 21100
Generation 220: Best Fitness = 1.008689e+02, FES = 22100
Generation 230: Best Fitness = 1.003780e+02, FES = 23100
Generation 240: Best Fitness = 1.002092e+02, FES = 24100
Generation 250: Best Fitness = 1.000812e+02, FES = 25100
Generation 260: Best Fitness = 1.000282e+02, FES = 26100
Generation 270: Best Fitness = 1.000120e+02, FES = 27100
Generation 280: Best Fitness = 1.000055e+02, FES = 28100
Generation 290: Best Fitness = 1.000013e+02, FES = 29100
Generation 300: Best Fitness = 1.000012e+02, FES = 30100
Generation 310: Best Fitness = 1.000003e+02, FES = 31100
Generation 320: Best Fitness = 1.000002e+02, FES = 32100
Generation 330: Best Fitness = 1.000001e+02, FES = 33100
Generation 340: Best Fitness = 1.000000e+02, FES = 34100
Generation 350: Best Fitness = 1.000000e+02, FES = 35100
Generation 360: Best Fitness = 1.000000e+02, FES = 36100
Generation 370: Best Fitness = 1.000000e+02, FES = 37100
Generation 380: Best Fitness = 1.000000e+02, FES = 38100
Generation 390: Best Fitness = 1.000000e+02, FES = 39100
Generation 400: Best Fitness = 1.000000e+02, FES = 40100
Generation 410: Best Fitness = 1.000000e+02, FES = 41100
Generation 420: Best Fitness = 1.000000e+02, FES = 42100
Generation 430: Best Fitness = 1.000000e+02, FES = 43100
```

```
Generation 440: Best Fitness = 1.000000e+02, FES = 44100
Generation 450: Best Fitness = 1.000000e+02, FES = 45100
Generation 460: Best Fitness = 1.000000e+02, FES = 46100
Generation 470: Best Fitness = 1.000000e+02, FES = 47100
Generation 480: Best Fitness = 1.000000e+02, FES = 48100
Generation 490: Best Fitness = 1.000000e+02, FES = 49100
Generation 500: Best Fitness = 1.000000e+02, FES = 50100
Generation 510: Best Fitness = 1.000000e+02, FES = 51100
Generation 520: Best Fitness = 1.000000e+02, FES = 52100
Generation 530: Best Fitness = 1.000000e+02, FES = 53100
Generation 540: Best Fitness = 1.000000e+02, FES = 54100
Generation 550: Best Fitness = 1.000000e+02, FES = 55100
Generation 560: Best Fitness = 1.000000e+02, FES = 56100
Generation 570: Best Fitness = 1.000000e+02, FES = 57100
Generation 580: Best Fitness = 1.000000e+02, FES = 58100
Generation 590: Best Fitness = 1.000000e+02, FES = 59100
Generation 600: Best Fitness = 1.000000e+02, FES = 60100
Generation 610: Best Fitness = 1.000000e+02, FES = 61100
Generation 620: Best Fitness = 1.000000e+02, FES = 62100
Generation 630: Best Fitness = 1.000000e+02, FES = 63100
Generation 640: Best Fitness = 1.000000e+02, FES = 64100
Generation 650: Best Fitness = 1.000000e+02, FES = 65100
Generation 660: Best Fitness = 1.000000e+02, FES = 66100
Generation 670: Best Fitness = 1.000000e+02, FES = 67100
Generation 680: Best Fitness = 1.000000e+02, FES = 68100
Generation 690: Best Fitness = 1.000000e+02, FES = 69100
Generation 700: Best Fitness = 1.000000e+02, FES = 70100
Generation 710: Best Fitness = 1.000000e+02, FES = 71100
Generation 720: Best Fitness = 1.000000e+02, FES = 72100
Generation 730: Best Fitness = 1.000000e+02, FES = 73100
Generation 740: Best Fitness = 1.000000e+02, FES = 74100
Generation 750: Best Fitness = 1.000000e+02, FES = 75100
Generation 760: Best Fitness = 1.000000e+02, FES = 76100
Generation 770: Best Fitness = 1.000000e+02, FES = 77100
Generation 780: Best Fitness = 1.000000e+02, FES = 78100
Generation 790: Best Fitness = 1.000000e+02, FES = 79100
Generation 800: Best Fitness = 1.000000e+02, FES = 80100
Generation 810: Best Fitness = 1.000000e+02, FES = 81100
Generation 820: Best Fitness = 1.000000e+02, FES = 82100
Generation 830: Best Fitness = 1.000000e+02, FES = 83100
Generation 840: Best Fitness = 1.000000e+02, FES = 84100
Generation 850: Best Fitness = 1.000000e+02, FES = 85100
Generation 860: Best Fitness = 1.000000e+02, FES = 86100
Generation 870: Best Fitness = 1.000000e+02, FES = 87100
Generation 880: Best Fitness = 1.000000e+02, FES = 88100
Generation 890: Best Fitness = 1.000000e+02, FES = 89100
Generation 900: Best Fitness = 1.000000e+02, FES = 90100
Generation 910: Best Fitness = 1.000000e+02, FES = 91100
Generation 920: Best Fitness = 1.000000e+02, FES = 92100
```

```
Generation 930: Best Fitness = 1.000000e+02, FES = 93100
Generation 940: Best Fitness = 1.000000e+02, FES = 94100
Generation 950: Best Fitness = 1.000000e+02, FES = 95100
Generation 960: Best Fitness = 1.000000e+02, FES = 96100
Generation 970: Best Fitness = 1.000000e+02, FES = 97100
Generation 980: Best Fitness = 1.000000e+02, FES = 98100
Generation 990: Best Fitness = 1.000000e+02, FES = 99100

Optimization completed!
Final Best Fitness = 1.000000e+02
Total Function Evaluations = 100000

================================================================================
Results:
  Best Fitness           : 1.000000e+02
  Global Optimum         : 1.000000e+02
  Error (|best-optimum|) : 0.000000e+00
  Best Solution (5 dim)  : [-55.27639851 -70.42955972 -29.61018187 -58.32676328  22.08960188]
  D^(0) (init. diversity): 0.0000e+00
================================================================================
```



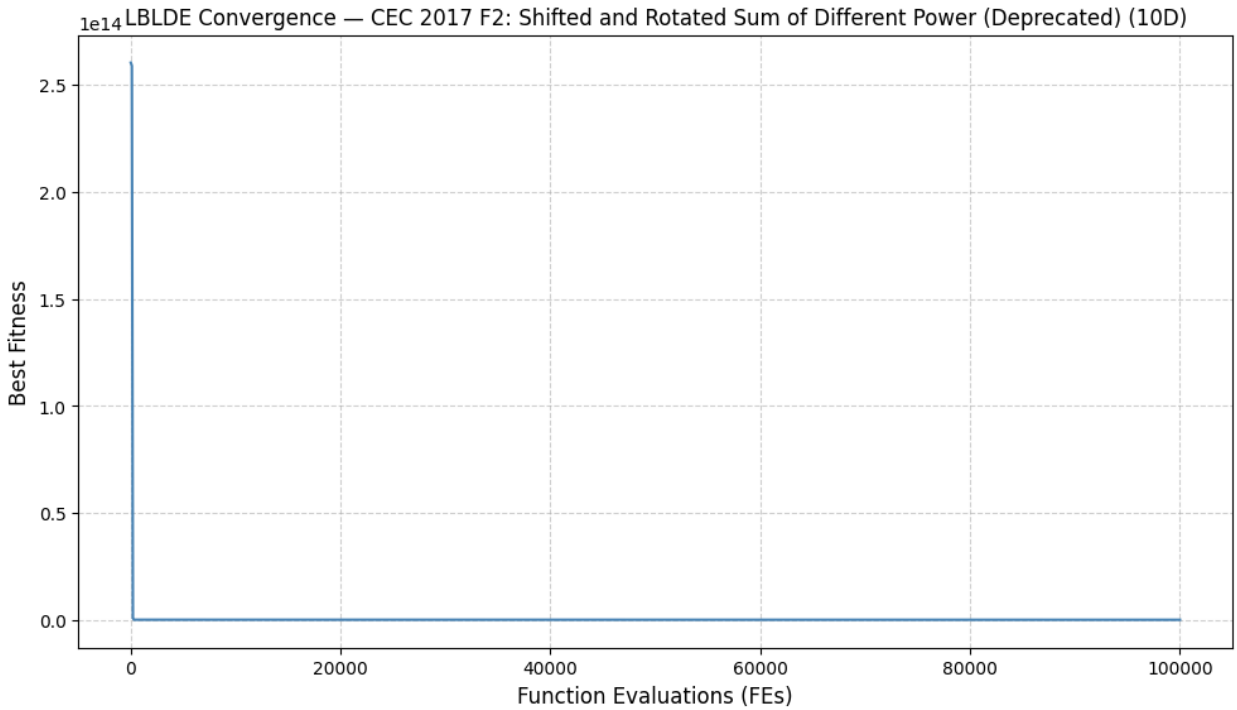

```
Graph saved as convergence_F1.png
```

```
# ==============================================================
# F2 — Shifted and Rotated Sum of Different Power
# ==============================================================
run_and_plot('F2', make_wrapper(simple.f2), 200.0, 'Shifted and
Rotated Sum of Different Power (Deprecated)')
```

```
=========================================================================
==========
LBLDE on CEC 2017 F2 — Single Run
=========================================================================
==========
Function       : Shifted and Rotated Sum of Different Power
(Deprecated) (10D)
Bounds         : [-100, 100]
Global optimum : 200.0
WARNING: f2 has been deprecated from the CEC 2017 benchmark suite
Generation 0: Best Fitness = 2.603565e+14
Generation 10: Best Fitness = 5.753243e+09, FES = 1100
Generation 20: Best Fitness = 2.839999e+07, FES = 2100
Generation 30: Best Fitness = 3.919296e+05, FES = 3100
Generation 40: Best Fitness = 2.283454e+04, FES = 4100
Generation 50: Best Fitness = 4.411867e+03, FES = 5100
Generation 60: Best Fitness = 7.655686e+02, FES = 6100
Generation 70: Best Fitness = 3.632613e+02, FES = 7100
Generation 80: Best Fitness = 2.127564e+02, FES = 8100
Generation 90: Best Fitness = 2.067742e+02, FES = 9100
Generation 100: Best Fitness = 2.008420e+02, FES = 10100
Generation 110: Best Fitness = 2.000087e+02, FES = 11100
Generation 120: Best Fitness = 2.000060e+02, FES = 12100
Generation 130: Best Fitness = 2.000011e+02, FES = 13100
Generation 140: Best Fitness = 2.000003e+02, FES = 14100
Generation 150: Best Fitness = 2.000001e+02, FES = 15100
Generation 160: Best Fitness = 2.000000e+02, FES = 16100
Generation 170: Best Fitness = 2.000000e+02, FES = 17100
Generation 180: Best Fitness = 2.000000e+02, FES = 18100
Generation 190: Best Fitness = 2.000000e+02, FES = 19100
Generation 200: Best Fitness = 2.000000e+02, FES = 20100
Generation 210: Best Fitness = 2.000000e+02, FES = 21100
Generation 220: Best Fitness = 2.000000e+02, FES = 22100
Generation 230: Best Fitness = 2.000000e+02, FES = 23100
Generation 240: Best Fitness = 2.000000e+02, FES = 24100
Generation 250: Best Fitness = 2.000000e+02, FES = 25100
Generation 260: Best Fitness = 2.000000e+02, FES = 26100
Generation 270: Best Fitness = 2.000000e+02, FES = 27100
Generation 280: Best Fitness = 2.000000e+02, FES = 28100
Generation 290: Best Fitness = 2.000000e+02, FES = 29100
Generation 300: Best Fitness = 2.000000e+02, FES = 30100
Generation 310: Best Fitness = 2.000000e+02, FES = 31100
Generation 320: Best Fitness = 2.000000e+02, FES = 32100
Generation 330: Best Fitness = 2.000000e+02, FES = 33100
```

```
Generation 340: Best Fitness = 2.000000e+02, FES = 34100
Generation 350: Best Fitness = 2.000000e+02, FES = 35100
Generation 360: Best Fitness = 2.000000e+02, FES = 36100
Generation 370: Best Fitness = 2.000000e+02, FES = 37100
Generation 380: Best Fitness = 2.000000e+02, FES = 38100
Generation 390: Best Fitness = 2.000000e+02, FES = 39100
Generation 400: Best Fitness = 2.000000e+02, FES = 40100
Generation 410: Best Fitness = 2.000000e+02, FES = 41100
Generation 420: Best Fitness = 2.000000e+02, FES = 42100
Generation 430: Best Fitness = 2.000000e+02, FES = 43100
Generation 440: Best Fitness = 2.000000e+02, FES = 44100
Generation 450: Best Fitness = 2.000000e+02, FES = 45100
Generation 460: Best Fitness = 2.000000e+02, FES = 46100
Generation 470: Best Fitness = 2.000000e+02, FES = 47100
Generation 480: Best Fitness = 2.000000e+02, FES = 48100
Generation 490: Best Fitness = 2.000000e+02, FES = 49100
Generation 500: Best Fitness = 2.000000e+02, FES = 50100
Generation 510: Best Fitness = 2.000000e+02, FES = 51100
Generation 520: Best Fitness = 2.000000e+02, FES = 52100
Generation 530: Best Fitness = 2.000000e+02, FES = 53100
Generation 540: Best Fitness = 2.000000e+02, FES = 54100
Generation 550: Best Fitness = 2.000000e+02, FES = 55100
Generation 560: Best Fitness = 2.000000e+02, FES = 56100
Generation 570: Best Fitness = 2.000000e+02, FES = 57100
Generation 580: Best Fitness = 2.000000e+02, FES = 58100
Generation 590: Best Fitness = 2.000000e+02, FES = 59100
Generation 600: Best Fitness = 2.000000e+02, FES = 60100
Generation 610: Best Fitness = 2.000000e+02, FES = 61100
Generation 620: Best Fitness = 2.000000e+02, FES = 62100
Generation 630: Best Fitness = 2.000000e+02, FES = 63100
Generation 640: Best Fitness = 2.000000e+02, FES = 64100
Generation 650: Best Fitness = 2.000000e+02, FES = 65100
Generation 660: Best Fitness = 2.000000e+02, FES = 66100
Generation 670: Best Fitness = 2.000000e+02, FES = 67100
Generation 680: Best Fitness = 2.000000e+02, FES = 68100
Generation 690: Best Fitness = 2.000000e+02, FES = 69100
Generation 700: Best Fitness = 2.000000e+02, FES = 70100
Generation 710: Best Fitness = 2.000000e+02, FES = 71100
Generation 720: Best Fitness = 2.000000e+02, FES = 72100
Generation 730: Best Fitness = 2.000000e+02, FES = 73100
Generation 740: Best Fitness = 2.000000e+02, FES = 74100
Generation 750: Best Fitness = 2.000000e+02, FES = 75100
Generation 760: Best Fitness = 2.000000e+02, FES = 76100
Generation 770: Best Fitness = 2.000000e+02, FES = 77100
Generation 780: Best Fitness = 2.000000e+02, FES = 78100
Generation 790: Best Fitness = 2.000000e+02, FES = 79100
Generation 800: Best Fitness = 2.000000e+02, FES = 80100
Generation 810: Best Fitness = 2.000000e+02, FES = 81100
Generation 820: Best Fitness = 2.000000e+02, FES = 82100
```

```
Generation 830: Best Fitness = 2.000000e+02, FES = 83100
Generation 840: Best Fitness = 2.000000e+02, FES = 84100
Generation 850: Best Fitness = 2.000000e+02, FES = 85100
Generation 860: Best Fitness = 2.000000e+02, FES = 86100
Generation 870: Best Fitness = 2.000000e+02, FES = 87100
Generation 880: Best Fitness = 2.000000e+02, FES = 88100
Generation 890: Best Fitness = 2.000000e+02, FES = 89100
Generation 900: Best Fitness = 2.000000e+02, FES = 90100
Generation 910: Best Fitness = 2.000000e+02, FES = 91100
Generation 920: Best Fitness = 2.000000e+02, FES = 92100
Generation 930: Best Fitness = 2.000000e+02, FES = 93100
Generation 940: Best Fitness = 2.000000e+02, FES = 94100
Generation 950: Best Fitness = 2.000000e+02, FES = 95100
Generation 960: Best Fitness = 2.000000e+02, FES = 96100
Generation 970: Best Fitness = 2.000000e+02, FES = 97100
Generation 980: Best Fitness = 2.000000e+02, FES = 98100
Generation 990: Best Fitness = 2.000000e+02, FES = 99100

Optimization completed!
Final Best Fitness = 2.000000e+02
Total Function Evaluations = 100000

=====================================================================
==========
Results:
  Best Fitness           : 2.000000e+02
  Global Optimum         : 2.000000e+02
  Error (|best-optimum|) : 0.000000e+00
  Best Solution (5 dim)  : [-29.3408606  -17.04649484 -75.4827044   -
0.67549881   4.25250816]
  D^(0) (init. diversity): 0.0000e+00
=====================================================================
==========
```

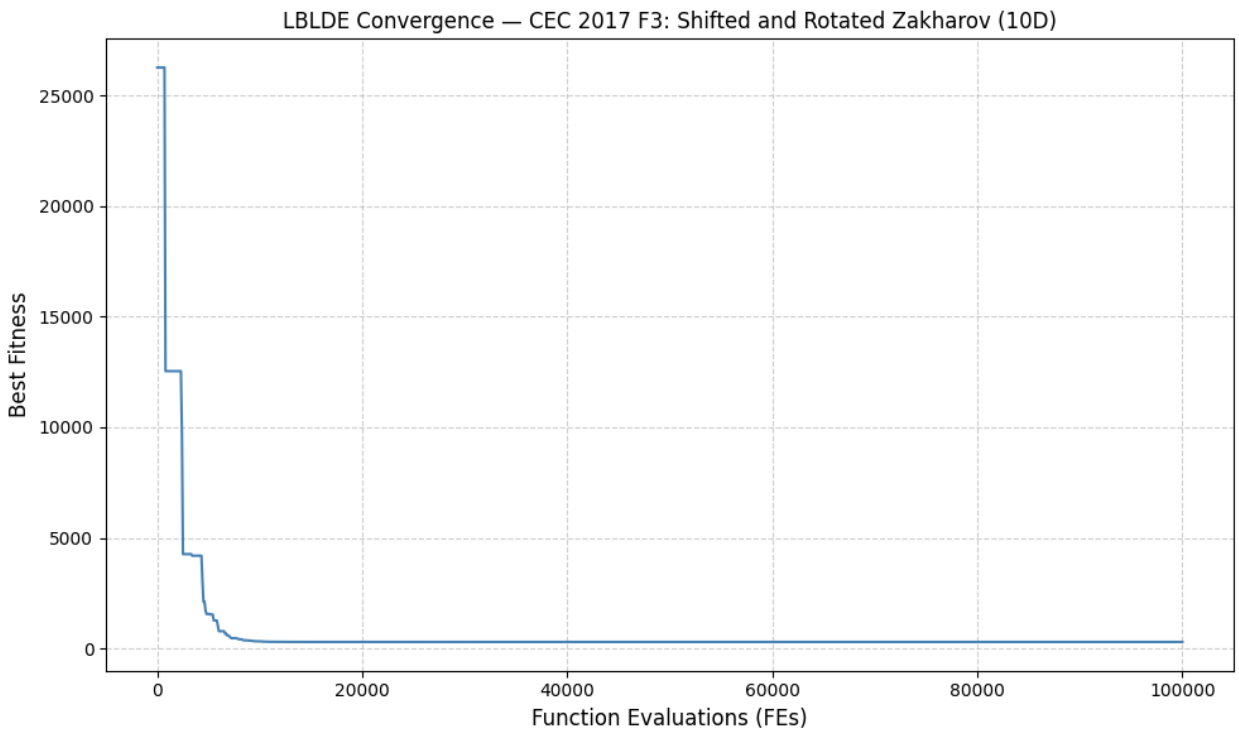



```
Graph saved as convergence_F2.png


# ============================================================
# F3 — Shifted and Rotated Zakharov
# ============================================================
run_and_plot('F3', make_wrapper(simple.f3), 300.0, 'Shifted and
Rotated Zakharov')

======================================================================
==========
LBLDE on CEC 2017 F3 — Single Run
======================================================================
==========
Function       : Shifted and Rotated Zakharov (10D)
Bounds         : [-100, 100]
Global optimum : 300.0
Generation 0: Best Fitness = 2.625937e+04
Generation 10: Best Fitness = 1.253936e+04, FES = 1100
Generation 20: Best Fitness = 1.253936e+04, FES = 2100
Generation 30: Best Fitness = 4.269488e+03, FES = 3100
Generation 40: Best Fitness = 4.197544e+03, FES = 4100
Generation 50: Best Fitness = 1.561216e+03, FES = 5100
Generation 60: Best Fitness = 7.862199e+02, FES = 6100
Generation 70: Best Fitness = 5.720962e+02, FES = 7100
Generation 80: Best Fitness = 4.191860e+02, FES = 8100
Generation 90: Best Fitness = 3.594736e+02, FES = 9100
Generation 100: Best Fitness = 3.266259e+02, FES = 10100
```

```
Generation 110: Best Fitness = 3.076009e+02, FES = 11100
Generation 120: Best Fitness = 3.030825e+02, FES = 12100
Generation 130: Best Fitness = 3.013697e+02, FES = 13100
Generation 140: Best Fitness = 3.004407e+02, FES = 14100
Generation 150: Best Fitness = 3.001708e+02, FES = 15100
Generation 160: Best Fitness = 3.000466e+02, FES = 16100
Generation 170: Best Fitness = 3.000109e+02, FES = 17100
Generation 180: Best Fitness = 3.000050e+02, FES = 18100
Generation 190: Best Fitness = 3.000015e+02, FES = 19100
Generation 200: Best Fitness = 3.000004e+02, FES = 20100
Generation 210: Best Fitness = 3.000001e+02, FES = 21100
Generation 220: Best Fitness = 3.000001e+02, FES = 22100
Generation 230: Best Fitness = 3.000000e+02, FES = 23100
Generation 240: Best Fitness = 3.000000e+02, FES = 24100
Generation 250: Best Fitness = 3.000000e+02, FES = 25100
Generation 260: Best Fitness = 3.000000e+02, FES = 26100
Generation 270: Best Fitness = 3.000000e+02, FES = 27100
Generation 280: Best Fitness = 3.000000e+02, FES = 28100
Generation 290: Best Fitness = 3.000000e+02, FES = 29100
Generation 300: Best Fitness = 3.000000e+02, FES = 30100
Generation 310: Best Fitness = 3.000000e+02, FES = 31100
Generation 320: Best Fitness = 3.000000e+02, FES = 32100
Generation 330: Best Fitness = 3.000000e+02, FES = 33100
Generation 340: Best Fitness = 3.000000e+02, FES = 34100
Generation 350: Best Fitness = 3.000000e+02, FES = 35100
Generation 360: Best Fitness = 3.000000e+02, FES = 36100
Generation 370: Best Fitness = 3.000000e+02, FES = 37100
Generation 380: Best Fitness = 3.000000e+02, FES = 38100
Generation 390: Best Fitness = 3.000000e+02, FES = 39100
Generation 400: Best Fitness = 3.000000e+02, FES = 40100
Generation 410: Best Fitness = 3.000000e+02, FES = 41100
Generation 420: Best Fitness = 3.000000e+02, FES = 42100
Generation 430: Best Fitness = 3.000000e+02, FES = 43100
Generation 440: Best Fitness = 3.000000e+02, FES = 44100
Generation 450: Best Fitness = 3.000000e+02, FES = 45100
Generation 460: Best Fitness = 3.000000e+02, FES = 46100
Generation 470: Best Fitness = 3.000000e+02, FES = 47100
Generation 480: Best Fitness = 3.000000e+02, FES = 48100
Generation 490: Best Fitness = 3.000000e+02, FES = 49100
Generation 500: Best Fitness = 3.000000e+02, FES = 50100
Generation 510: Best Fitness = 3.000000e+02, FES = 51100
Generation 520: Best Fitness = 3.000000e+02, FES = 52100
Generation 530: Best Fitness = 3.000000e+02, FES = 53100
Generation 540: Best Fitness = 3.000000e+02, FES = 54100
Generation 550: Best Fitness = 3.000000e+02, FES = 55100
Generation 560: Best Fitness = 3.000000e+02, FES = 56100
Generation 570: Best Fitness = 3.000000e+02, FES = 57100
Generation 580: Best Fitness = 3.000000e+02, FES = 58100
Generation 590: Best Fitness = 3.000000e+02, FES = 59100
```

```
Generation 600: Best Fitness = 3.000000e+02, FES = 60100
Generation 610: Best Fitness = 3.000000e+02, FES = 61100
Generation 620: Best Fitness = 3.000000e+02, FES = 62100
Generation 630: Best Fitness = 3.000000e+02, FES = 63100
Generation 640: Best Fitness = 3.000000e+02, FES = 64100
Generation 650: Best Fitness = 3.000000e+02, FES = 65100
Generation 660: Best Fitness = 3.000000e+02, FES = 66100
Generation 670: Best Fitness = 3.000000e+02, FES = 67100
Generation 680: Best Fitness = 3.000000e+02, FES = 68100
Generation 690: Best Fitness = 3.000000e+02, FES = 69100
Generation 700: Best Fitness = 3.000000e+02, FES = 70100
Generation 710: Best Fitness = 3.000000e+02, FES = 71100
Generation 720: Best Fitness = 3.000000e+02, FES = 72100
Generation 730: Best Fitness = 3.000000e+02, FES = 73100
Generation 740: Best Fitness = 3.000000e+02, FES = 74100
Generation 750: Best Fitness = 3.000000e+02, FES = 75100
Generation 760: Best Fitness = 3.000000e+02, FES = 76100
Generation 770: Best Fitness = 3.000000e+02, FES = 77100
Generation 780: Best Fitness = 3.000000e+02, FES = 78100
Generation 790: Best Fitness = 3.000000e+02, FES = 79100
Generation 800: Best Fitness = 3.000000e+02, FES = 80100
Generation 810: Best Fitness = 3.000000e+02, FES = 81100
Generation 820: Best Fitness = 3.000000e+02, FES = 82100
Generation 830: Best Fitness = 3.000000e+02, FES = 83100
Generation 840: Best Fitness = 3.000000e+02, FES = 84100
Generation 850: Best Fitness = 3.000000e+02, FES = 85100
Generation 860: Best Fitness = 3.000000e+02, FES = 86100
Generation 870: Best Fitness = 3.000000e+02, FES = 87100
Generation 880: Best Fitness = 3.000000e+02, FES = 88100
Generation 890: Best Fitness = 3.000000e+02, FES = 89100
Generation 900: Best Fitness = 3.000000e+02, FES = 90100
Generation 910: Best Fitness = 3.000000e+02, FES = 91100
Generation 920: Best Fitness = 3.000000e+02, FES = 92100
Generation 930: Best Fitness = 3.000000e+02, FES = 93100
Generation 940: Best Fitness = 3.000000e+02, FES = 94100
Generation 950: Best Fitness = 3.000000e+02, FES = 95100
Generation 960: Best Fitness = 3.000000e+02, FES = 96100
Generation 970: Best Fitness = 3.000000e+02, FES = 97100
Generation 980: Best Fitness = 3.000000e+02, FES = 98100
Generation 990: Best Fitness = 3.000000e+02, FES = 99100

Optimization completed!
Final Best Fitness = 3.000000e+02
Total Function Evaluations = 100000

=====================================================================
==========
Results:
  Best Fitness           : 3.000000e+02
```

```
  Global Optimum         : 3.000000e+02
  Error (|best-optimum|) : 0.000000e+00
  Best Solution (5 dim)  : [-55.9383267    4.54306531  35.30007018
8.27944093 -47.43694287]
  D^(0) (init. diversity): 0.0000e+00
======================================================================
==========
```




```
Graph saved as convergence_F3.png
```

6.2 Multimodal Functions (F4-F10)

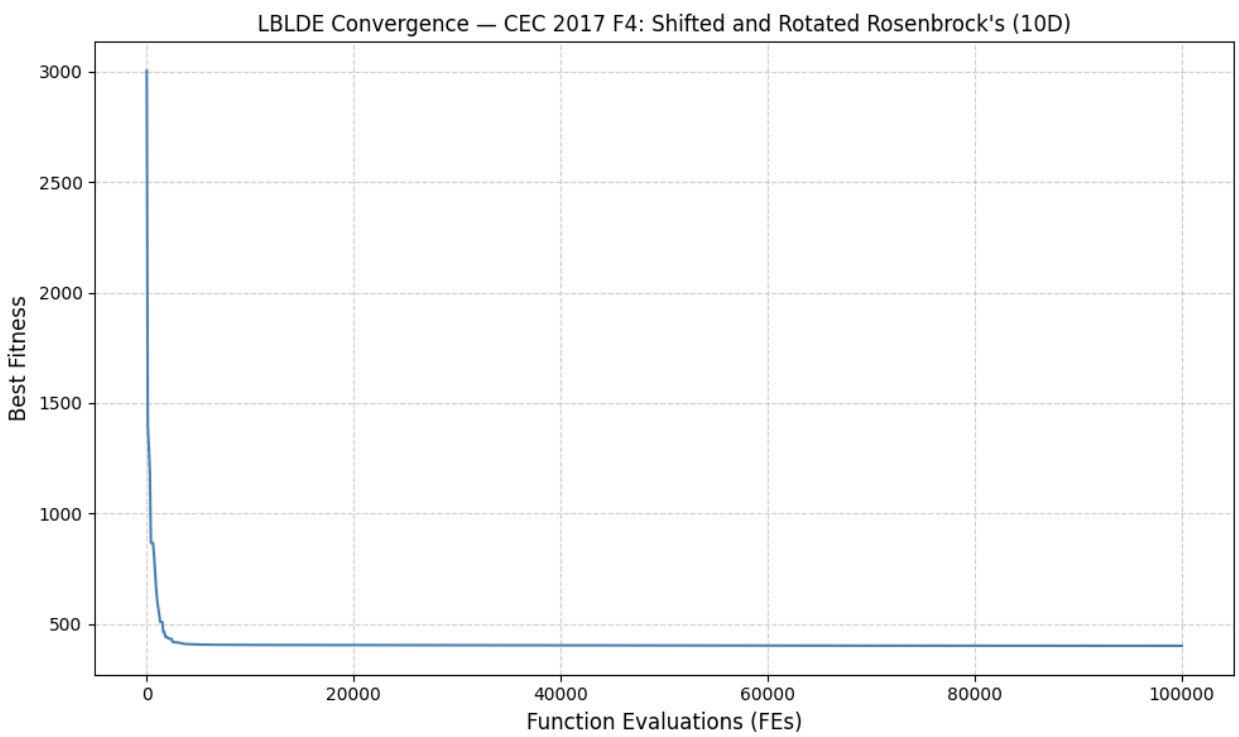
```
# ============================================================
# F4 — Shifted and Rotated Rosenbrock
# ============================================================
run_and_plot('F4', make_wrapper(simple.f4), 400.0, "Shifted and
Rotated Rosenbrock's")
```

```
======================================================================
==========
LBLDE on CEC 2017 F4 — Single Run
======================================================================
==========
Function       : Shifted and Rotated Rosenbrock's (10D)
Bounds         : [-100, 100]
```

```
Global optimum : 400.0
Generation 0: Best Fitness = 3.004346e+03
Generation 10: Best Fitness = 6.072825e+02, FES = 1100
Generation 20: Best Fitness = 4.365366e+02, FES = 2100
Generation 30: Best Fitness = 4.155978e+02, FES = 3100
Generation 40: Best Fitness = 4.079644e+02, FES = 4100
Generation 50: Best Fitness = 4.062035e+02, FES = 5100
Generation 60: Best Fitness = 4.053927e+02, FES = 6100
Generation 70: Best Fitness = 4.049429e+02, FES = 7100
Generation 80: Best Fitness = 4.047050e+02, FES = 8100
Generation 90: Best Fitness = 4.045633e+02, FES = 9100
Generation 100: Best Fitness = 4.044465e+02, FES = 10100
Generation 110: Best Fitness = 4.043073e+02, FES = 11100
Generation 120: Best Fitness = 4.042242e+02, FES = 12100
Generation 130: Best Fitness = 4.040990e+02, FES = 13100
Generation 140: Best Fitness = 4.040495e+02, FES = 14100
Generation 150: Best Fitness = 4.039706e+02, FES = 15100
Generation 160: Best Fitness = 4.039081e+02, FES = 16100
Generation 170: Best Fitness = 4.038634e+02, FES = 17100
Generation 180: Best Fitness = 4.038260e+02, FES = 18100
Generation 190: Best Fitness = 4.037777e+02, FES = 19100
Generation 200: Best Fitness = 4.037404e+02, FES = 20100
Generation 210: Best Fitness = 4.036892e+02, FES = 21100
Generation 220: Best Fitness = 4.036401e+02, FES = 22100
Generation 230: Best Fitness = 4.035908e+02, FES = 23100
Generation 240: Best Fitness = 4.035421e+02, FES = 24100
Generation 250: Best Fitness = 4.034975e+02, FES = 25100
Generation 260: Best Fitness = 4.034215e+02, FES = 26100
Generation 270: Best Fitness = 4.033418e+02, FES = 27100
Generation 280: Best Fitness = 4.032696e+02, FES = 28100
Generation 290: Best Fitness = 4.032064e+02, FES = 29100
Generation 300: Best Fitness = 4.031517e+02, FES = 30100
Generation 310: Best Fitness = 4.030892e+02, FES = 31100
Generation 320: Best Fitness = 4.030177e+02, FES = 32100
Generation 330: Best Fitness = 4.029280e+02, FES = 33100
Generation 340: Best Fitness = 4.028513e+02, FES = 34100
Generation 350: Best Fitness = 4.027940e+02, FES = 35100
Generation 360: Best Fitness = 4.027328e+02, FES = 36100
Generation 370: Best Fitness = 4.026935e+02, FES = 37100
Generation 380: Best Fitness = 4.026609e+02, FES = 38100
Generation 390: Best Fitness = 4.026130e+02, FES = 39100
Generation 400: Best Fitness = 4.025622e+02, FES = 40100
Generation 410: Best Fitness = 4.025091e+02, FES = 41100
Generation 420: Best Fitness = 4.024593e+02, FES = 42100
Generation 430: Best Fitness = 4.024182e+02, FES = 43100
Generation 440: Best Fitness = 4.023662e+02, FES = 44100
Generation 450: Best Fitness = 4.023209e+02, FES = 45100
Generation 460: Best Fitness = 4.022530e+02, FES = 46100
Generation 470: Best Fitness = 4.021899e+02, FES = 47100
```

```
Generation 480: Best Fitness = 4.020956e+02, FES = 48100
Generation 490: Best Fitness = 4.020057e+02, FES = 49100
Generation 500: Best Fitness = 4.019154e+02, FES = 50100
Generation 510: Best Fitness = 4.018556e+02, FES = 51100
Generation 520: Best Fitness = 4.017655e+02, FES = 52100
Generation 530: Best Fitness = 4.017330e+02, FES = 53100
Generation 540: Best Fitness = 4.016841e+02, FES = 54100
Generation 550: Best Fitness = 4.016463e+02, FES = 55100
Generation 560: Best Fitness = 4.016173e+02, FES = 56100
Generation 570: Best Fitness = 4.015847e+02, FES = 57100
Generation 580: Best Fitness = 4.015468e+02, FES = 58100
Generation 590: Best Fitness = 4.014928e+02, FES = 59100
Generation 600: Best Fitness = 4.014623e+02, FES = 60100
Generation 610: Best Fitness = 4.014297e+02, FES = 61100
Generation 620: Best Fitness = 4.014004e+02, FES = 62100
Generation 630: Best Fitness = 4.013744e+02, FES = 63100
Generation 640: Best Fitness = 4.013480e+02, FES = 64100
Generation 650: Best Fitness = 4.013113e+02, FES = 65100
Generation 660: Best Fitness = 4.012890e+02, FES = 66100
Generation 670: Best Fitness = 4.012609e+02, FES = 67100
Generation 680: Best Fitness = 4.012267e+02, FES = 68100
Generation 690: Best Fitness = 4.011998e+02, FES = 69100
Generation 700: Best Fitness = 4.011661e+02, FES = 70100
Generation 710: Best Fitness = 4.011447e+02, FES = 71100
Generation 720: Best Fitness = 4.011235e+02, FES = 72100
Generation 730: Best Fitness = 4.010988e+02, FES = 73100
Generation 740: Best Fitness = 4.010752e+02, FES = 74100
Generation 750: Best Fitness = 4.010455e+02, FES = 75100
Generation 760: Best Fitness = 4.010237e+02, FES = 76100
Generation 770: Best Fitness = 4.010048e+02, FES = 77100
Generation 780: Best Fitness = 4.009805e+02, FES = 78100
Generation 790: Best Fitness = 4.009618e+02, FES = 79100
Generation 800: Best Fitness = 4.009218e+02, FES = 80100
Generation 810: Best Fitness = 4.008711e+02, FES = 81100
Generation 820: Best Fitness = 4.008215e+02, FES = 82100
Generation 830: Best Fitness = 4.007737e+02, FES = 83100
Generation 840: Best Fitness = 4.007342e+02, FES = 84100
Generation 850: Best Fitness = 4.007094e+02, FES = 85100
Generation 860: Best Fitness = 4.006876e+02, FES = 86100
Generation 870: Best Fitness = 4.006742e+02, FES = 87100
Generation 880: Best Fitness = 4.006624e+02, FES = 88100
Generation 890: Best Fitness = 4.006469e+02, FES = 89100
Generation 900: Best Fitness = 4.006314e+02, FES = 90100
Generation 910: Best Fitness = 4.006161e+02, FES = 91100
Generation 920: Best Fitness = 4.006020e+02, FES = 92100
Generation 930: Best Fitness = 4.005904e+02, FES = 93100
Generation 940: Best Fitness = 4.005714e+02, FES = 94100
Generation 950: Best Fitness = 4.005503e+02, FES = 95100
Generation 960: Best Fitness = 4.005217e+02, FES = 96100
```

```
Generation 970: Best Fitness = 4.004969e+02, FES = 97100
Generation 980: Best Fitness = 4.004663e+02, FES = 98100
Generation 990: Best Fitness = 4.004402e+02, FES = 99100

Optimization completed!
Final Best Fitness = 4.004217e+02
Total Function Evaluations = 100000

======================================================================
==========
Results:
  Best Fitness           : 4.004217e+02
  Global Optimum         : 4.000000e+02
  Error (|best-optimum|) : 4.217135e-01
  Best Solution (5 dim)  : [ 47.18956019  17.04882326 -64.06688438
66.03425724  24.43446955]
  D^(0) (init. diversity): 0.0000e+00
======================================================================
==========
```



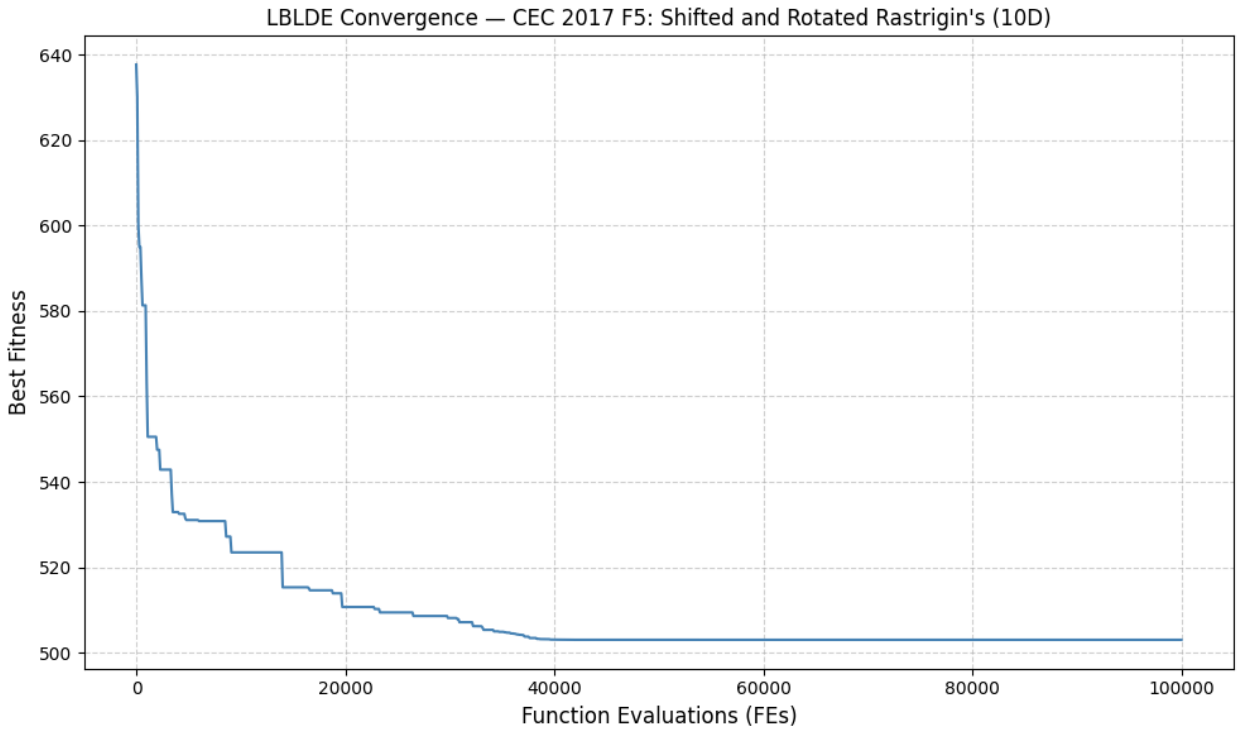

```
Graph saved as convergence_F4.png


# ============================================================
# F5 — Shifted and Rotated Rastrigin
# ============================================================
```

```
run_and_plot('F5', make_wrapper(simple.f5), 500.0, "Shifted and
Rotated Rastrigin's")
```

```
==========================================================================
==========
LBLDE on CEC 2017 F5 — Single Run
==========================================================================
==========
Function       : Shifted and Rotated Rastrigin's (10D)
Bounds         : [-100, 100]
Global optimum : 500.0
Generation 0: Best Fitness = 6.377184e+02
Generation 10: Best Fitness = 5.633495e+02, FES = 1100
Generation 20: Best Fitness = 5.475008e+02, FES = 2100
Generation 30: Best Fitness = 5.428465e+02, FES = 3100
Generation 40: Best Fitness = 5.329017e+02, FES = 4100
Generation 50: Best Fitness = 5.310467e+02, FES = 5100
Generation 60: Best Fitness = 5.307979e+02, FES = 6100
Generation 70: Best Fitness = 5.307979e+02, FES = 7100
Generation 80: Best Fitness = 5.307979e+02, FES = 8100
Generation 90: Best Fitness = 5.271632e+02, FES = 9100
Generation 100: Best Fitness = 5.234647e+02, FES = 10100
Generation 110: Best Fitness = 5.234647e+02, FES = 11100
Generation 120: Best Fitness = 5.234647e+02, FES = 12100
Generation 130: Best Fitness = 5.234647e+02, FES = 13100
Generation 140: Best Fitness = 5.152848e+02, FES = 14100
Generation 150: Best Fitness = 5.152848e+02, FES = 15100
Generation 160: Best Fitness = 5.152848e+02, FES = 16100
Generation 170: Best Fitness = 5.145850e+02, FES = 17100
Generation 180: Best Fitness = 5.145850e+02, FES = 18100
Generation 190: Best Fitness = 5.138741e+02, FES = 19100
Generation 200: Best Fitness = 5.106769e+02, FES = 20100
Generation 210: Best Fitness = 5.106769e+02, FES = 21100
Generation 220: Best Fitness = 5.106769e+02, FES = 22100
Generation 230: Best Fitness = 5.102088e+02, FES = 23100
Generation 240: Best Fitness = 5.093980e+02, FES = 24100
Generation 250: Best Fitness = 5.093980e+02, FES = 25100
Generation 260: Best Fitness = 5.093980e+02, FES = 26100
Generation 270: Best Fitness = 5.085689e+02, FES = 27100
Generation 280: Best Fitness = 5.085689e+02, FES = 28100
Generation 290: Best Fitness = 5.085689e+02, FES = 29100
Generation 300: Best Fitness = 5.080962e+02, FES = 30100
Generation 310: Best Fitness = 5.071064e+02, FES = 31100
Generation 320: Best Fitness = 5.071064e+02, FES = 32100
Generation 330: Best Fitness = 5.061637e+02, FES = 33100
Generation 340: Best Fitness = 5.053184e+02, FES = 34100
Generation 350: Best Fitness = 5.048488e+02, FES = 35100
Generation 360: Best Fitness = 5.044738e+02, FES = 36100
Generation 370: Best Fitness = 5.041126e+02, FES = 37100
Generation 380: Best Fitness = 5.034063e+02, FES = 38100
```

```
Generation 390: Best Fitness = 5.031508e+02, FES = 39100
Generation 400: Best Fitness = 5.030407e+02, FES = 40100
Generation 410: Best Fitness = 5.029993e+02, FES = 41100
Generation 420: Best Fitness = 5.029898e+02, FES = 42100
Generation 430: Best Fitness = 5.029859e+02, FES = 43100
Generation 440: Best Fitness = 5.029851e+02, FES = 44100
Generation 450: Best Fitness = 5.029849e+02, FES = 45100
Generation 460: Best Fitness = 5.029849e+02, FES = 46100
Generation 470: Best Fitness = 5.029849e+02, FES = 47100
Generation 480: Best Fitness = 5.029849e+02, FES = 48100
Generation 490: Best Fitness = 5.029849e+02, FES = 49100
Generation 500: Best Fitness = 5.029849e+02, FES = 50100
Generation 510: Best Fitness = 5.029849e+02, FES = 51100
Generation 520: Best Fitness = 5.029849e+02, FES = 52100
Generation 530: Best Fitness = 5.029849e+02, FES = 53100
Generation 540: Best Fitness = 5.029849e+02, FES = 54100
Generation 550: Best Fitness = 5.029849e+02, FES = 55100
Generation 560: Best Fitness = 5.029849e+02, FES = 56100
Generation 570: Best Fitness = 5.029849e+02, FES = 57100
Generation 580: Best Fitness = 5.029849e+02, FES = 58100
Generation 590: Best Fitness = 5.029849e+02, FES = 59100
Generation 600: Best Fitness = 5.029849e+02, FES = 60100
Generation 610: Best Fitness = 5.029849e+02, FES = 61100
Generation 620: Best Fitness = 5.029849e+02, FES = 62100
Generation 630: Best Fitness = 5.029849e+02, FES = 63100
Generation 640: Best Fitness = 5.029849e+02, FES = 64100
Generation 650: Best Fitness = 5.029849e+02, FES = 65100
Generation 660: Best Fitness = 5.029849e+02, FES = 66100
Generation 670: Best Fitness = 5.029849e+02, FES = 67100
Generation 680: Best Fitness = 5.029849e+02, FES = 68100
Generation 690: Best Fitness = 5.029849e+02, FES = 69100
Generation 700: Best Fitness = 5.029849e+02, FES = 70100
Generation 710: Best Fitness = 5.029849e+02, FES = 71100
Generation 720: Best Fitness = 5.029849e+02, FES = 72100
Generation 730: Best Fitness = 5.029849e+02, FES = 73100
Generation 740: Best Fitness = 5.029849e+02, FES = 74100
Generation 750: Best Fitness = 5.029849e+02, FES = 75100
Generation 760: Best Fitness = 5.029849e+02, FES = 76100
Generation 770: Best Fitness = 5.029849e+02, FES = 77100
Generation 780: Best Fitness = 5.029849e+02, FES = 78100
Generation 790: Best Fitness = 5.029849e+02, FES = 79100
Generation 800: Best Fitness = 5.029849e+02, FES = 80100
Generation 810: Best Fitness = 5.029849e+02, FES = 81100
Generation 820: Best Fitness = 5.029849e+02, FES = 82100
Generation 830: Best Fitness = 5.029849e+02, FES = 83100
Generation 840: Best Fitness = 5.029849e+02, FES = 84100
Generation 850: Best Fitness = 5.029849e+02, FES = 85100
Generation 860: Best Fitness = 5.029849e+02, FES = 86100
Generation 870: Best Fitness = 5.029849e+02, FES = 87100
```

```
Generation 880: Best Fitness = 5.029849e+02, FES = 88100
Generation 890: Best Fitness = 5.029849e+02, FES = 89100
Generation 900: Best Fitness = 5.029849e+02, FES = 90100
Generation 910: Best Fitness = 5.029849e+02, FES = 91100
Generation 920: Best Fitness = 5.029849e+02, FES = 92100
Generation 930: Best Fitness = 5.029849e+02, FES = 93100
Generation 940: Best Fitness = 5.029849e+02, FES = 94100
Generation 950: Best Fitness = 5.029849e+02, FES = 95100
Generation 960: Best Fitness = 5.029849e+02, FES = 96100
Generation 970: Best Fitness = 5.029849e+02, FES = 97100
Generation 980: Best Fitness = 5.029849e+02, FES = 98100
Generation 990: Best Fitness = 5.029849e+02, FES = 99100

Optimization completed!
Final Best Fitness = 5.029849e+02
Total Function Evaluations = 100000

==========================================================================
==========
Results:
  Best Fitness           : 5.029849e+02
  Global Optimum         : 5.000000e+02
  Error (|best-optimum|) : 2.984877e+00
  Best Solution (5 dim)  : [  4.76871232  67.32445466 -16.2787446  -
56.75631143  16.67395225]
  D^(0) (init. diversity): 0.0000e+00
==========================================================================
==========
```

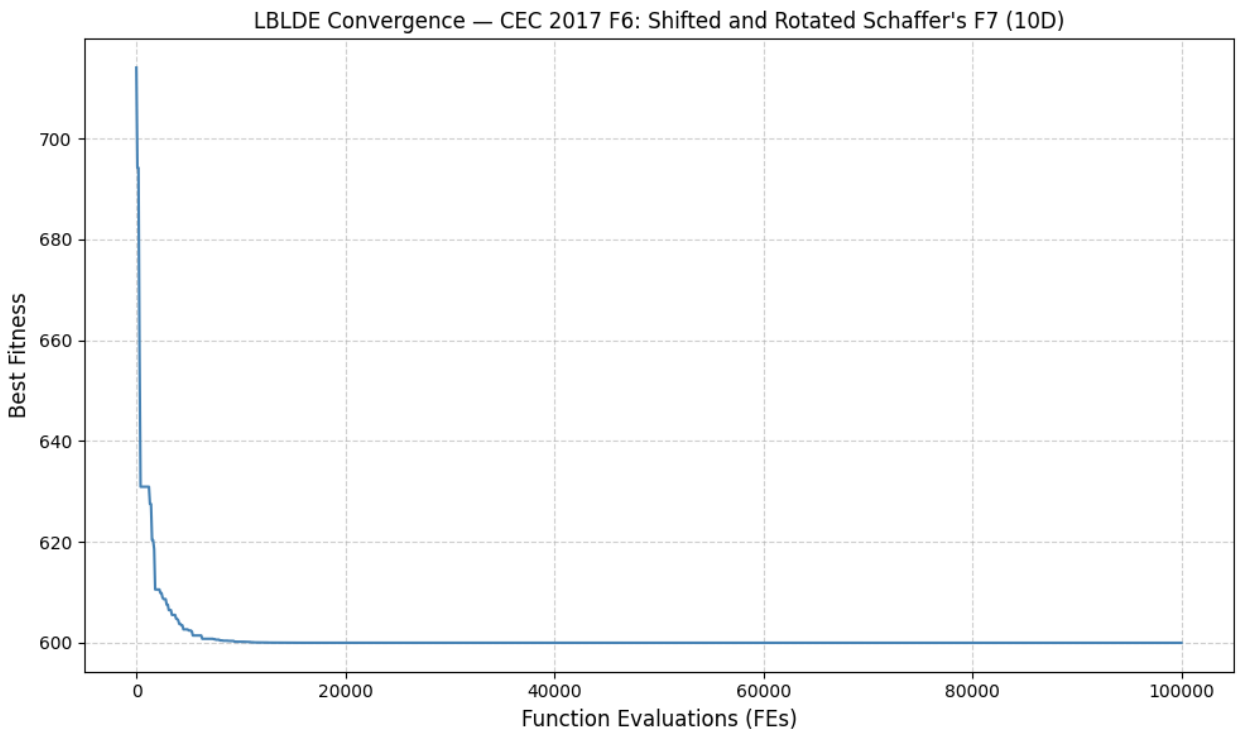



```
Graph saved as convergence_F5.png


# ============================================================
# F6 — Shifted and Rotated Schaffer's F7
# ============================================================
run_and_plot('F6', make_wrapper(simple.f6), 600.0, "Shifted and
Rotated Schaffer's F7")

======================================================================
==========
LBLDE on CEC 2017 F6 — Single Run
======================================================================
==========
Function       : Shifted and Rotated Schaffer's F7 (10D)
Bounds         : [-100, 100]
Global optimum : 600.0
Generation 0: Best Fitness = 7.140452e+02
Generation 10: Best Fitness = 6.309437e+02, FES = 1100
Generation 20: Best Fitness = 6.105705e+02, FES = 2100
Generation 30: Best Fitness = 6.075569e+02, FES = 3100
Generation 40: Best Fitness = 6.044679e+02, FES = 4100
Generation 50: Best Fitness = 6.024440e+02, FES = 5100
Generation 60: Best Fitness = 6.014529e+02, FES = 6100
Generation 70: Best Fitness = 6.007976e+02, FES = 7100
Generation 80: Best Fitness = 6.004775e+02, FES = 8100
Generation 90: Best Fitness = 6.003825e+02, FES = 9100
```

```
Generation 100: Best Fitness = 6.002205e+02, FES = 10100
Generation 110: Best Fitness = 6.001107e+02, FES = 11100
Generation 120: Best Fitness = 6.000680e+02, FES = 12100
Generation 130: Best Fitness = 6.000403e+02, FES = 13100
Generation 140: Best Fitness = 6.000252e+02, FES = 14100
Generation 150: Best Fitness = 6.000168e+02, FES = 15100
Generation 160: Best Fitness = 6.000085e+02, FES = 16100
Generation 170: Best Fitness = 6.000056e+02, FES = 17100
Generation 180: Best Fitness = 6.000035e+02, FES = 18100
Generation 190: Best Fitness = 6.000023e+02, FES = 19100
Generation 200: Best Fitness = 6.000013e+02, FES = 20100
Generation 210: Best Fitness = 6.000008e+02, FES = 21100
Generation 220: Best Fitness = 6.000007e+02, FES = 22100
Generation 230: Best Fitness = 6.000004e+02, FES = 23100
Generation 240: Best Fitness = 6.000002e+02, FES = 24100
Generation 250: Best Fitness = 6.000002e+02, FES = 25100
Generation 260: Best Fitness = 6.000001e+02, FES = 26100
Generation 270: Best Fitness = 6.000001e+02, FES = 27100
Generation 280: Best Fitness = 6.000001e+02, FES = 28100
Generation 290: Best Fitness = 6.000000e+02, FES = 29100
Generation 300: Best Fitness = 6.000000e+02, FES = 30100
Generation 310: Best Fitness = 6.000000e+02, FES = 31100
Generation 320: Best Fitness = 6.000000e+02, FES = 32100
Generation 330: Best Fitness = 6.000000e+02, FES = 33100
Generation 340: Best Fitness = 6.000000e+02, FES = 34100
Generation 350: Best Fitness = 6.000000e+02, FES = 35100
Generation 360: Best Fitness = 6.000000e+02, FES = 36100
Generation 370: Best Fitness = 6.000000e+02, FES = 37100
Generation 380: Best Fitness = 6.000000e+02, FES = 38100
Generation 390: Best Fitness = 6.000000e+02, FES = 39100
Generation 400: Best Fitness = 6.000000e+02, FES = 40100
Generation 410: Best Fitness = 6.000000e+02, FES = 41100
Generation 420: Best Fitness = 6.000000e+02, FES = 42100
Generation 430: Best Fitness = 6.000000e+02, FES = 43100
Generation 440: Best Fitness = 6.000000e+02, FES = 44100
Generation 450: Best Fitness = 6.000000e+02, FES = 45100
Generation 460: Best Fitness = 6.000000e+02, FES = 46100
Generation 470: Best Fitness = 6.000000e+02, FES = 47100
Generation 480: Best Fitness = 6.000000e+02, FES = 48100
Generation 490: Best Fitness = 6.000000e+02, FES = 49100
Generation 500: Best Fitness = 6.000000e+02, FES = 50100
Generation 510: Best Fitness = 6.000000e+02, FES = 51100
Generation 520: Best Fitness = 6.000000e+02, FES = 52100
Generation 530: Best Fitness = 6.000000e+02, FES = 53100
Generation 540: Best Fitness = 6.000000e+02, FES = 54100
Generation 550: Best Fitness = 6.000000e+02, FES = 55100
Generation 560: Best Fitness = 6.000000e+02, FES = 56100
Generation 570: Best Fitness = 6.000000e+02, FES = 57100
Generation 580: Best Fitness = 6.000000e+02, FES = 58100
```

```
Generation 590: Best Fitness = 6.000000e+02, FES = 59100
Generation 600: Best Fitness = 6.000000e+02, FES = 60100
Generation 610: Best Fitness = 6.000000e+02, FES = 61100
Generation 620: Best Fitness = 6.000000e+02, FES = 62100
Generation 630: Best Fitness = 6.000000e+02, FES = 63100
Generation 640: Best Fitness = 6.000000e+02, FES = 64100
Generation 650: Best Fitness = 6.000000e+02, FES = 65100
Generation 660: Best Fitness = 6.000000e+02, FES = 66100
Generation 670: Best Fitness = 6.000000e+02, FES = 67100
Generation 680: Best Fitness = 6.000000e+02, FES = 68100
Generation 690: Best Fitness = 6.000000e+02, FES = 69100
Generation 700: Best Fitness = 6.000000e+02, FES = 70100
Generation 710: Best Fitness = 6.000000e+02, FES = 71100
Generation 720: Best Fitness = 6.000000e+02, FES = 72100
Generation 730: Best Fitness = 6.000000e+02, FES = 73100
Generation 740: Best Fitness = 6.000000e+02, FES = 74100
Generation 750: Best Fitness = 6.000000e+02, FES = 75100
Generation 760: Best Fitness = 6.000000e+02, FES = 76100
Generation 770: Best Fitness = 6.000000e+02, FES = 77100
Generation 780: Best Fitness = 6.000000e+02, FES = 78100
Generation 790: Best Fitness = 6.000000e+02, FES = 79100
Generation 800: Best Fitness = 6.000000e+02, FES = 80100
Generation 810: Best Fitness = 6.000000e+02, FES = 81100
Generation 820: Best Fitness = 6.000000e+02, FES = 82100
Generation 830: Best Fitness = 6.000000e+02, FES = 83100
Generation 840: Best Fitness = 6.000000e+02, FES = 84100
Generation 850: Best Fitness = 6.000000e+02, FES = 85100
Generation 860: Best Fitness = 6.000000e+02, FES = 86100
Generation 870: Best Fitness = 6.000000e+02, FES = 87100
Generation 880: Best Fitness = 6.000000e+02, FES = 88100
Generation 890: Best Fitness = 6.000000e+02, FES = 89100
Generation 900: Best Fitness = 6.000000e+02, FES = 90100
Generation 910: Best Fitness = 6.000000e+02, FES = 91100
Generation 920: Best Fitness = 6.000000e+02, FES = 92100
Generation 930: Best Fitness = 6.000000e+02, FES = 93100
Generation 940: Best Fitness = 6.000000e+02, FES = 94100
Generation 950: Best Fitness = 6.000000e+02, FES = 95100
Generation 960: Best Fitness = 6.000000e+02, FES = 96100
Generation 970: Best Fitness = 6.000000e+02, FES = 97100
Generation 980: Best Fitness = 6.000000e+02, FES = 98100
Generation 990: Best Fitness = 6.000000e+02, FES = 99100

Optimization completed!
Final Best Fitness = 6.000000e+02
Total Function Evaluations = 100000

==========================================================================
==========
Results:
```

```
  Best Fitness           : 6.000000e+02
  Global Optimum         : 6.000000e+02
  Error (|best-optimum|) : 0.000000e+00
  Best Solution (5 dim)  : [ 79.08939294 -24.57277765 -34.67392644 -
44.68162088  75.99787965]
  D^(0) (init. diversity): 0.0000e+00
==================================================================
==========
```

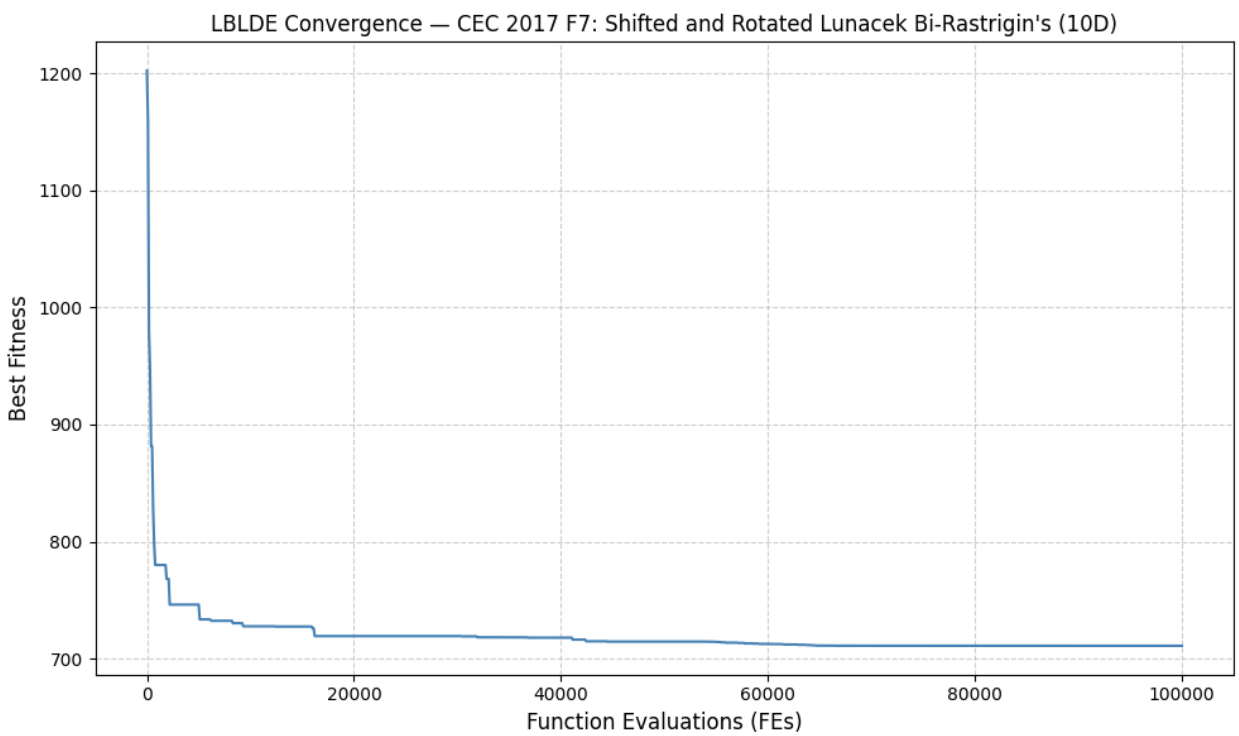



```
Graph saved as convergence_F6.png


# ============================================================
# F7 — Shifted and Rotated Lunacek Bi-Rastrigin
# ============================================================
run_and_plot('F7', make_wrapper(simple.f7), 700.0, "Shifted and
Rotated Lunacek Bi-Rastrigin's")

==================================================================
==========
LBLDE on CEC 2017 F7 — Single Run
==================================================================
==========
Function       : Shifted and Rotated Lunacek Bi-Rastrigin's (10D)
Bounds         : [-100, 100]
Global optimum : 700.0
Generation 0: Best Fitness = 1.202264e+03
```

```
Generation 10: Best Fitness = 7.800995e+02, FES = 1100
Generation 20: Best Fitness = 7.682080e+02, FES = 2100
Generation 30: Best Fitness = 7.462106e+02, FES = 3100
Generation 40: Best Fitness = 7.462106e+02, FES = 4100
Generation 50: Best Fitness = 7.462106e+02, FES = 5100
Generation 60: Best Fitness = 7.335947e+02, FES = 6100
Generation 70: Best Fitness = 7.323744e+02, FES = 7100
Generation 80: Best Fitness = 7.323744e+02, FES = 8100
Generation 90: Best Fitness = 7.303443e+02, FES = 9100
Generation 100: Best Fitness = 7.276989e+02, FES = 10100
Generation 110: Best Fitness = 7.276989e+02, FES = 11100
Generation 120: Best Fitness = 7.276989e+02, FES = 12100
Generation 130: Best Fitness = 7.274892e+02, FES = 13100
Generation 140: Best Fitness = 7.274892e+02, FES = 14100
Generation 150: Best Fitness = 7.274892e+02, FES = 15100
Generation 160: Best Fitness = 7.258532e+02, FES = 16100
Generation 170: Best Fitness = 7.193502e+02, FES = 17100
Generation 180: Best Fitness = 7.193502e+02, FES = 18100
Generation 190: Best Fitness = 7.193502e+02, FES = 19100
Generation 200: Best Fitness = 7.193502e+02, FES = 20100
Generation 210: Best Fitness = 7.193502e+02, FES = 21100
Generation 220: Best Fitness = 7.193502e+02, FES = 22100
Generation 230: Best Fitness = 7.193502e+02, FES = 23100
Generation 240: Best Fitness = 7.193502e+02, FES = 24100
Generation 250: Best Fitness = 7.193502e+02, FES = 25100
Generation 260: Best Fitness = 7.193502e+02, FES = 26100
Generation 270: Best Fitness = 7.193502e+02, FES = 27100
Generation 280: Best Fitness = 7.193502e+02, FES = 28100
Generation 290: Best Fitness = 7.193502e+02, FES = 29100
Generation 300: Best Fitness = 7.193502e+02, FES = 30100
Generation 310: Best Fitness = 7.191316e+02, FES = 31100
Generation 320: Best Fitness = 7.183362e+02, FES = 32100
Generation 330: Best Fitness = 7.183362e+02, FES = 33100
Generation 340: Best Fitness = 7.182300e+02, FES = 34100
Generation 350: Best Fitness = 7.182300e+02, FES = 35100
Generation 360: Best Fitness = 7.182300e+02, FES = 36100
Generation 370: Best Fitness = 7.179862e+02, FES = 37100
Generation 380: Best Fitness = 7.179862e+02, FES = 38100
Generation 390: Best Fitness = 7.179862e+02, FES = 39100
Generation 400: Best Fitness = 7.179862e+02, FES = 40100
Generation 410: Best Fitness = 7.179862e+02, FES = 41100
Generation 420: Best Fitness = 7.163058e+02, FES = 42100
Generation 430: Best Fitness = 7.148988e+02, FES = 43100
Generation 440: Best Fitness = 7.148988e+02, FES = 44100
Generation 450: Best Fitness = 7.146401e+02, FES = 45100
Generation 460: Best Fitness = 7.146401e+02, FES = 46100
Generation 470: Best Fitness = 7.146401e+02, FES = 47100
Generation 480: Best Fitness = 7.146401e+02, FES = 48100
Generation 490: Best Fitness = 7.146401e+02, FES = 49100
```

```
Generation 500: Best Fitness = 7.146401e+02, FES = 50100
Generation 510: Best Fitness = 7.146401e+02, FES = 51100
Generation 520: Best Fitness = 7.146401e+02, FES = 52100
Generation 530: Best Fitness = 7.146401e+02, FES = 53100
Generation 540: Best Fitness = 7.146401e+02, FES = 54100
Generation 550: Best Fitness = 7.143223e+02, FES = 55100
Generation 560: Best Fitness = 7.137018e+02, FES = 56100
Generation 570: Best Fitness = 7.137015e+02, FES = 57100
Generation 580: Best Fitness = 7.130601e+02, FES = 58100
Generation 590: Best Fitness = 7.129346e+02, FES = 59100
Generation 600: Best Fitness = 7.125117e+02, FES = 60100
Generation 610: Best Fitness = 7.125117e+02, FES = 61100
Generation 620: Best Fitness = 7.121558e+02, FES = 62100
Generation 630: Best Fitness = 7.118815e+02, FES = 63100
Generation 640: Best Fitness = 7.116571e+02, FES = 64100
Generation 650: Best Fitness = 7.112185e+02, FES = 65100
Generation 660: Best Fitness = 7.111784e+02, FES = 66100
Generation 670: Best Fitness = 7.110847e+02, FES = 67100
Generation 680: Best Fitness = 7.110709e+02, FES = 68100
Generation 690: Best Fitness = 7.110335e+02, FES = 69100
Generation 700: Best Fitness = 7.110281e+02, FES = 70100
Generation 710: Best Fitness = 7.110269e+02, FES = 71100
Generation 720: Best Fitness = 7.110266e+02, FES = 72100
Generation 730: Best Fitness = 7.110265e+02, FES = 73100
Generation 740: Best Fitness = 7.110265e+02, FES = 74100
Generation 750: Best Fitness = 7.110264e+02, FES = 75100
Generation 760: Best Fitness = 7.110264e+02, FES = 76100
Generation 770: Best Fitness = 7.110264e+02, FES = 77100
Generation 780: Best Fitness = 7.110264e+02, FES = 78100
Generation 790: Best Fitness = 7.110264e+02, FES = 79100
Generation 800: Best Fitness = 7.110264e+02, FES = 80100
Generation 810: Best Fitness = 7.110264e+02, FES = 81100
Generation 820: Best Fitness = 7.110264e+02, FES = 82100
Generation 830: Best Fitness = 7.110264e+02, FES = 83100
Generation 840: Best Fitness = 7.110264e+02, FES = 84100
Generation 850: Best Fitness = 7.110264e+02, FES = 85100
Generation 860: Best Fitness = 7.110264e+02, FES = 86100
Generation 870: Best Fitness = 7.110264e+02, FES = 87100
Generation 880: Best Fitness = 7.110264e+02, FES = 88100
Generation 890: Best Fitness = 7.110264e+02, FES = 89100
Generation 900: Best Fitness = 7.110264e+02, FES = 90100
Generation 910: Best Fitness = 7.110264e+02, FES = 91100
Generation 920: Best Fitness = 7.110264e+02, FES = 92100
Generation 930: Best Fitness = 7.110264e+02, FES = 93100
Generation 940: Best Fitness = 7.110264e+02, FES = 94100
Generation 950: Best Fitness = 7.110264e+02, FES = 95100
Generation 960: Best Fitness = 7.110264e+02, FES = 96100
Generation 970: Best Fitness = 7.110264e+02, FES = 97100
Generation 980: Best Fitness = 7.110264e+02, FES = 98100
```

```
Generation 990: Best Fitness = 7.110264e+02, FES = 99100

Optimization completed!
Final Best Fitness = 7.110264e+02
Total Function Evaluations = 100000

======================================================================
==========
Results:
  Best Fitness           : 7.110264e+02
  Global Optimum         : 7.000000e+02
  Error (|best-optimum|) : 1.102645e+01
  Best Solution (5 dim)  : [-19.60699923  18.29193118 -23.87741717 -
10.17684314  18.93075077]
  D^(0) (init. diversity): 0.0000e+00
======================================================================
==========
```

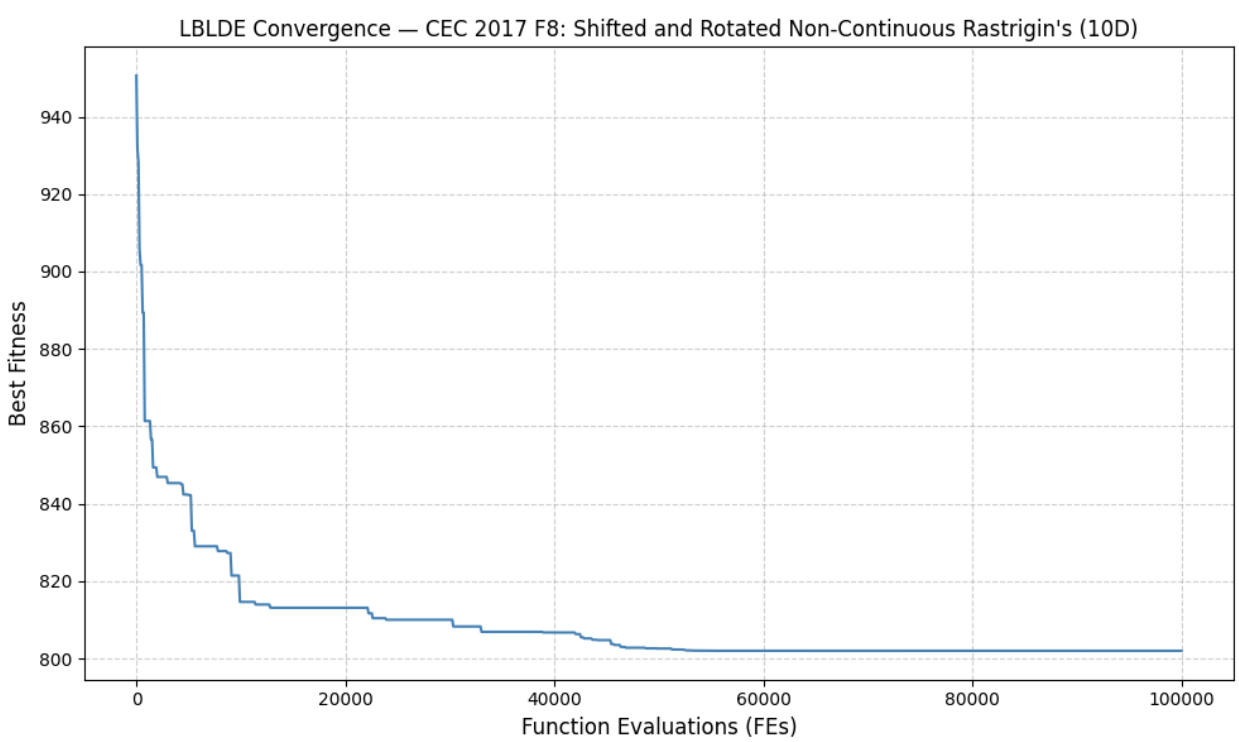



```
Graph saved as convergence_F7.png


# ============================================================
# F8 — Shifted and Rotated Non-Continuous Rastrigin
# ============================================================
run_and_plot('F8', make_wrapper(simple.f8), 800.0, "Shifted and
Rotated Non-Continuous Rastrigin's")
```

```
==========================================================================
==========
LBLDE on CEC 2017 F8 — Single Run
==========================================================================
==========
Function       : Shifted and Rotated Non-Continuous Rastrigin's (10D)
Bounds         : [-100, 100]
Global optimum : 800.0
Generation 0: Best Fitness = 9.505966e+02
Generation 10: Best Fitness = 8.613592e+02, FES = 1100
Generation 20: Best Fitness = 8.469195e+02, FES = 2100
Generation 30: Best Fitness = 8.453314e+02, FES = 3100
Generation 40: Best Fitness = 8.453314e+02, FES = 4100
Generation 50: Best Fitness = 8.422175e+02, FES = 5100
Generation 60: Best Fitness = 8.289991e+02, FES = 6100
Generation 70: Best Fitness = 8.289991e+02, FES = 7100
Generation 80: Best Fitness = 8.277575e+02, FES = 8100
Generation 90: Best Fitness = 8.272605e+02, FES = 9100
Generation 100: Best Fitness = 8.146235e+02, FES = 10100
Generation 110: Best Fitness = 8.146235e+02, FES = 11100
Generation 120: Best Fitness = 8.139525e+02, FES = 12100
Generation 130: Best Fitness = 8.131115e+02, FES = 13100
Generation 140: Best Fitness = 8.131115e+02, FES = 14100
Generation 150: Best Fitness = 8.131115e+02, FES = 15100
Generation 160: Best Fitness = 8.131115e+02, FES = 16100
Generation 170: Best Fitness = 8.131115e+02, FES = 17100
Generation 180: Best Fitness = 8.131115e+02, FES = 18100
Generation 190: Best Fitness = 8.131115e+02, FES = 19100
Generation 200: Best Fitness = 8.131115e+02, FES = 20100
Generation 210: Best Fitness = 8.131115e+02, FES = 21100
Generation 220: Best Fitness = 8.131115e+02, FES = 22100
Generation 230: Best Fitness = 8.104203e+02, FES = 23100
Generation 240: Best Fitness = 8.100117e+02, FES = 24100
Generation 250: Best Fitness = 8.100117e+02, FES = 25100
Generation 260: Best Fitness = 8.100117e+02, FES = 26100
Generation 270: Best Fitness = 8.100117e+02, FES = 27100
Generation 280: Best Fitness = 8.100117e+02, FES = 28100
Generation 290: Best Fitness = 8.100117e+02, FES = 29100
Generation 300: Best Fitness = 8.100117e+02, FES = 30100
Generation 310: Best Fitness = 8.082786e+02, FES = 31100
Generation 320: Best Fitness = 8.082786e+02, FES = 32100
Generation 330: Best Fitness = 8.068922e+02, FES = 33100
Generation 340: Best Fitness = 8.068922e+02, FES = 34100
Generation 350: Best Fitness = 8.068922e+02, FES = 35100
Generation 360: Best Fitness = 8.068922e+02, FES = 36100
Generation 370: Best Fitness = 8.068922e+02, FES = 37100
Generation 380: Best Fitness = 8.068922e+02, FES = 38100
Generation 390: Best Fitness = 8.067366e+02, FES = 39100
Generation 400: Best Fitness = 8.067230e+02, FES = 40100
Generation 410: Best Fitness = 8.067230e+02, FES = 41100
```

```
Generation 420: Best Fitness = 8.063090e+02, FES = 42100
Generation 430: Best Fitness = 8.051870e+02, FES = 43100
Generation 440: Best Fitness = 8.048439e+02, FES = 44100
Generation 450: Best Fitness = 8.047578e+02, FES = 45100
Generation 460: Best Fitness = 8.035057e+02, FES = 46100
Generation 470: Best Fitness = 8.027861e+02, FES = 47100
Generation 480: Best Fitness = 8.027861e+02, FES = 48100
Generation 490: Best Fitness = 8.026320e+02, FES = 49100
Generation 500: Best Fitness = 8.025824e+02, FES = 50100
Generation 510: Best Fitness = 8.025824e+02, FES = 51100
Generation 520: Best Fitness = 8.023403e+02, FES = 52100
Generation 530: Best Fitness = 8.020896e+02, FES = 53100
Generation 540: Best Fitness = 8.020240e+02, FES = 54100
Generation 550: Best Fitness = 8.020022e+02, FES = 55100
Generation 560: Best Fitness = 8.019948e+02, FES = 56100
Generation 570: Best Fitness = 8.019913e+02, FES = 57100
Generation 580: Best Fitness = 8.019901e+02, FES = 58100
Generation 590: Best Fitness = 8.019900e+02, FES = 59100
Generation 600: Best Fitness = 8.019899e+02, FES = 60100
Generation 610: Best Fitness = 8.019899e+02, FES = 61100
Generation 620: Best Fitness = 8.019899e+02, FES = 62100
Generation 630: Best Fitness = 8.019899e+02, FES = 63100
Generation 640: Best Fitness = 8.019899e+02, FES = 64100
Generation 650: Best Fitness = 8.019899e+02, FES = 65100
Generation 660: Best Fitness = 8.019899e+02, FES = 66100
Generation 670: Best Fitness = 8.019899e+02, FES = 67100
Generation 680: Best Fitness = 8.019899e+02, FES = 68100
Generation 690: Best Fitness = 8.019899e+02, FES = 69100
Generation 700: Best Fitness = 8.019899e+02, FES = 70100
Generation 710: Best Fitness = 8.019899e+02, FES = 71100
Generation 720: Best Fitness = 8.019899e+02, FES = 72100
Generation 730: Best Fitness = 8.019899e+02, FES = 73100
Generation 740: Best Fitness = 8.019899e+02, FES = 74100
Generation 750: Best Fitness = 8.019899e+02, FES = 75100
Generation 760: Best Fitness = 8.019899e+02, FES = 76100
Generation 770: Best Fitness = 8.019899e+02, FES = 77100
Generation 780: Best Fitness = 8.019899e+02, FES = 78100
Generation 790: Best Fitness = 8.019899e+02, FES = 79100
Generation 800: Best Fitness = 8.019899e+02, FES = 80100
Generation 810: Best Fitness = 8.019899e+02, FES = 81100
Generation 820: Best Fitness = 8.019899e+02, FES = 82100
Generation 830: Best Fitness = 8.019899e+02, FES = 83100
Generation 840: Best Fitness = 8.019899e+02, FES = 84100
Generation 850: Best Fitness = 8.019899e+02, FES = 85100
Generation 860: Best Fitness = 8.019899e+02, FES = 86100
Generation 870: Best Fitness = 8.019899e+02, FES = 87100
Generation 880: Best Fitness = 8.019899e+02, FES = 88100
Generation 890: Best Fitness = 8.019899e+02, FES = 89100
Generation 900: Best Fitness = 8.019899e+02, FES = 90100
Generation 910: Best Fitness = 8.019899e+02, FES = 91100
```

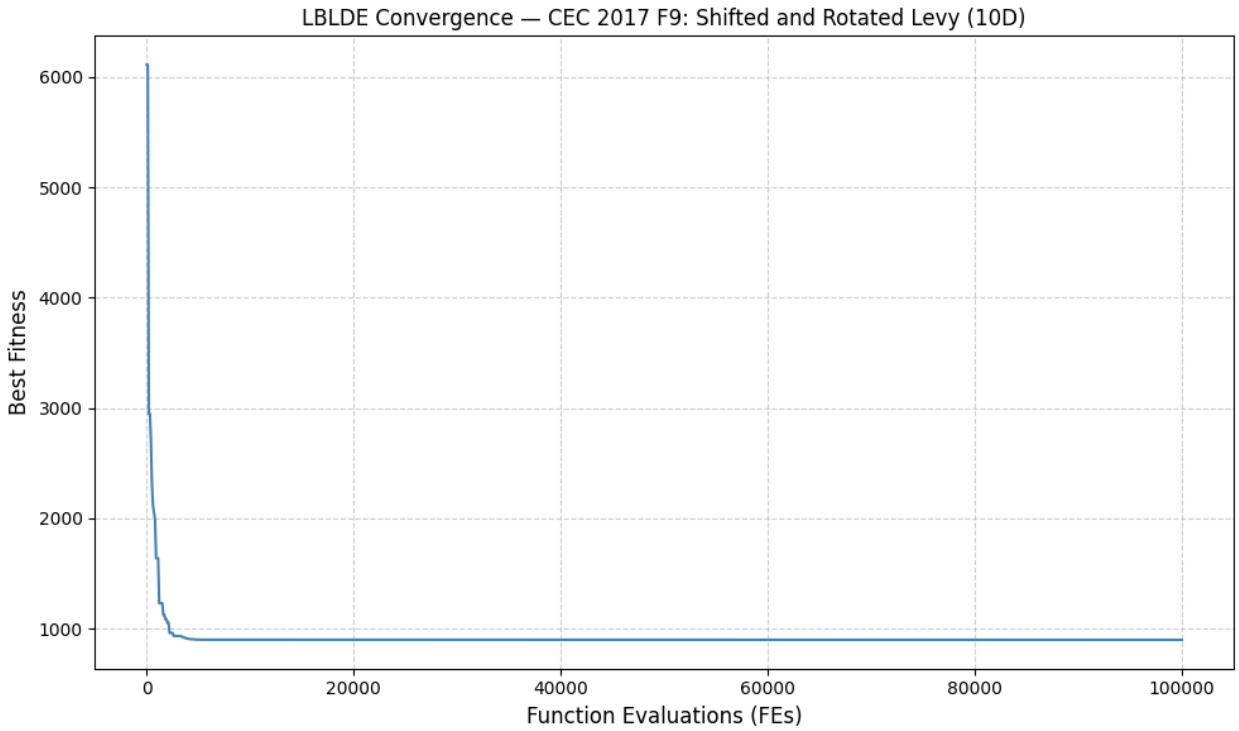
```
Generation 920: Best Fitness = 8.019899e+02, FES = 92100
Generation 930: Best Fitness = 8.019899e+02, FES = 93100
Generation 940: Best Fitness = 8.019899e+02, FES = 94100
Generation 950: Best Fitness = 8.019899e+02, FES = 95100
Generation 960: Best Fitness = 8.019899e+02, FES = 96100
Generation 970: Best Fitness = 8.019899e+02, FES = 97100
Generation 980: Best Fitness = 8.019899e+02, FES = 98100
Generation 990: Best Fitness = 8.019899e+02, FES = 99100

Optimization completed!
Final Best Fitness = 8.019899e+02
Total Function Evaluations = 100000

======================================================================
==========
Results:
  Best Fitness           : 8.019899e+02
  Global Optimum         : 8.000000e+02
  Error (|best-optimum|) : 1.989918e+00
  Best Solution (5 dim)  : [ 32.16421931 -55.96501375  36.01071255 -
64.8428469  -34.97768531]
  D^(0) (init. diversity): 0.0000e+00
======================================================================
==========
```



```
Graph saved as convergence_F8.png


# ============================================================
# F9 — Shifted and Rotated Levy
# ============================================================
run_and_plot('F9', make_wrapper(simple.f9), 900.0, 'Shifted and
Rotated Levy')

======================================================================
==========
LBLDE on CEC 2017 F9 — Single Run
======================================================================
==========
Function       : Shifted and Rotated Levy (10D)
Bounds         : [-100, 100]
Global optimum : 900.0
Generation 0: Best Fitness = 6.112557e+03
Generation 10: Best Fitness = 1.640249e+03, FES = 1100
Generation 20: Best Fitness = 1.054603e+03, FES = 2100
Generation 30: Best Fitness = 9.347282e+02, FES = 3100
Generation 40: Best Fitness = 9.081663e+02, FES = 4100
Generation 50: Best Fitness = 9.017169e+02, FES = 5100
Generation 60: Best Fitness = 9.005595e+02, FES = 6100
Generation 70: Best Fitness = 9.001371e+02, FES = 7100
Generation 80: Best Fitness = 9.000337e+02, FES = 8100
Generation 90: Best Fitness = 9.000116e+02, FES = 9100
Generation 100: Best Fitness = 9.000012e+02, FES = 10100
Generation 110: Best Fitness = 9.000002e+02, FES = 11100
Generation 120: Best Fitness = 9.000001e+02, FES = 12100
Generation 130: Best Fitness = 9.000000e+02, FES = 13100
Generation 140: Best Fitness = 9.000000e+02, FES = 14100
Generation 150: Best Fitness = 9.000000e+02, FES = 15100
Generation 160: Best Fitness = 9.000000e+02, FES = 16100
Generation 170: Best Fitness = 9.000000e+02, FES = 17100
Generation 180: Best Fitness = 9.000000e+02, FES = 18100
Generation 190: Best Fitness = 9.000000e+02, FES = 19100
Generation 200: Best Fitness = 9.000000e+02, FES = 20100
Generation 210: Best Fitness = 9.000000e+02, FES = 21100
Generation 220: Best Fitness = 9.000000e+02, FES = 22100
Generation 230: Best Fitness = 9.000000e+02, FES = 23100
Generation 240: Best Fitness = 9.000000e+02, FES = 24100
Generation 250: Best Fitness = 9.000000e+02, FES = 25100
Generation 260: Best Fitness = 9.000000e+02, FES = 26100
Generation 270: Best Fitness = 9.000000e+02, FES = 27100
Generation 280: Best Fitness = 9.000000e+02, FES = 28100
Generation 290: Best Fitness = 9.000000e+02, FES = 29100
Generation 300: Best Fitness = 9.000000e+02, FES = 30100
Generation 310: Best Fitness = 9.000000e+02, FES = 31100
Generation 320: Best Fitness = 9.000000e+02, FES = 32100
```

```
Generation 330: Best Fitness = 9.000000e+02, FES = 33100
Generation 340: Best Fitness = 9.000000e+02, FES = 34100
Generation 350: Best Fitness = 9.000000e+02, FES = 35100
Generation 360: Best Fitness = 9.000000e+02, FES = 36100
Generation 370: Best Fitness = 9.000000e+02, FES = 37100
Generation 380: Best Fitness = 9.000000e+02, FES = 38100
Generation 390: Best Fitness = 9.000000e+02, FES = 39100
Generation 400: Best Fitness = 9.000000e+02, FES = 40100
Generation 410: Best Fitness = 9.000000e+02, FES = 41100
Generation 420: Best Fitness = 9.000000e+02, FES = 42100
Generation 430: Best Fitness = 9.000000e+02, FES = 43100
Generation 440: Best Fitness = 9.000000e+02, FES = 44100
Generation 450: Best Fitness = 9.000000e+02, FES = 45100
Generation 460: Best Fitness = 9.000000e+02, FES = 46100
Generation 470: Best Fitness = 9.000000e+02, FES = 47100
Generation 480: Best Fitness = 9.000000e+02, FES = 48100
Generation 490: Best Fitness = 9.000000e+02, FES = 49100
Generation 500: Best Fitness = 9.000000e+02, FES = 50100
Generation 510: Best Fitness = 9.000000e+02, FES = 51100
Generation 520: Best Fitness = 9.000000e+02, FES = 52100
Generation 530: Best Fitness = 9.000000e+02, FES = 53100
Generation 540: Best Fitness = 9.000000e+02, FES = 54100
Generation 550: Best Fitness = 9.000000e+02, FES = 55100
Generation 560: Best Fitness = 9.000000e+02, FES = 56100
Generation 570: Best Fitness = 9.000000e+02, FES = 57100
Generation 580: Best Fitness = 9.000000e+02, FES = 58100
Generation 590: Best Fitness = 9.000000e+02, FES = 59100
Generation 600: Best Fitness = 9.000000e+02, FES = 60100
Generation 610: Best Fitness = 9.000000e+02, FES = 61100
Generation 620: Best Fitness = 9.000000e+02, FES = 62100
Generation 630: Best Fitness = 9.000000e+02, FES = 63100
Generation 640: Best Fitness = 9.000000e+02, FES = 64100
Generation 650: Best Fitness = 9.000000e+02, FES = 65100
Generation 660: Best Fitness = 9.000000e+02, FES = 66100
Generation 670: Best Fitness = 9.000000e+02, FES = 67100
Generation 680: Best Fitness = 9.000000e+02, FES = 68100
Generation 690: Best Fitness = 9.000000e+02, FES = 69100
Generation 700: Best Fitness = 9.000000e+02, FES = 70100
Generation 710: Best Fitness = 9.000000e+02, FES = 71100
Generation 720: Best Fitness = 9.000000e+02, FES = 72100
Generation 730: Best Fitness = 9.000000e+02, FES = 73100
Generation 740: Best Fitness = 9.000000e+02, FES = 74100
Generation 750: Best Fitness = 9.000000e+02, FES = 75100
Generation 760: Best Fitness = 9.000000e+02, FES = 76100
Generation 770: Best Fitness = 9.000000e+02, FES = 77100
Generation 780: Best Fitness = 9.000000e+02, FES = 78100
Generation 790: Best Fitness = 9.000000e+02, FES = 79100
Generation 800: Best Fitness = 9.000000e+02, FES = 80100
Generation 810: Best Fitness = 9.000000e+02, FES = 81100
```

```
Generation 820: Best Fitness = 9.000000e+02, FES = 82100
Generation 830: Best Fitness = 9.000000e+02, FES = 83100
Generation 840: Best Fitness = 9.000000e+02, FES = 84100
Generation 850: Best Fitness = 9.000000e+02, FES = 85100
Generation 860: Best Fitness = 9.000000e+02, FES = 86100
Generation 870: Best Fitness = 9.000000e+02, FES = 87100
Generation 880: Best Fitness = 9.000000e+02, FES = 88100
Generation 890: Best Fitness = 9.000000e+02, FES = 89100
Generation 900: Best Fitness = 9.000000e+02, FES = 90100
Generation 910: Best Fitness = 9.000000e+02, FES = 91100
Generation 920: Best Fitness = 9.000000e+02, FES = 92100
Generation 930: Best Fitness = 9.000000e+02, FES = 93100
Generation 940: Best Fitness = 9.000000e+02, FES = 94100
Generation 950: Best Fitness = 9.000000e+02, FES = 95100
Generation 960: Best Fitness = 9.000000e+02, FES = 96100
Generation 970: Best Fitness = 9.000000e+02, FES = 97100
Generation 980: Best Fitness = 9.000000e+02, FES = 98100
Generation 990: Best Fitness = 9.000000e+02, FES = 99100

Optimization completed!
Final Best Fitness = 9.000000e+02
Total Function Evaluations = 100000

=====================================================================
==========
Results:
  Best Fitness           : 9.000000e+02
  Global Optimum         : 9.000000e+02
  Error (|best-optimum|) : 0.000000e+00
  Best Solution (5 dim)  : [-23.59314007   0.12032421  20.8326407
52.54992626 -18.22787193]
  D^(0) (init. diversity): 0.0000e+00
=====================================================================
==========
```

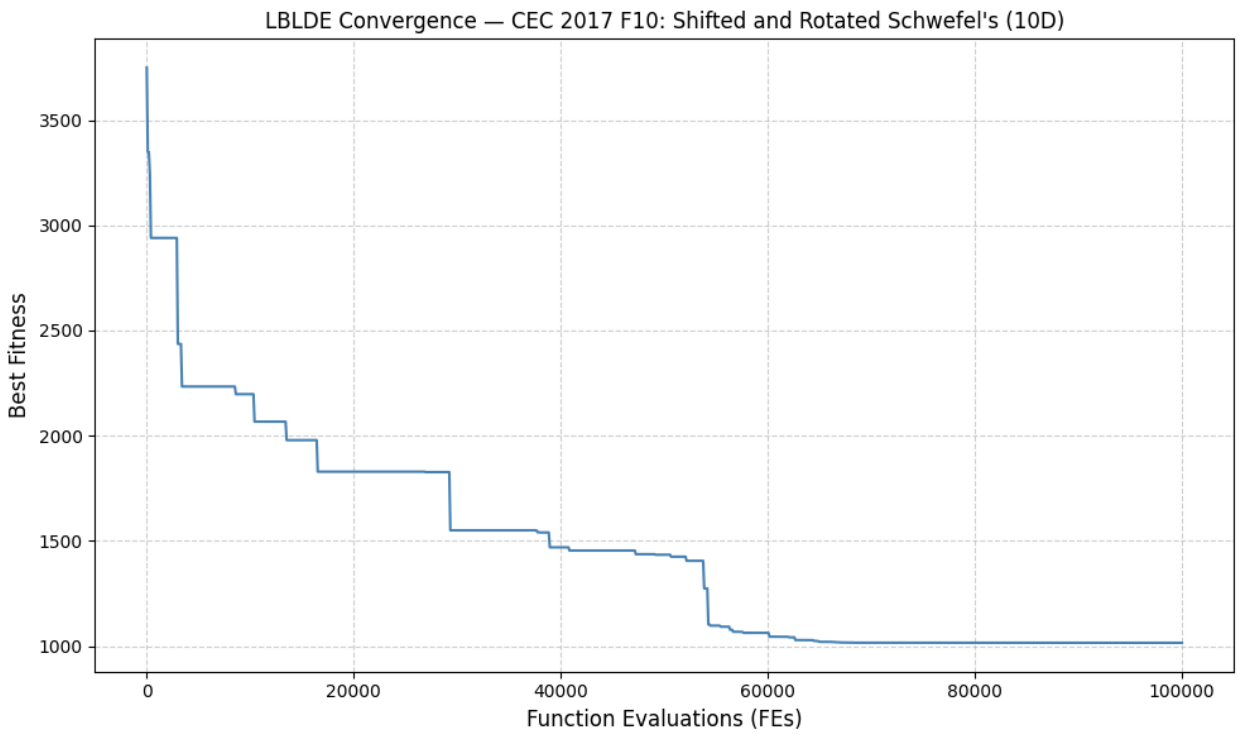



```
Graph saved as convergence_F9.png


# ============================================================
# F10 — Shifted and Rotated Schwefel
# ============================================================
run_and_plot('F10', make_wrapper(simple.f10), 1000.0, "Shifted and
Rotated Schwefel's")

======================================================================
==========
LBLDE on CEC 2017 F10 — Single Run
======================================================================
==========
Function       : Shifted and Rotated Schwefel's (10D)
Bounds         : [-100, 100]
Global optimum : 1000.0
Generation 0: Best Fitness = 3.749402e+03
Generation 10: Best Fitness = 2.939820e+03, FES = 1100
Generation 20: Best Fitness = 2.939820e+03, FES = 2100
Generation 30: Best Fitness = 2.435899e+03, FES = 3100
Generation 40: Best Fitness = 2.233415e+03, FES = 4100
Generation 50: Best Fitness = 2.233415e+03, FES = 5100
Generation 60: Best Fitness = 2.233415e+03, FES = 6100
Generation 70: Best Fitness = 2.233415e+03, FES = 7100
Generation 80: Best Fitness = 2.233415e+03, FES = 8100
Generation 90: Best Fitness = 2.197199e+03, FES = 9100
```

```
Generation 100: Best Fitness = 2.197199e+03, FES = 10100
Generation 110: Best Fitness = 2.066474e+03, FES = 11100
Generation 120: Best Fitness = 2.066474e+03, FES = 12100
Generation 130: Best Fitness = 2.066474e+03, FES = 13100
Generation 140: Best Fitness = 1.978695e+03, FES = 14100
Generation 150: Best Fitness = 1.978695e+03, FES = 15100
Generation 160: Best Fitness = 1.978695e+03, FES = 16100
Generation 170: Best Fitness = 1.828695e+03, FES = 17100
Generation 180: Best Fitness = 1.828695e+03, FES = 18100
Generation 190: Best Fitness = 1.828695e+03, FES = 19100
Generation 200: Best Fitness = 1.828695e+03, FES = 20100
Generation 210: Best Fitness = 1.828695e+03, FES = 21100
Generation 220: Best Fitness = 1.828695e+03, FES = 22100
Generation 230: Best Fitness = 1.828695e+03, FES = 23100
Generation 240: Best Fitness = 1.828695e+03, FES = 24100
Generation 250: Best Fitness = 1.828695e+03, FES = 25100
Generation 260: Best Fitness = 1.828695e+03, FES = 26100
Generation 270: Best Fitness = 1.826584e+03, FES = 27100
Generation 280: Best Fitness = 1.826584e+03, FES = 28100
Generation 290: Best Fitness = 1.826584e+03, FES = 29100
Generation 300: Best Fitness = 1.549953e+03, FES = 30100
Generation 310: Best Fitness = 1.549953e+03, FES = 31100
Generation 320: Best Fitness = 1.549953e+03, FES = 32100
Generation 330: Best Fitness = 1.549953e+03, FES = 33100
Generation 340: Best Fitness = 1.549953e+03, FES = 34100
Generation 350: Best Fitness = 1.549953e+03, FES = 35100
Generation 360: Best Fitness = 1.549953e+03, FES = 36100
Generation 370: Best Fitness = 1.549953e+03, FES = 37100
Generation 380: Best Fitness = 1.539725e+03, FES = 38100
Generation 390: Best Fitness = 1.469067e+03, FES = 39100
Generation 400: Best Fitness = 1.469067e+03, FES = 40100
Generation 410: Best Fitness = 1.453790e+03, FES = 41100
Generation 420: Best Fitness = 1.453790e+03, FES = 42100
Generation 430: Best Fitness = 1.453790e+03, FES = 43100
Generation 440: Best Fitness = 1.453790e+03, FES = 44100
Generation 450: Best Fitness = 1.453790e+03, FES = 45100
Generation 460: Best Fitness = 1.453790e+03, FES = 46100
Generation 470: Best Fitness = 1.453790e+03, FES = 47100
Generation 480: Best Fitness = 1.436492e+03, FES = 48100
Generation 490: Best Fitness = 1.436492e+03, FES = 49100
Generation 500: Best Fitness = 1.433865e+03, FES = 50100
Generation 510: Best Fitness = 1.424572e+03, FES = 51100
Generation 520: Best Fitness = 1.424572e+03, FES = 52100
Generation 530: Best Fitness = 1.405072e+03, FES = 53100
Generation 540: Best Fitness = 1.273662e+03, FES = 54100
Generation 550: Best Fitness = 1.096802e+03, FES = 55100
Generation 560: Best Fitness = 1.091742e+03, FES = 56100
Generation 570: Best Fitness = 1.067366e+03, FES = 57100
Generation 580: Best Fitness = 1.062645e+03, FES = 58100
```

```
Generation 590: Best Fitness = 1.062645e+03, FES = 59100
Generation 600: Best Fitness = 1.062645e+03, FES = 60100
Generation 610: Best Fitness = 1.043899e+03, FES = 61100
Generation 620: Best Fitness = 1.041340e+03, FES = 62100
Generation 630: Best Fitness = 1.028369e+03, FES = 63100
Generation 640: Best Fitness = 1.028178e+03, FES = 64100
Generation 650: Best Fitness = 1.019795e+03, FES = 65100
Generation 660: Best Fitness = 1.019580e+03, FES = 66100
Generation 670: Best Fitness = 1.016737e+03, FES = 67100
Generation 680: Best Fitness = 1.016214e+03, FES = 68100
Generation 690: Best Fitness = 1.015505e+03, FES = 69100
Generation 700: Best Fitness = 1.015322e+03, FES = 70100
Generation 710: Best Fitness = 1.015281e+03, FES = 71100
Generation 720: Best Fitness = 1.015207e+03, FES = 72100
Generation 730: Best Fitness = 1.015190e+03, FES = 73100
Generation 740: Best Fitness = 1.015185e+03, FES = 74100
Generation 750: Best Fitness = 1.015183e+03, FES = 75100
Generation 760: Best Fitness = 1.015182e+03, FES = 76100
Generation 770: Best Fitness = 1.015182e+03, FES = 77100
Generation 780: Best Fitness = 1.015182e+03, FES = 78100
Generation 790: Best Fitness = 1.015182e+03, FES = 79100
Generation 800: Best Fitness = 1.015182e+03, FES = 80100
Generation 810: Best Fitness = 1.015182e+03, FES = 81100
Generation 820: Best Fitness = 1.015182e+03, FES = 82100
Generation 830: Best Fitness = 1.015182e+03, FES = 83100
Generation 840: Best Fitness = 1.015182e+03, FES = 84100
Generation 850: Best Fitness = 1.015182e+03, FES = 85100
Generation 860: Best Fitness = 1.015182e+03, FES = 86100
Generation 870: Best Fitness = 1.015182e+03, FES = 87100
Generation 880: Best Fitness = 1.015182e+03, FES = 88100
Generation 890: Best Fitness = 1.015182e+03, FES = 89100
Generation 900: Best Fitness = 1.015182e+03, FES = 90100
Generation 910: Best Fitness = 1.015182e+03, FES = 91100
Generation 920: Best Fitness = 1.015182e+03, FES = 92100
Generation 930: Best Fitness = 1.015182e+03, FES = 93100
Generation 940: Best Fitness = 1.015182e+03, FES = 94100
Generation 950: Best Fitness = 1.015182e+03, FES = 95100
Generation 960: Best Fitness = 1.015182e+03, FES = 96100
Generation 970: Best Fitness = 1.015182e+03, FES = 97100
Generation 980: Best Fitness = 1.015182e+03, FES = 98100
Generation 990: Best Fitness = 1.015182e+03, FES = 99100

Optimization completed!
Final Best Fitness = 1.015182e+03
Total Function Evaluations = 100000

==========================================================================
==========
Results:
```

```
  Best Fitness           : 1.015182e+03
  Global Optimum         : 1.000000e+03
  Error (|best-optimum|) : 1.518216e+01
  Best Solution (5 dim)  : [-17.17634423 -43.52193548  57.70144694
47.57030029  13.53058168]
  D^(0) (init. diversity): 0.0000e+00
======================================================================
==========
```




```
Graph saved as convergence_F10.png
```

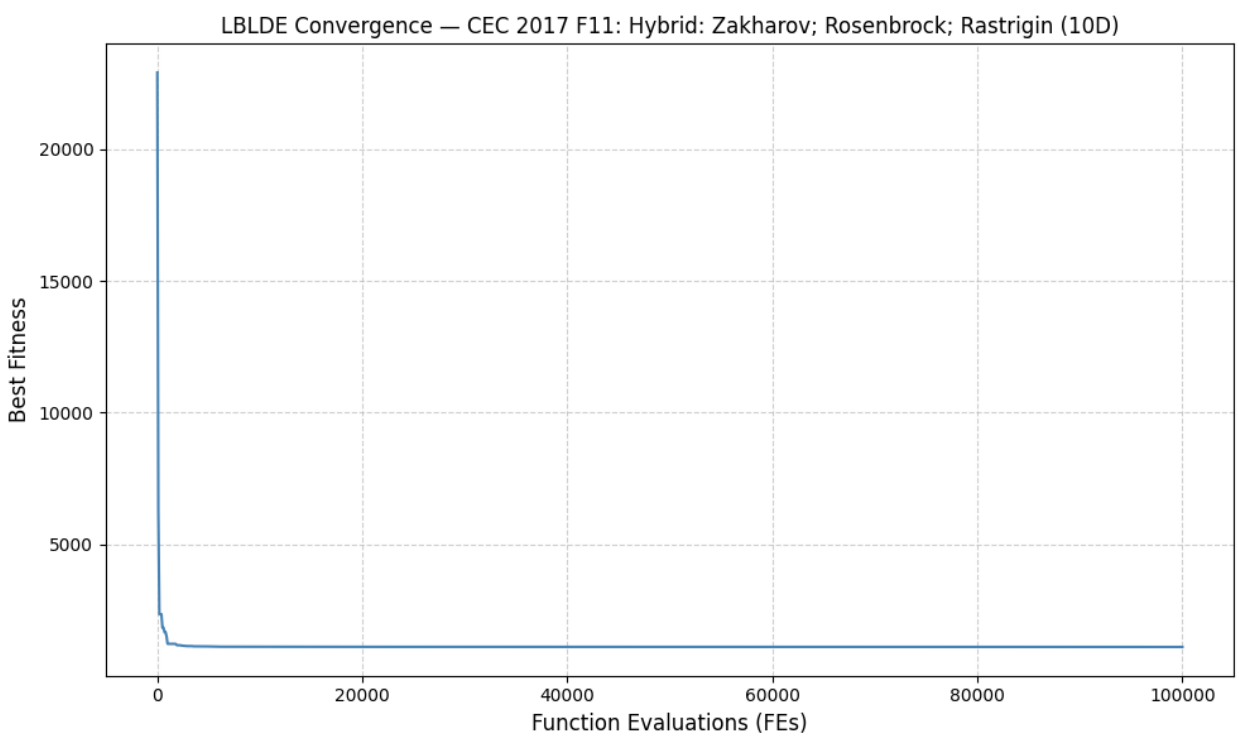
6.3 Hybrid Functions (F11-F20)

```
# ============================================================
# F11 — Hybrid Function 1 (Zakharov; Rosenbrock; Rastrigin)
# ============================================================
run_and_plot('F11', make_wrapper(hybrid.f11), 1100.0, 'Hybrid:
Zakharov; Rosenbrock; Rastrigin')

======================================================================
==========
LBLDE on CEC 2017 F11 — Single Run
======================================================================
==========
Function       : Hybrid: Zakharov; Rosenbrock; Rastrigin (10D)
```

```
Bounds         : [-100, 100]
Global optimum : 1100.0
Generation 0: Best Fitness = 2.292232e+04
Generation 10: Best Fitness = 1.220696e+03, FES = 1100
Generation 20: Best Fitness = 1.166953e+03, FES = 2100
Generation 30: Best Fitness = 1.132450e+03, FES = 3100
Generation 40: Best Fitness = 1.124102e+03, FES = 4100
Generation 50: Best Fitness = 1.119533e+03, FES = 5100
Generation 60: Best Fitness = 1.109540e+03, FES = 6100
Generation 70: Best Fitness = 1.109540e+03, FES = 7100
Generation 80: Best Fitness = 1.109540e+03, FES = 8100
Generation 90: Best Fitness = 1.109333e+03, FES = 9100
Generation 100: Best Fitness = 1.108608e+03, FES = 10100
Generation 110: Best Fitness = 1.107315e+03, FES = 11100
Generation 120: Best Fitness = 1.107315e+03, FES = 12100
Generation 130: Best Fitness = 1.106577e+03, FES = 13100
Generation 140: Best Fitness = 1.106338e+03, FES = 14100
Generation 150: Best Fitness = 1.105330e+03, FES = 15100
Generation 160: Best Fitness = 1.105141e+03, FES = 16100
Generation 170: Best Fitness = 1.105140e+03, FES = 17100
Generation 180: Best Fitness = 1.104280e+03, FES = 18100
Generation 190: Best Fitness = 1.104280e+03, FES = 19100
Generation 200: Best Fitness = 1.104280e+03, FES = 20100
Generation 210: Best Fitness = 1.104006e+03, FES = 21100
Generation 220: Best Fitness = 1.103678e+03, FES = 22100
Generation 230: Best Fitness = 1.103619e+03, FES = 23100
Generation 240: Best Fitness = 1.103390e+03, FES = 24100
Generation 250: Best Fitness = 1.103318e+03, FES = 25100
Generation 260: Best Fitness = 1.103098e+03, FES = 26100
Generation 270: Best Fitness = 1.102814e+03, FES = 27100
Generation 280: Best Fitness = 1.102801e+03, FES = 28100
Generation 290: Best Fitness = 1.102567e+03, FES = 29100
Generation 300: Best Fitness = 1.101767e+03, FES = 30100
Generation 310: Best Fitness = 1.101250e+03, FES = 31100
Generation 320: Best Fitness = 1.100928e+03, FES = 32100
Generation 330: Best Fitness = 1.100829e+03, FES = 33100
Generation 340: Best Fitness = 1.100659e+03, FES = 34100
Generation 350: Best Fitness = 1.100636e+03, FES = 35100
Generation 360: Best Fitness = 1.100549e+03, FES = 36100
Generation 370: Best Fitness = 1.100434e+03, FES = 37100
Generation 380: Best Fitness = 1.100316e+03, FES = 38100
Generation 390: Best Fitness = 1.100221e+03, FES = 39100
Generation 400: Best Fitness = 1.100159e+03, FES = 40100
Generation 410: Best Fitness = 1.100106e+03, FES = 41100
Generation 420: Best Fitness = 1.100090e+03, FES = 42100
Generation 430: Best Fitness = 1.100068e+03, FES = 43100
Generation 440: Best Fitness = 1.100042e+03, FES = 44100
Generation 450: Best Fitness = 1.100034e+03, FES = 45100
Generation 460: Best Fitness = 1.100019e+03, FES = 46100
```

```
Generation 470: Best Fitness = 1.100011e+03, FES = 47100
Generation 480: Best Fitness = 1.100011e+03, FES = 48100
Generation 490: Best Fitness = 1.100007e+03, FES = 49100
Generation 500: Best Fitness = 1.100004e+03, FES = 50100
Generation 510: Best Fitness = 1.100003e+03, FES = 51100
Generation 520: Best Fitness = 1.100002e+03, FES = 52100
Generation 530: Best Fitness = 1.100001e+03, FES = 53100
Generation 540: Best Fitness = 1.100001e+03, FES = 54100
Generation 550: Best Fitness = 1.100000e+03, FES = 55100
Generation 560: Best Fitness = 1.100000e+03, FES = 56100
Generation 570: Best Fitness = 1.100000e+03, FES = 57100
Generation 580: Best Fitness = 1.100000e+03, FES = 58100
Generation 590: Best Fitness = 1.100000e+03, FES = 59100
Generation 600: Best Fitness = 1.100000e+03, FES = 60100
Generation 610: Best Fitness = 1.100000e+03, FES = 61100
Generation 620: Best Fitness = 1.100000e+03, FES = 62100
Generation 630: Best Fitness = 1.100000e+03, FES = 63100
Generation 640: Best Fitness = 1.100000e+03, FES = 64100
Generation 650: Best Fitness = 1.100000e+03, FES = 65100
Generation 660: Best Fitness = 1.100000e+03, FES = 66100
Generation 670: Best Fitness = 1.100000e+03, FES = 67100
Generation 680: Best Fitness = 1.100000e+03, FES = 68100
Generation 690: Best Fitness = 1.100000e+03, FES = 69100
Generation 700: Best Fitness = 1.100000e+03, FES = 70100
Generation 710: Best Fitness = 1.100000e+03, FES = 71100
Generation 720: Best Fitness = 1.100000e+03, FES = 72100
Generation 730: Best Fitness = 1.100000e+03, FES = 73100
Generation 740: Best Fitness = 1.100000e+03, FES = 74100
Generation 750: Best Fitness = 1.100000e+03, FES = 75100
Generation 760: Best Fitness = 1.100000e+03, FES = 76100
Generation 770: Best Fitness = 1.100000e+03, FES = 77100
Generation 780: Best Fitness = 1.100000e+03, FES = 78100
Generation 790: Best Fitness = 1.100000e+03, FES = 79100
Generation 800: Best Fitness = 1.100000e+03, FES = 80100
Generation 810: Best Fitness = 1.100000e+03, FES = 81100
Generation 820: Best Fitness = 1.100000e+03, FES = 82100
Generation 830: Best Fitness = 1.100000e+03, FES = 83100
Generation 840: Best Fitness = 1.100000e+03, FES = 84100
Generation 850: Best Fitness = 1.100000e+03, FES = 85100
Generation 860: Best Fitness = 1.100000e+03, FES = 86100
Generation 870: Best Fitness = 1.100000e+03, FES = 87100
Generation 880: Best Fitness = 1.100000e+03, FES = 88100
Generation 890: Best Fitness = 1.100000e+03, FES = 89100
Generation 900: Best Fitness = 1.100000e+03, FES = 90100
Generation 910: Best Fitness = 1.100000e+03, FES = 91100
Generation 920: Best Fitness = 1.100000e+03, FES = 92100
Generation 930: Best Fitness = 1.100000e+03, FES = 93100
Generation 940: Best Fitness = 1.100000e+03, FES = 94100
Generation 950: Best Fitness = 1.100000e+03, FES = 95100
```

```
Generation 960: Best Fitness = 1.100000e+03, FES = 96100
Generation 970: Best Fitness = 1.100000e+03, FES = 97100
Generation 980: Best Fitness = 1.100000e+03, FES = 98100
Generation 990: Best Fitness = 1.100000e+03, FES = 99100

Optimization completed!
Final Best Fitness = 1.100000e+03
Total Function Evaluations = 100000

======================================================================
==========
Results:
  Best Fitness           : 1.100000e+03
  Global Optimum         : 1.100000e+03
  Error (|best-optimum|) : 0.000000e+00
  Best Solution (5 dim)  : [-24.75259026 -16.78697025 -58.62265222
9.08309142  19.51774121]
  D^(0) (init. diversity): 0.0000e+00
======================================================================
==========
```



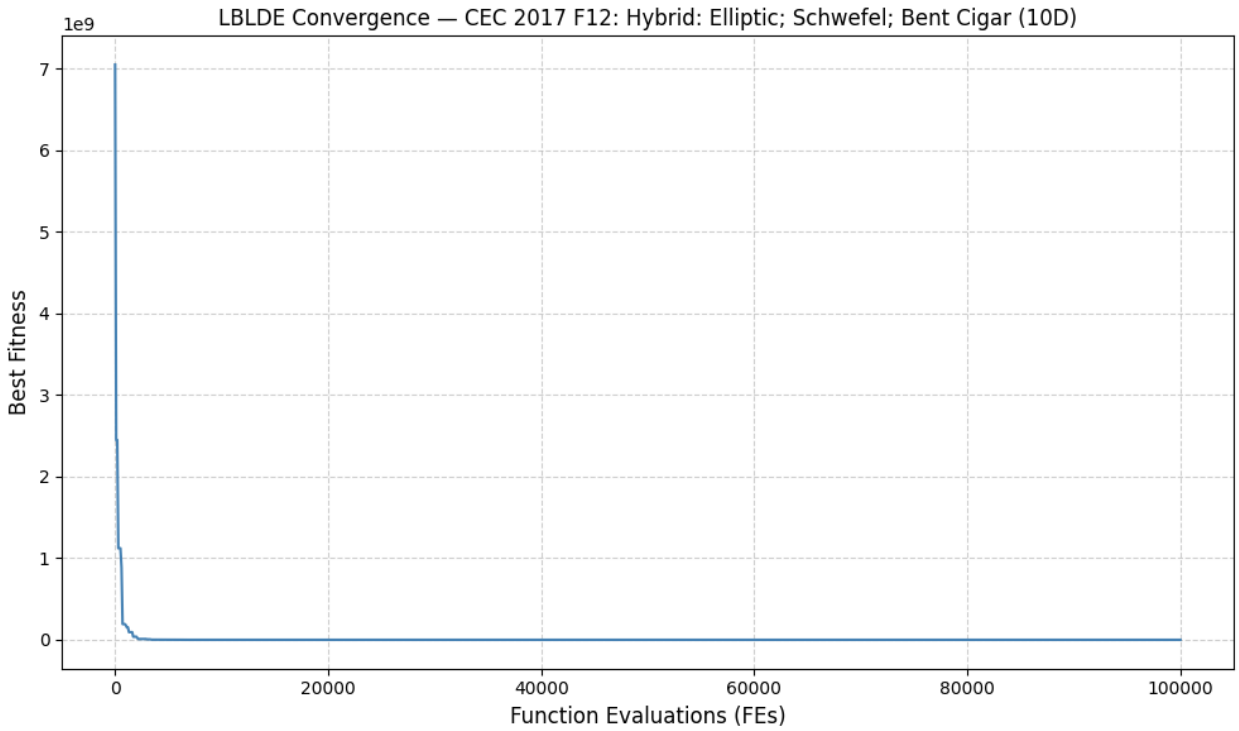

```
Graph saved as convergence_F11.png


# ============================================================
# F12 — Hybrid Function 2 (Elliptic; Schwefel; Bent Cigar)
# ============================================================
```

```
run_and_plot('F12', make_wrapper(hybrid.f12), 1200.0, 'Hybrid:
Elliptic; Schwefel; Bent Cigar')
```

```
======================================================================
==========
LBLDE on CEC 2017 F12 — Single Run
======================================================================
==========
Function       : Hybrid: Elliptic; Schwefel; Bent Cigar (10D)
Bounds         : [-100, 100]
Global optimum : 1200.0
Generation 0: Best Fitness = 7.052719e+09
Generation 10: Best Fitness = 1.800351e+08, FES = 1100
Generation 20: Best Fitness = 3.810025e+07, FES = 2100
Generation 30: Best Fitness = 6.348359e+06, FES = 3100
Generation 40: Best Fitness = 1.863698e+06, FES = 4100
Generation 50: Best Fitness = 1.214815e+06, FES = 5100
Generation 60: Best Fitness = 4.683876e+05, FES = 6100
Generation 70: Best Fitness = 9.175683e+04, FES = 7100
Generation 80: Best Fitness = 4.540789e+04, FES = 8100
Generation 90: Best Fitness = 3.083798e+04, FES = 9100
Generation 100: Best Fitness = 9.379111e+03, FES = 10100
Generation 110: Best Fitness = 8.200349e+03, FES = 11100
Generation 120: Best Fitness = 3.984329e+03, FES = 12100
Generation 130: Best Fitness = 3.984329e+03, FES = 13100
Generation 140: Best Fitness = 2.753864e+03, FES = 14100
Generation 150: Best Fitness = 2.475806e+03, FES = 15100
Generation 160: Best Fitness = 2.002916e+03, FES = 16100
Generation 170: Best Fitness = 1.894129e+03, FES = 17100
Generation 180: Best Fitness = 1.492443e+03, FES = 18100
Generation 190: Best Fitness = 1.428081e+03, FES = 19100
Generation 200: Best Fitness = 1.282643e+03, FES = 20100
Generation 210: Best Fitness = 1.241950e+03, FES = 21100
Generation 220: Best Fitness = 1.230234e+03, FES = 22100
Generation 230: Best Fitness = 1.221447e+03, FES = 23100
Generation 240: Best Fitness = 1.215475e+03, FES = 24100
Generation 250: Best Fitness = 1.213525e+03, FES = 25100
Generation 260: Best Fitness = 1.212660e+03, FES = 26100
Generation 270: Best Fitness = 1.212067e+03, FES = 27100
Generation 280: Best Fitness = 1.211760e+03, FES = 28100
Generation 290: Best Fitness = 1.211602e+03, FES = 29100
Generation 300: Best Fitness = 1.211479e+03, FES = 30100
Generation 310: Best Fitness = 1.211414e+03, FES = 31100
Generation 320: Best Fitness = 1.211395e+03, FES = 32100
Generation 330: Best Fitness = 1.211386e+03, FES = 33100
Generation 340: Best Fitness = 1.211383e+03, FES = 34100
Generation 350: Best Fitness = 1.211382e+03, FES = 35100
Generation 360: Best Fitness = 1.211381e+03, FES = 36100
Generation 370: Best Fitness = 1.211381e+03, FES = 37100
Generation 380: Best Fitness = 1.211381e+03, FES = 38100
```

```
Generation 390: Best Fitness = 1.211381e+03, FES = 39100
Generation 400: Best Fitness = 1.211381e+03, FES = 40100
Generation 410: Best Fitness = 1.211381e+03, FES = 41100
Generation 420: Best Fitness = 1.211381e+03, FES = 42100
Generation 430: Best Fitness = 1.211381e+03, FES = 43100
Generation 440: Best Fitness = 1.211381e+03, FES = 44100
Generation 450: Best Fitness = 1.211381e+03, FES = 45100
Generation 460: Best Fitness = 1.211381e+03, FES = 46100
Generation 470: Best Fitness = 1.211381e+03, FES = 47100
Generation 480: Best Fitness = 1.211381e+03, FES = 48100
Generation 490: Best Fitness = 1.211381e+03, FES = 49100
Generation 500: Best Fitness = 1.211381e+03, FES = 50100
Generation 510: Best Fitness = 1.211381e+03, FES = 51100
Generation 520: Best Fitness = 1.211381e+03, FES = 52100
Generation 530: Best Fitness = 1.211381e+03, FES = 53100
Generation 540: Best Fitness = 1.211381e+03, FES = 54100
Generation 550: Best Fitness = 1.211381e+03, FES = 55100
Generation 560: Best Fitness = 1.211381e+03, FES = 56100
Generation 570: Best Fitness = 1.211381e+03, FES = 57100
Generation 580: Best Fitness = 1.211381e+03, FES = 58100
Generation 590: Best Fitness = 1.211381e+03, FES = 59100
Generation 600: Best Fitness = 1.211381e+03, FES = 60100
Generation 610: Best Fitness = 1.211381e+03, FES = 61100
Generation 620: Best Fitness = 1.211381e+03, FES = 62100
Generation 630: Best Fitness = 1.211381e+03, FES = 63100
Generation 640: Best Fitness = 1.211381e+03, FES = 64100
Generation 650: Best Fitness = 1.211381e+03, FES = 65100
Generation 660: Best Fitness = 1.211381e+03, FES = 66100
Generation 670: Best Fitness = 1.211381e+03, FES = 67100
Generation 680: Best Fitness = 1.211381e+03, FES = 68100
Generation 690: Best Fitness = 1.211381e+03, FES = 69100
Generation 700: Best Fitness = 1.211381e+03, FES = 70100
Generation 710: Best Fitness = 1.211381e+03, FES = 71100
Generation 720: Best Fitness = 1.211381e+03, FES = 72100
Generation 730: Best Fitness = 1.211381e+03, FES = 73100
Generation 740: Best Fitness = 1.211381e+03, FES = 74100
Generation 750: Best Fitness = 1.211381e+03, FES = 75100
Generation 760: Best Fitness = 1.211381e+03, FES = 76100
Generation 770: Best Fitness = 1.211381e+03, FES = 77100
Generation 780: Best Fitness = 1.211381e+03, FES = 78100
Generation 790: Best Fitness = 1.211381e+03, FES = 79100
Generation 800: Best Fitness = 1.211381e+03, FES = 80100
Generation 810: Best Fitness = 1.211381e+03, FES = 81100
Generation 820: Best Fitness = 1.211381e+03, FES = 82100
Generation 830: Best Fitness = 1.211381e+03, FES = 83100
Generation 840: Best Fitness = 1.211381e+03, FES = 84100
Generation 850: Best Fitness = 1.211381e+03, FES = 85100
Generation 860: Best Fitness = 1.211381e+03, FES = 86100
Generation 870: Best Fitness = 1.211381e+03, FES = 87100
```

```
Generation 880: Best Fitness = 1.211381e+03, FES = 88100
Generation 890: Best Fitness = 1.211381e+03, FES = 89100
Generation 900: Best Fitness = 1.211381e+03, FES = 90100
Generation 910: Best Fitness = 1.211381e+03, FES = 91100
Generation 920: Best Fitness = 1.211381e+03, FES = 92100
Generation 930: Best Fitness = 1.211381e+03, FES = 93100
Generation 940: Best Fitness = 1.211381e+03, FES = 94100
Generation 950: Best Fitness = 1.211381e+03, FES = 95100
Generation 960: Best Fitness = 1.211381e+03, FES = 96100
Generation 970: Best Fitness = 1.211381e+03, FES = 97100
Generation 980: Best Fitness = 1.211381e+03, FES = 98100
Generation 990: Best Fitness = 1.211381e+03, FES = 99100

Optimization completed!
Final Best Fitness = 1.211381e+03
Total Function Evaluations = 100000

==========================================================================
==========
Results:
  Best Fitness           : 1.211381e+03
  Global Optimum         : 1.200000e+03
  Error (|best-optimum|) : 1.138114e+01
  Best Solution (5 dim)  : [ 65.25883255  61.19520288 -52.04281874
59.16740537 -34.47152768]
  D^(0) (init. diversity): 0.0000e+00
==========================================================================
==========
```

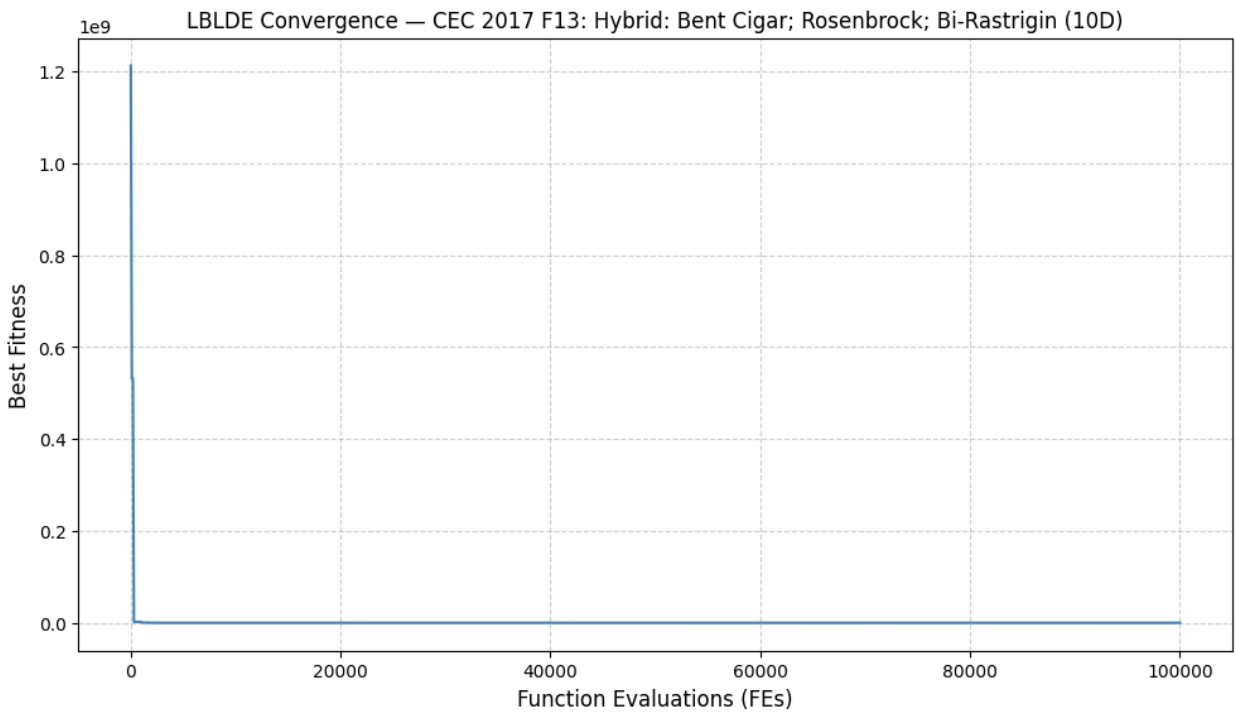



```
Graph saved as convergence_F12.png


# ==============================================================
# F13 — Hybrid Function 3 (Bent Cigar; Rosenbrock; Bi-Rastrigin)
# ==============================================================
run_and_plot('F13', make_wrapper(hybrid.f13), 1300.0, 'Hybrid: Bent
Cigar; Rosenbrock; Bi-Rastrigin')

=========================================================================
==========
LBLDE on CEC 2017 F13 — Single Run
=========================================================================
==========
Function       : Hybrid: Bent Cigar; Rosenbrock; Bi-Rastrigin (10D)
Bounds         : [-100, 100]
Global optimum : 1300.0
Generation 0: Best Fitness = 1.212051e+09
Generation 10: Best Fitness = 1.965735e+06, FES = 1100
Generation 20: Best Fitness = 2.699847e+04, FES = 2100
Generation 30: Best Fitness = 7.476611e+03, FES = 3100
Generation 40: Best Fitness = 3.400077e+03, FES = 4100
Generation 50: Best Fitness = 1.842104e+03, FES = 5100
Generation 60: Best Fitness = 1.514055e+03, FES = 6100
Generation 70: Best Fitness = 1.400366e+03, FES = 7100
Generation 80: Best Fitness = 1.380238e+03, FES = 8100
Generation 90: Best Fitness = 1.349926e+03, FES = 9100
```

```
Generation 100: Best Fitness = 1.334588e+03, FES = 10100
Generation 110: Best Fitness = 1.334588e+03, FES = 11100
Generation 120: Best Fitness = 1.327438e+03, FES = 12100
Generation 130: Best Fitness = 1.323838e+03, FES = 13100
Generation 140: Best Fitness = 1.323838e+03, FES = 14100
Generation 150: Best Fitness = 1.323838e+03, FES = 15100
Generation 160: Best Fitness = 1.316780e+03, FES = 16100
Generation 170: Best Fitness = 1.316780e+03, FES = 17100
Generation 180: Best Fitness = 1.316780e+03, FES = 18100
Generation 190: Best Fitness = 1.316356e+03, FES = 19100
Generation 200: Best Fitness = 1.316055e+03, FES = 20100
Generation 210: Best Fitness = 1.315553e+03, FES = 21100
Generation 220: Best Fitness = 1.315126e+03, FES = 22100
Generation 230: Best Fitness = 1.315126e+03, FES = 23100
Generation 240: Best Fitness = 1.312213e+03, FES = 24100
Generation 250: Best Fitness = 1.312117e+03, FES = 25100
Generation 260: Best Fitness = 1.312117e+03, FES = 26100
Generation 270: Best Fitness = 1.312117e+03, FES = 27100
Generation 280: Best Fitness = 1.312117e+03, FES = 28100
Generation 290: Best Fitness = 1.311401e+03, FES = 29100
Generation 300: Best Fitness = 1.311401e+03, FES = 30100
Generation 310: Best Fitness = 1.311401e+03, FES = 31100
Generation 320: Best Fitness = 1.310802e+03, FES = 32100
Generation 330: Best Fitness = 1.309516e+03, FES = 33100
Generation 340: Best Fitness = 1.309516e+03, FES = 34100
Generation 350: Best Fitness = 1.309448e+03, FES = 35100
Generation 360: Best Fitness = 1.309448e+03, FES = 36100
Generation 370: Best Fitness = 1.309448e+03, FES = 37100
Generation 380: Best Fitness = 1.309448e+03, FES = 38100
Generation 390: Best Fitness = 1.309263e+03, FES = 39100
Generation 400: Best Fitness = 1.308787e+03, FES = 40100
Generation 410: Best Fitness = 1.308771e+03, FES = 41100
Generation 420: Best Fitness = 1.308771e+03, FES = 42100
Generation 430: Best Fitness = 1.308771e+03, FES = 43100
Generation 440: Best Fitness = 1.308687e+03, FES = 44100
Generation 450: Best Fitness = 1.308315e+03, FES = 45100
Generation 460: Best Fitness = 1.307857e+03, FES = 46100
Generation 470: Best Fitness = 1.307857e+03, FES = 47100
Generation 480: Best Fitness = 1.307857e+03, FES = 48100
Generation 490: Best Fitness = 1.307554e+03, FES = 49100
Generation 500: Best Fitness = 1.307524e+03, FES = 50100
Generation 510: Best Fitness = 1.307266e+03, FES = 51100
Generation 520: Best Fitness = 1.307266e+03, FES = 52100
Generation 530: Best Fitness = 1.306949e+03, FES = 53100
Generation 540: Best Fitness = 1.306843e+03, FES = 54100
Generation 550: Best Fitness = 1.306843e+03, FES = 55100
Generation 560: Best Fitness = 1.306574e+03, FES = 56100
Generation 570: Best Fitness = 1.306453e+03, FES = 57100
Generation 580: Best Fitness = 1.306453e+03, FES = 58100
```

```
Generation 590: Best Fitness = 1.306418e+03, FES = 59100
Generation 600: Best Fitness = 1.306385e+03, FES = 60100
Generation 610: Best Fitness = 1.306332e+03, FES = 61100
Generation 620: Best Fitness = 1.306301e+03, FES = 62100
Generation 630: Best Fitness = 1.306264e+03, FES = 63100
Generation 640: Best Fitness = 1.306206e+03, FES = 64100
Generation 650: Best Fitness = 1.306183e+03, FES = 65100
Generation 660: Best Fitness = 1.306177e+03, FES = 66100
Generation 670: Best Fitness = 1.306168e+03, FES = 67100
Generation 680: Best Fitness = 1.306125e+03, FES = 68100
Generation 690: Best Fitness = 1.306072e+03, FES = 69100
Generation 700: Best Fitness = 1.306062e+03, FES = 70100
Generation 710: Best Fitness = 1.306043e+03, FES = 71100
Generation 720: Best Fitness = 1.306008e+03, FES = 72100
Generation 730: Best Fitness = 1.305992e+03, FES = 73100
Generation 740: Best Fitness = 1.305991e+03, FES = 74100
Generation 750: Best Fitness = 1.305980e+03, FES = 75100
Generation 760: Best Fitness = 1.305966e+03, FES = 76100
Generation 770: Best Fitness = 1.305960e+03, FES = 77100
Generation 780: Best Fitness = 1.305958e+03, FES = 78100
Generation 790: Best Fitness = 1.305955e+03, FES = 79100
Generation 800: Best Fitness = 1.305953e+03, FES = 80100
Generation 810: Best Fitness = 1.305952e+03, FES = 81100
Generation 820: Best Fitness = 1.305952e+03, FES = 82100
Generation 830: Best Fitness = 1.305951e+03, FES = 83100
Generation 840: Best Fitness = 1.305951e+03, FES = 84100
Generation 850: Best Fitness = 1.305951e+03, FES = 85100
Generation 860: Best Fitness = 1.305951e+03, FES = 86100
Generation 870: Best Fitness = 1.305951e+03, FES = 87100
Generation 880: Best Fitness = 1.305951e+03, FES = 88100
Generation 890: Best Fitness = 1.305951e+03, FES = 89100
Generation 900: Best Fitness = 1.305951e+03, FES = 90100
Generation 910: Best Fitness = 1.305951e+03, FES = 91100
Generation 920: Best Fitness = 1.305951e+03, FES = 92100
Generation 930: Best Fitness = 1.305951e+03, FES = 93100
Generation 940: Best Fitness = 1.305951e+03, FES = 94100
Generation 950: Best Fitness = 1.305951e+03, FES = 95100
Generation 960: Best Fitness = 1.305951e+03, FES = 96100
Generation 970: Best Fitness = 1.305951e+03, FES = 97100
Generation 980: Best Fitness = 1.305951e+03, FES = 98100
Generation 990: Best Fitness = 1.305951e+03, FES = 99100

Optimization completed!
Final Best Fitness = 1.305951e+03
Total Function Evaluations = 100000

==========================================================================
==========
Results:
```

```
  Best Fitness           : 1.305951e+03
  Global Optimum         : 1.300000e+03
  Error (|best-optimum|) : 5.951113e+00
  Best Solution (5 dim)  : [ 10.29015325 -42.97113395  -6.24587965 -
14.50725504  25.61557025]
  D^(0) (init. diversity): 0.0000e+00
==================================================================
==========
```

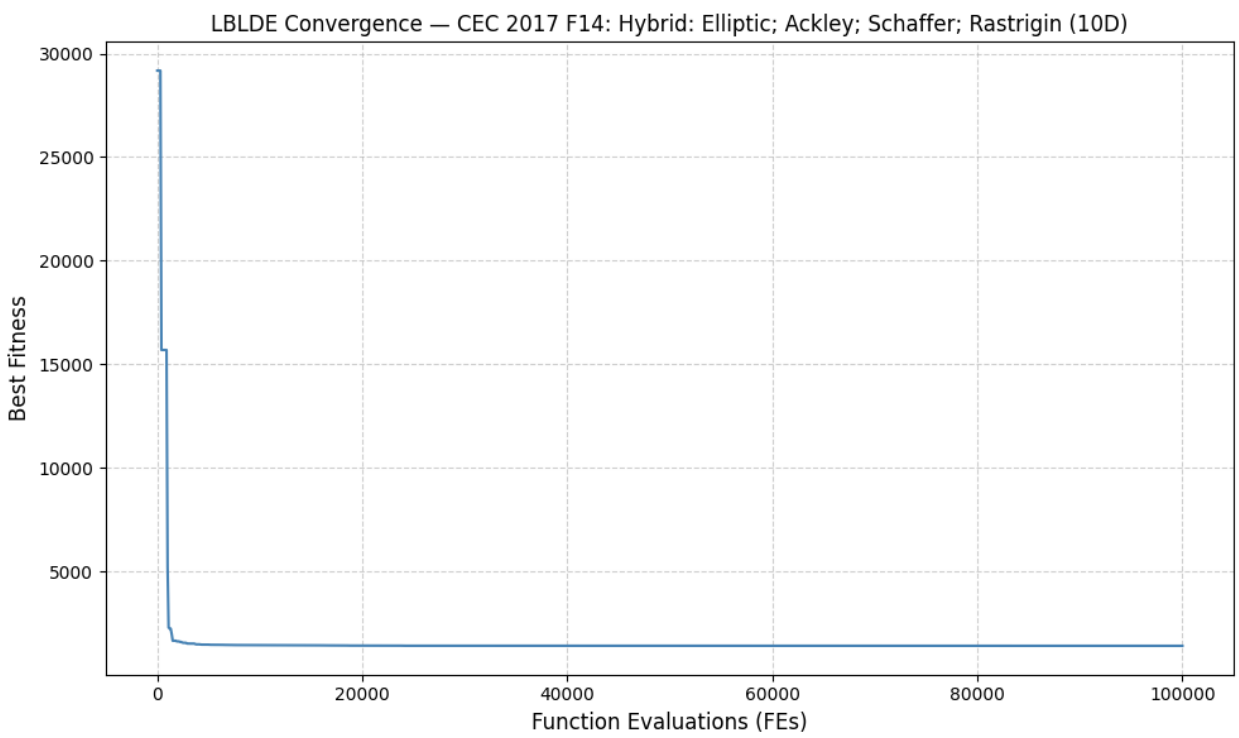



```
Graph saved as convergence_F13.png


# ============================================================
# F14 — Hybrid Function 4 (Elliptic; Ackley; Schaffer; Rastrigin)
# ============================================================
run_and_plot('F14', make_wrapper(hybrid.f14), 1400.0, 'Hybrid:
Elliptic; Ackley; Schaffer; Rastrigin')

==================================================================
==========
LBLDE on CEC 2017 F14 — Single Run
==================================================================
==========
Function       : Hybrid: Elliptic; Ackley; Schaffer; Rastrigin (10D)
Bounds         : [-100, 100]
Global optimum : 1400.0
Generation 0: Best Fitness = 2.917787e+04
```

```
Generation 10: Best Fitness = 5.213946e+03, FES = 1100
Generation 20: Best Fitness = 1.614055e+03, FES = 2100
Generation 30: Best Fitness = 1.507531e+03, FES = 3100
Generation 40: Best Fitness = 1.472085e+03, FES = 4100
Generation 50: Best Fitness = 1.456220e+03, FES = 5100
Generation 60: Best Fitness = 1.446816e+03, FES = 6100
Generation 70: Best Fitness = 1.438691e+03, FES = 7100
Generation 80: Best Fitness = 1.435691e+03, FES = 8100
Generation 90: Best Fitness = 1.430194e+03, FES = 9100
Generation 100: Best Fitness = 1.429546e+03, FES = 10100
Generation 110: Best Fitness = 1.429407e+03, FES = 11100
Generation 120: Best Fitness = 1.428347e+03, FES = 12100
Generation 130: Best Fitness = 1.427266e+03, FES = 13100
Generation 140: Best Fitness = 1.426262e+03, FES = 14100
Generation 150: Best Fitness = 1.423473e+03, FES = 15100
Generation 160: Best Fitness = 1.421475e+03, FES = 16100
Generation 170: Best Fitness = 1.418886e+03, FES = 17100
Generation 180: Best Fitness = 1.416402e+03, FES = 18100
Generation 190: Best Fitness = 1.411658e+03, FES = 19100
Generation 200: Best Fitness = 1.411658e+03, FES = 20100
Generation 210: Best Fitness = 1.408835e+03, FES = 21100
Generation 220: Best Fitness = 1.408632e+03, FES = 22100
Generation 230: Best Fitness = 1.407484e+03, FES = 23100
Generation 240: Best Fitness = 1.404122e+03, FES = 24100
Generation 250: Best Fitness = 1.404122e+03, FES = 25100
Generation 260: Best Fitness = 1.404029e+03, FES = 26100
Generation 270: Best Fitness = 1.404029e+03, FES = 27100
Generation 280: Best Fitness = 1.402910e+03, FES = 28100
Generation 290: Best Fitness = 1.402640e+03, FES = 29100
Generation 300: Best Fitness = 1.402640e+03, FES = 30100
Generation 310: Best Fitness = 1.401891e+03, FES = 31100
Generation 320: Best Fitness = 1.401375e+03, FES = 32100
Generation 330: Best Fitness = 1.401282e+03, FES = 33100
Generation 340: Best Fitness = 1.400765e+03, FES = 34100
Generation 350: Best Fitness = 1.400715e+03, FES = 35100
Generation 360: Best Fitness = 1.400616e+03, FES = 36100
Generation 370: Best Fitness = 1.400427e+03, FES = 37100
Generation 380: Best Fitness = 1.400279e+03, FES = 38100
Generation 390: Best Fitness = 1.400223e+03, FES = 39100
Generation 400: Best Fitness = 1.400142e+03, FES = 40100
Generation 410: Best Fitness = 1.400034e+03, FES = 41100
Generation 420: Best Fitness = 1.400026e+03, FES = 42100
Generation 430: Best Fitness = 1.400010e+03, FES = 43100
Generation 440: Best Fitness = 1.400008e+03, FES = 44100
Generation 450: Best Fitness = 1.400005e+03, FES = 45100
Generation 460: Best Fitness = 1.400001e+03, FES = 46100
Generation 470: Best Fitness = 1.400001e+03, FES = 47100
Generation 480: Best Fitness = 1.400001e+03, FES = 48100
Generation 490: Best Fitness = 1.400000e+03, FES = 49100
```

```
Generation 500: Best Fitness = 1.400000e+03, FES = 50100
Generation 510: Best Fitness = 1.400000e+03, FES = 51100
Generation 520: Best Fitness = 1.400000e+03, FES = 52100
Generation 530: Best Fitness = 1.400000e+03, FES = 53100
Generation 540: Best Fitness = 1.400000e+03, FES = 54100
Generation 550: Best Fitness = 1.400000e+03, FES = 55100
Generation 560: Best Fitness = 1.400000e+03, FES = 56100
Generation 570: Best Fitness = 1.400000e+03, FES = 57100
Generation 580: Best Fitness = 1.400000e+03, FES = 58100
Generation 590: Best Fitness = 1.400000e+03, FES = 59100
Generation 600: Best Fitness = 1.400000e+03, FES = 60100
Generation 610: Best Fitness = 1.400000e+03, FES = 61100
Generation 620: Best Fitness = 1.400000e+03, FES = 62100
Generation 630: Best Fitness = 1.400000e+03, FES = 63100
Generation 640: Best Fitness = 1.400000e+03, FES = 64100
Generation 650: Best Fitness = 1.400000e+03, FES = 65100
Generation 660: Best Fitness = 1.400000e+03, FES = 66100
Generation 670: Best Fitness = 1.400000e+03, FES = 67100
Generation 680: Best Fitness = 1.400000e+03, FES = 68100
Generation 690: Best Fitness = 1.400000e+03, FES = 69100
Generation 700: Best Fitness = 1.400000e+03, FES = 70100
Generation 710: Best Fitness = 1.400000e+03, FES = 71100
Generation 720: Best Fitness = 1.400000e+03, FES = 72100
Generation 730: Best Fitness = 1.400000e+03, FES = 73100
Generation 740: Best Fitness = 1.400000e+03, FES = 74100
Generation 750: Best Fitness = 1.400000e+03, FES = 75100
Generation 760: Best Fitness = 1.400000e+03, FES = 76100
Generation 770: Best Fitness = 1.400000e+03, FES = 77100
Generation 780: Best Fitness = 1.400000e+03, FES = 78100
Generation 790: Best Fitness = 1.400000e+03, FES = 79100
Generation 800: Best Fitness = 1.400000e+03, FES = 80100
Generation 810: Best Fitness = 1.400000e+03, FES = 81100
Generation 820: Best Fitness = 1.400000e+03, FES = 82100
Generation 830: Best Fitness = 1.400000e+03, FES = 83100
Generation 840: Best Fitness = 1.400000e+03, FES = 84100
Generation 850: Best Fitness = 1.400000e+03, FES = 85100
Generation 860: Best Fitness = 1.400000e+03, FES = 86100
Generation 870: Best Fitness = 1.400000e+03, FES = 87100
Generation 880: Best Fitness = 1.400000e+03, FES = 88100
Generation 890: Best Fitness = 1.400000e+03, FES = 89100
Generation 900: Best Fitness = 1.400000e+03, FES = 90100
Generation 910: Best Fitness = 1.400000e+03, FES = 91100
Generation 920: Best Fitness = 1.400000e+03, FES = 92100
Generation 930: Best Fitness = 1.400000e+03, FES = 93100
Generation 940: Best Fitness = 1.400000e+03, FES = 94100
Generation 950: Best Fitness = 1.400000e+03, FES = 95100
Generation 960: Best Fitness = 1.400000e+03, FES = 96100
Generation 970: Best Fitness = 1.400000e+03, FES = 97100
Generation 980: Best Fitness = 1.400000e+03, FES = 98100
```

```
Generation 990: Best Fitness = 1.400000e+03, FES = 99100

Optimization completed!
Final Best Fitness = 1.400000e+03
Total Function Evaluations = 100000

======================================================================
==========
Results:
  Best Fitness           : 1.400000e+03
  Global Optimum         : 1.400000e+03
  Error (|best-optimum|) : 0.000000e+00
  Best Solution (5 dim)  : [-36.61752681 -70.77608488  66.34141    -
39.92255214  61.66827387]
  D^(0) (init. diversity): 0.0000e+00
======================================================================
==========
```

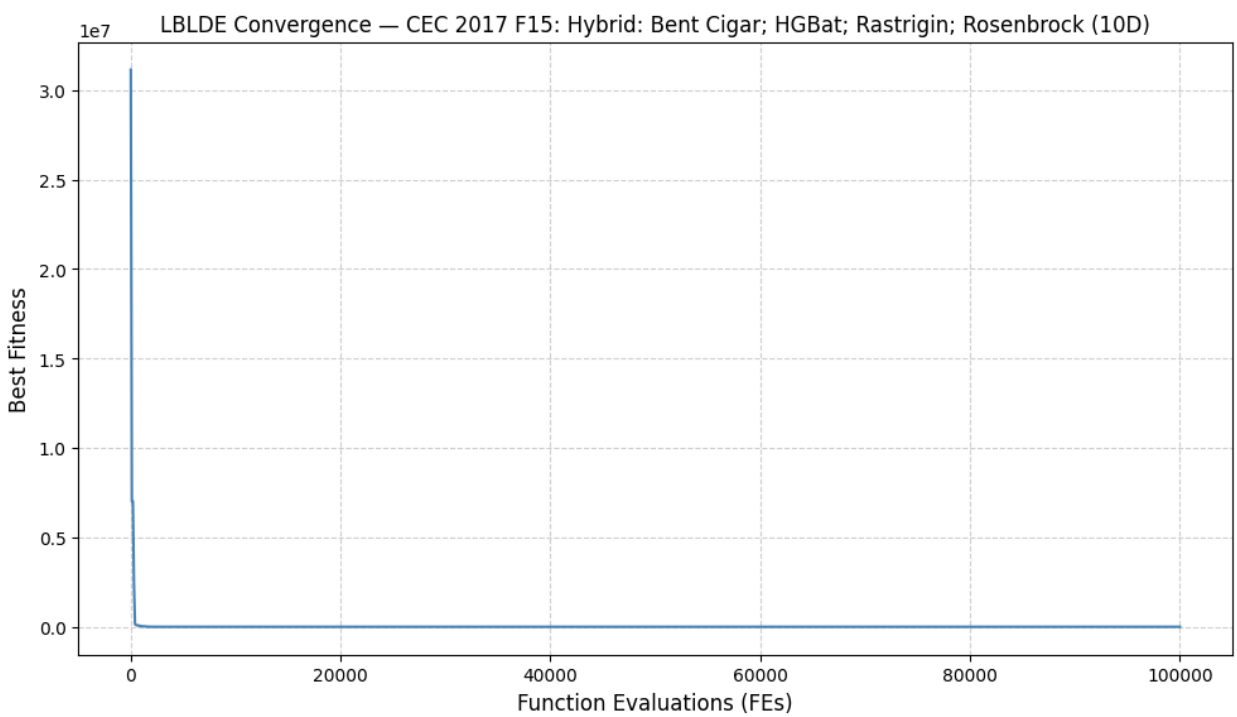



```
Graph saved as convergence_F14.png


# ============================================================
# F15 — Hybrid Function 5 (Bent Cigar; HGBat; Rastrigin; Rosenbrock)
# ============================================================
run_and_plot('F15', make_wrapper(hybrid.f15), 1500.0, 'Hybrid: Bent
Cigar; HGBat; Rastrigin; Rosenbrock')
```

```
==========================================================================
==========
LBLDE on CEC 2017 F15 — Single Run
==========================================================================
==========
Function       : Hybrid: Bent Cigar; HGBat; Rastrigin; Rosenbrock
(10D)
Bounds         : [-100, 100]
Global optimum : 1500.0
Generation 0: Best Fitness = 3.114909e+07
Generation 10: Best Fitness = 3.985958e+04, FES = 1100
Generation 20: Best Fitness = 4.625174e+03, FES = 2100
Generation 30: Best Fitness = 1.709629e+03, FES = 3100
Generation 40: Best Fitness = 1.573825e+03, FES = 4100
Generation 50: Best Fitness = 1.523564e+03, FES = 5100
Generation 60: Best Fitness = 1.520810e+03, FES = 6100
Generation 70: Best Fitness = 1.508489e+03, FES = 7100
Generation 80: Best Fitness = 1.507340e+03, FES = 8100
Generation 90: Best Fitness = 1.507340e+03, FES = 9100
Generation 100: Best Fitness = 1.505644e+03, FES = 10100
Generation 110: Best Fitness = 1.505108e+03, FES = 11100
Generation 120: Best Fitness = 1.505088e+03, FES = 12100
Generation 130: Best Fitness = 1.504469e+03, FES = 13100
Generation 140: Best Fitness = 1.504469e+03, FES = 14100
Generation 150: Best Fitness = 1.503484e+03, FES = 15100
Generation 160: Best Fitness = 1.503190e+03, FES = 16100
Generation 170: Best Fitness = 1.503190e+03, FES = 17100
Generation 180: Best Fitness = 1.503190e+03, FES = 18100
Generation 190: Best Fitness = 1.502736e+03, FES = 19100
Generation 200: Best Fitness = 1.502328e+03, FES = 20100
Generation 210: Best Fitness = 1.502299e+03, FES = 21100
Generation 220: Best Fitness = 1.502140e+03, FES = 22100
Generation 230: Best Fitness = 1.501807e+03, FES = 23100
Generation 240: Best Fitness = 1.501807e+03, FES = 24100
Generation 250: Best Fitness = 1.501758e+03, FES = 25100
Generation 260: Best Fitness = 1.501282e+03, FES = 26100
Generation 270: Best Fitness = 1.501075e+03, FES = 27100
Generation 280: Best Fitness = 1.501075e+03, FES = 28100
Generation 290: Best Fitness = 1.500937e+03, FES = 29100
Generation 300: Best Fitness = 1.500693e+03, FES = 30100
Generation 310: Best Fitness = 1.500641e+03, FES = 31100
Generation 320: Best Fitness = 1.500633e+03, FES = 32100
Generation 330: Best Fitness = 1.500498e+03, FES = 33100
Generation 340: Best Fitness = 1.500429e+03, FES = 34100
Generation 350: Best Fitness = 1.500429e+03, FES = 35100
Generation 360: Best Fitness = 1.500333e+03, FES = 36100
Generation 370: Best Fitness = 1.500333e+03, FES = 37100
Generation 380: Best Fitness = 1.500331e+03, FES = 38100
Generation 390: Best Fitness = 1.500312e+03, FES = 39100
Generation 400: Best Fitness = 1.500275e+03, FES = 40100
```

```
Generation 410: Best Fitness = 1.500222e+03, FES = 41100
Generation 420: Best Fitness = 1.500220e+03, FES = 42100
Generation 430: Best Fitness = 1.500186e+03, FES = 43100
Generation 440: Best Fitness = 1.500122e+03, FES = 44100
Generation 450: Best Fitness = 1.500113e+03, FES = 45100
Generation 460: Best Fitness = 1.500094e+03, FES = 46100
Generation 470: Best Fitness = 1.500089e+03, FES = 47100
Generation 480: Best Fitness = 1.500089e+03, FES = 48100
Generation 490: Best Fitness = 1.500059e+03, FES = 49100
Generation 500: Best Fitness = 1.500058e+03, FES = 50100
Generation 510: Best Fitness = 1.500058e+03, FES = 51100
Generation 520: Best Fitness = 1.500058e+03, FES = 52100
Generation 530: Best Fitness = 1.500051e+03, FES = 53100
Generation 540: Best Fitness = 1.500046e+03, FES = 54100
Generation 550: Best Fitness = 1.500031e+03, FES = 55100
Generation 560: Best Fitness = 1.500031e+03, FES = 56100
Generation 570: Best Fitness = 1.500031e+03, FES = 57100
Generation 580: Best Fitness = 1.500031e+03, FES = 58100
Generation 590: Best Fitness = 1.500031e+03, FES = 59100
Generation 600: Best Fitness = 1.500028e+03, FES = 60100
Generation 610: Best Fitness = 1.500028e+03, FES = 61100
Generation 620: Best Fitness = 1.500026e+03, FES = 62100
Generation 630: Best Fitness = 1.500015e+03, FES = 63100
Generation 640: Best Fitness = 1.500015e+03, FES = 64100
Generation 650: Best Fitness = 1.500012e+03, FES = 65100
Generation 660: Best Fitness = 1.500012e+03, FES = 66100
Generation 670: Best Fitness = 1.500011e+03, FES = 67100
Generation 680: Best Fitness = 1.500011e+03, FES = 68100
Generation 690: Best Fitness = 1.500007e+03, FES = 69100
Generation 700: Best Fitness = 1.500007e+03, FES = 70100
Generation 710: Best Fitness = 1.500007e+03, FES = 71100
Generation 720: Best Fitness = 1.500004e+03, FES = 72100
Generation 730: Best Fitness = 1.500004e+03, FES = 73100
Generation 740: Best Fitness = 1.500003e+03, FES = 74100
Generation 750: Best Fitness = 1.500003e+03, FES = 75100
Generation 760: Best Fitness = 1.500002e+03, FES = 76100
Generation 770: Best Fitness = 1.500002e+03, FES = 77100
Generation 780: Best Fitness = 1.500002e+03, FES = 78100
Generation 790: Best Fitness = 1.500002e+03, FES = 79100
Generation 800: Best Fitness = 1.500002e+03, FES = 80100
Generation 810: Best Fitness = 1.500002e+03, FES = 81100
Generation 820: Best Fitness = 1.500001e+03, FES = 82100
Generation 830: Best Fitness = 1.500001e+03, FES = 83100
Generation 840: Best Fitness = 1.500001e+03, FES = 84100
Generation 850: Best Fitness = 1.500001e+03, FES = 85100
Generation 860: Best Fitness = 1.500001e+03, FES = 86100
Generation 870: Best Fitness = 1.500001e+03, FES = 87100
Generation 880: Best Fitness = 1.500001e+03, FES = 88100
Generation 890: Best Fitness = 1.500001e+03, FES = 89100
Generation 900: Best Fitness = 1.500001e+03, FES = 90100
```

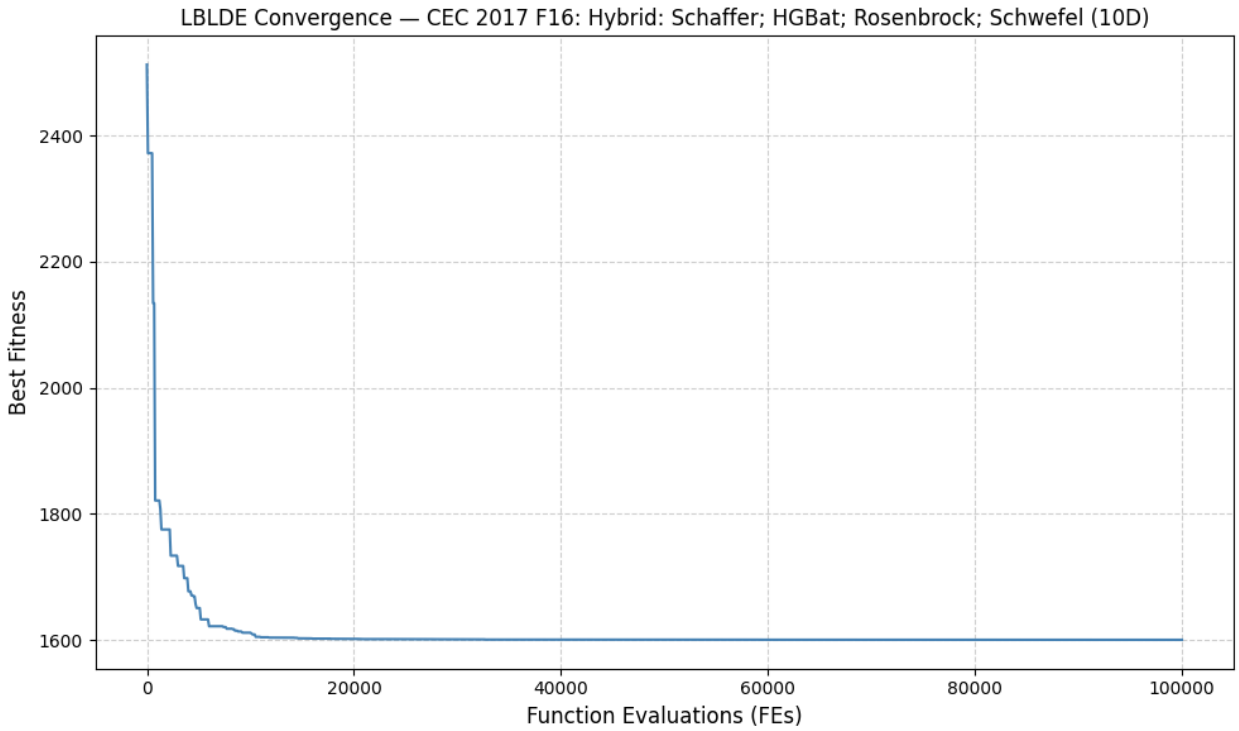
```
Generation 910: Best Fitness = 1.500001e+03, FES = 91100
Generation 920: Best Fitness = 1.500001e+03, FES = 92100
Generation 930: Best Fitness = 1.500001e+03, FES = 93100
Generation 940: Best Fitness = 1.500001e+03, FES = 94100
Generation 950: Best Fitness = 1.500001e+03, FES = 95100
Generation 960: Best Fitness = 1.500001e+03, FES = 96100
Generation 970: Best Fitness = 1.500001e+03, FES = 97100
Generation 980: Best Fitness = 1.500001e+03, FES = 98100
Generation 990: Best Fitness = 1.500001e+03, FES = 99100

Optimization completed!
Final Best Fitness = 1.500001e+03
Total Function Evaluations = 100000

================================================================================
Results:
  Best Fitness           : 1.500001e+03
  Global Optimum         : 1.500000e+03
  Error (|best-optimum|) : 7.984507e-04
  Best Solution (5 dim)  : [  5.50006048   1.96015317  65.4932222  -
42.02001783  73.01912095]
  D^(0) (init. diversity): 0.0000e+00
================================================================================
```



```
Graph saved as convergence_F15.png


# ==============================================================
# F16 — Hybrid Function 6 (Schaffer; HGBat; Rosenbrock; Schwefel)
# ==============================================================
run_and_plot('F16', make_wrapper(hybrid.f16), 1600.0, 'Hybrid:
Schaffer; HGBat; Rosenbrock; Schwefel')

=========================================================================
==========
LBLDE on CEC 2017 F16 — Single Run
=========================================================================
==========
Function       : Hybrid: Schaffer; HGBat; Rosenbrock; Schwefel (10D)
Bounds         : [-100, 100]
Global optimum : 1600.0
Generation 0: Best Fitness = 2.512498e+03
Generation 10: Best Fitness = 1.821167e+03, FES = 1100
Generation 20: Best Fitness = 1.775258e+03, FES = 2100
Generation 30: Best Fitness = 1.717383e+03, FES = 3100
Generation 40: Best Fitness = 1.677097e+03, FES = 4100
Generation 50: Best Fitness = 1.650616e+03, FES = 5100
Generation 60: Best Fitness = 1.621770e+03, FES = 6100
Generation 70: Best Fitness = 1.621770e+03, FES = 7100
Generation 80: Best Fitness = 1.617804e+03, FES = 8100
Generation 90: Best Fitness = 1.613498e+03, FES = 9100
Generation 100: Best Fitness = 1.611510e+03, FES = 10100
Generation 110: Best Fitness = 1.604274e+03, FES = 11100
Generation 120: Best Fitness = 1.603737e+03, FES = 12100
Generation 130: Best Fitness = 1.603730e+03, FES = 13100
Generation 140: Best Fitness = 1.603575e+03, FES = 14100
Generation 150: Best Fitness = 1.602602e+03, FES = 15100
Generation 160: Best Fitness = 1.602114e+03, FES = 16100
Generation 170: Best Fitness = 1.602093e+03, FES = 17100
Generation 180: Best Fitness = 1.601818e+03, FES = 18100
Generation 190: Best Fitness = 1.601778e+03, FES = 19100
Generation 200: Best Fitness = 1.601645e+03, FES = 20100
Generation 210: Best Fitness = 1.601302e+03, FES = 21100
Generation 220: Best Fitness = 1.601302e+03, FES = 22100
Generation 230: Best Fitness = 1.601302e+03, FES = 23100
Generation 240: Best Fitness = 1.601302e+03, FES = 24100
Generation 250: Best Fitness = 1.601266e+03, FES = 25100
Generation 260: Best Fitness = 1.601204e+03, FES = 26100
Generation 270: Best Fitness = 1.601173e+03, FES = 27100
Generation 280: Best Fitness = 1.601058e+03, FES = 28100
Generation 290: Best Fitness = 1.601058e+03, FES = 29100
Generation 300: Best Fitness = 1.600917e+03, FES = 30100
Generation 310: Best Fitness = 1.600917e+03, FES = 31100
Generation 320: Best Fitness = 1.600911e+03, FES = 32100
```

```
Generation 330: Best Fitness = 1.600610e+03, FES = 33100
Generation 340: Best Fitness = 1.600610e+03, FES = 34100
Generation 350: Best Fitness = 1.600610e+03, FES = 35100
Generation 360: Best Fitness = 1.600609e+03, FES = 36100
Generation 370: Best Fitness = 1.600609e+03, FES = 37100
Generation 380: Best Fitness = 1.600609e+03, FES = 38100
Generation 390: Best Fitness = 1.600609e+03, FES = 39100
Generation 400: Best Fitness = 1.600609e+03, FES = 40100
Generation 410: Best Fitness = 1.600609e+03, FES = 41100
Generation 420: Best Fitness = 1.600609e+03, FES = 42100
Generation 430: Best Fitness = 1.600532e+03, FES = 43100
Generation 440: Best Fitness = 1.600513e+03, FES = 44100
Generation 450: Best Fitness = 1.600513e+03, FES = 45100
Generation 460: Best Fitness = 1.600513e+03, FES = 46100
Generation 470: Best Fitness = 1.600508e+03, FES = 47100
Generation 480: Best Fitness = 1.600508e+03, FES = 48100
Generation 490: Best Fitness = 1.600508e+03, FES = 49100
Generation 500: Best Fitness = 1.600501e+03, FES = 50100
Generation 510: Best Fitness = 1.600501e+03, FES = 51100
Generation 520: Best Fitness = 1.600487e+03, FES = 52100
Generation 530: Best Fitness = 1.600487e+03, FES = 53100
Generation 540: Best Fitness = 1.600424e+03, FES = 54100
Generation 550: Best Fitness = 1.600424e+03, FES = 55100
Generation 560: Best Fitness = 1.600424e+03, FES = 56100
Generation 570: Best Fitness = 1.600424e+03, FES = 57100
Generation 580: Best Fitness = 1.600424e+03, FES = 58100
Generation 590: Best Fitness = 1.600376e+03, FES = 59100
Generation 600: Best Fitness = 1.600372e+03, FES = 60100
Generation 610: Best Fitness = 1.600372e+03, FES = 61100
Generation 620: Best Fitness = 1.600367e+03, FES = 62100
Generation 630: Best Fitness = 1.600367e+03, FES = 63100
Generation 640: Best Fitness = 1.600363e+03, FES = 64100
Generation 650: Best Fitness = 1.600329e+03, FES = 65100
Generation 660: Best Fitness = 1.600328e+03, FES = 66100
Generation 670: Best Fitness = 1.600316e+03, FES = 67100
Generation 680: Best Fitness = 1.600295e+03, FES = 68100
Generation 690: Best Fitness = 1.600295e+03, FES = 69100
Generation 700: Best Fitness = 1.600295e+03, FES = 70100
Generation 710: Best Fitness = 1.600295e+03, FES = 71100
Generation 720: Best Fitness = 1.600295e+03, FES = 72100
Generation 730: Best Fitness = 1.600294e+03, FES = 73100
Generation 740: Best Fitness = 1.600294e+03, FES = 74100
Generation 750: Best Fitness = 1.600290e+03, FES = 75100
Generation 760: Best Fitness = 1.600290e+03, FES = 76100
Generation 770: Best Fitness = 1.600290e+03, FES = 77100
Generation 780: Best Fitness = 1.600290e+03, FES = 78100
Generation 790: Best Fitness = 1.600249e+03, FES = 79100
Generation 800: Best Fitness = 1.600249e+03, FES = 80100
Generation 810: Best Fitness = 1.600249e+03, FES = 81100
```

```
Generation 820: Best Fitness = 1.600249e+03, FES = 82100
Generation 830: Best Fitness = 1.600249e+03, FES = 83100
Generation 840: Best Fitness = 1.600249e+03, FES = 84100
Generation 850: Best Fitness = 1.600249e+03, FES = 85100
Generation 860: Best Fitness = 1.600249e+03, FES = 86100
Generation 870: Best Fitness = 1.600249e+03, FES = 87100
Generation 880: Best Fitness = 1.600249e+03, FES = 88100
Generation 890: Best Fitness = 1.600249e+03, FES = 89100
Generation 900: Best Fitness = 1.600249e+03, FES = 90100
Generation 910: Best Fitness = 1.600249e+03, FES = 91100
Generation 920: Best Fitness = 1.600249e+03, FES = 92100
Generation 930: Best Fitness = 1.600249e+03, FES = 93100
Generation 940: Best Fitness = 1.600249e+03, FES = 94100
Generation 950: Best Fitness = 1.600249e+03, FES = 95100
Generation 960: Best Fitness = 1.600238e+03, FES = 96100
Generation 970: Best Fitness = 1.600238e+03, FES = 97100
Generation 980: Best Fitness = 1.600238e+03, FES = 98100
Generation 990: Best Fitness = 1.600235e+03, FES = 99100

Optimization completed!
Final Best Fitness = 1.600235e+03
Total Function Evaluations = 100000

=====================================================================
==========
Results:
  Best Fitness           : 1.600235e+03
  Global Optimum         : 1.600000e+03
  Error (|best-optimum|) : 2.353018e-01
  Best Solution (5 dim)  : [ 32.96415084  46.6483027   63.96441424 -
65.30016431 -42.42699788]
  D^(0) (init. diversity): 0.0000e+00
=====================================================================
==========
```

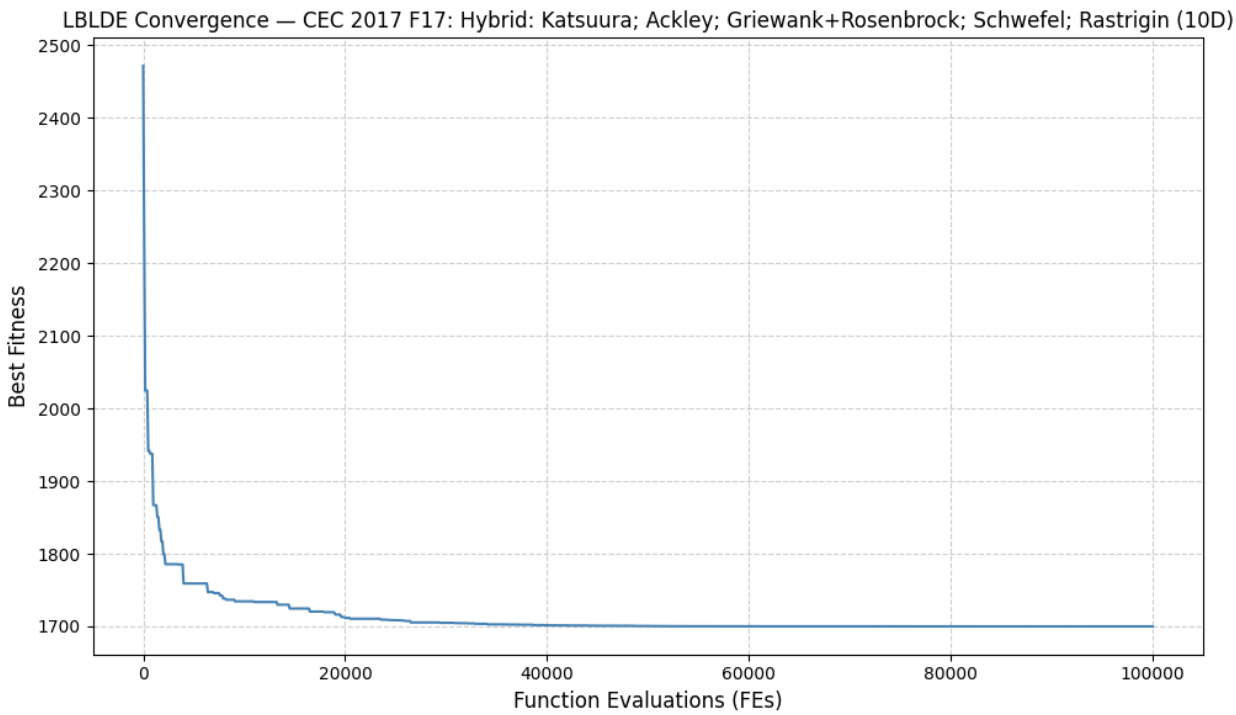



```
Graph saved as convergence_F16.png


# ============================================================
# F17 — Hybrid Function 7 (Katsuura; Ackley; Griewank+Rosenbrock;
Schwefel; Rastrigin)
# ============================================================
run_and_plot('F17', make_wrapper(hybrid.f17), 1700.0, 'Hybrid:
Katsuura; Ackley; Griewank+Rosenbrock; Schwefel; Rastrigin')

=====================================================================
==========
LBLDE on CEC 2017 F17 — Single Run
=====================================================================
==========
Function       : Hybrid: Katsuura; Ackley; Griewank+Rosenbrock;
Schwefel; Rastrigin (10D)
Bounds         : [-100, 100]
Global optimum : 1700.0
Generation 0: Best Fitness = 2.471803e+03
Generation 10: Best Fitness = 1.866862e+03, FES = 1100
Generation 20: Best Fitness = 1.799375e+03, FES = 2100
Generation 30: Best Fitness = 1.785687e+03, FES = 3100
Generation 40: Best Fitness = 1.759061e+03, FES = 4100
Generation 50: Best Fitness = 1.759061e+03, FES = 5100
Generation 60: Best Fitness = 1.759061e+03, FES = 6100
Generation 70: Best Fitness = 1.745790e+03, FES = 7100
```

```
Generation 80: Best Fitness = 1.738541e+03, FES = 8100
Generation 90: Best Fitness = 1.736659e+03, FES = 9100
Generation 100: Best Fitness = 1.734457e+03, FES = 10100
Generation 110: Best Fitness = 1.733613e+03, FES = 11100
Generation 120: Best Fitness = 1.733613e+03, FES = 12100
Generation 130: Best Fitness = 1.733525e+03, FES = 13100
Generation 140: Best Fitness = 1.729822e+03, FES = 14100
Generation 150: Best Fitness = 1.724596e+03, FES = 15100
Generation 160: Best Fitness = 1.724596e+03, FES = 16100
Generation 170: Best Fitness = 1.720397e+03, FES = 17100
Generation 180: Best Fitness = 1.719583e+03, FES = 18100
Generation 190: Best Fitness = 1.716636e+03, FES = 19100
Generation 200: Best Fitness = 1.711628e+03, FES = 20100
Generation 210: Best Fitness = 1.710549e+03, FES = 21100
Generation 220: Best Fitness = 1.710549e+03, FES = 22100
Generation 230: Best Fitness = 1.710549e+03, FES = 23100
Generation 240: Best Fitness = 1.709056e+03, FES = 24100
Generation 250: Best Fitness = 1.708476e+03, FES = 25100
Generation 260: Best Fitness = 1.707426e+03, FES = 26100
Generation 270: Best Fitness = 1.705429e+03, FES = 27100
Generation 280: Best Fitness = 1.705429e+03, FES = 28100
Generation 290: Best Fitness = 1.705429e+03, FES = 29100
Generation 300: Best Fitness = 1.704918e+03, FES = 30100
Generation 310: Best Fitness = 1.704590e+03, FES = 31100
Generation 320: Best Fitness = 1.704322e+03, FES = 32100
Generation 330: Best Fitness = 1.703594e+03, FES = 33100
Generation 340: Best Fitness = 1.703458e+03, FES = 34100
Generation 350: Best Fitness = 1.702797e+03, FES = 35100
Generation 360: Best Fitness = 1.702696e+03, FES = 36100
Generation 370: Best Fitness = 1.702427e+03, FES = 37100
Generation 380: Best Fitness = 1.702294e+03, FES = 38100
Generation 390: Best Fitness = 1.701729e+03, FES = 39100
Generation 400: Best Fitness = 1.701512e+03, FES = 40100
Generation 410: Best Fitness = 1.701512e+03, FES = 41100
Generation 420: Best Fitness = 1.701351e+03, FES = 42100
Generation 430: Best Fitness = 1.701195e+03, FES = 43100
Generation 440: Best Fitness = 1.701195e+03, FES = 44100
Generation 450: Best Fitness = 1.700809e+03, FES = 45100
Generation 460: Best Fitness = 1.700809e+03, FES = 46100
Generation 470: Best Fitness = 1.700801e+03, FES = 47100
Generation 480: Best Fitness = 1.700788e+03, FES = 48100
Generation 490: Best Fitness = 1.700599e+03, FES = 49100
Generation 500: Best Fitness = 1.700488e+03, FES = 50100
Generation 510: Best Fitness = 1.700308e+03, FES = 51100
Generation 520: Best Fitness = 1.700265e+03, FES = 52100
Generation 530: Best Fitness = 1.700257e+03, FES = 53100
Generation 540: Best Fitness = 1.700213e+03, FES = 54100
Generation 550: Best Fitness = 1.700208e+03, FES = 55100
Generation 560: Best Fitness = 1.700202e+03, FES = 56100
```

```
Generation 570: Best Fitness = 1.700198e+03, FES = 57100
Generation 580: Best Fitness = 1.700130e+03, FES = 58100
Generation 590: Best Fitness = 1.700126e+03, FES = 59100
Generation 600: Best Fitness = 1.700101e+03, FES = 60100
Generation 610: Best Fitness = 1.700097e+03, FES = 61100
Generation 620: Best Fitness = 1.700097e+03, FES = 62100
Generation 630: Best Fitness = 1.700076e+03, FES = 63100
Generation 640: Best Fitness = 1.700072e+03, FES = 64100
Generation 650: Best Fitness = 1.700047e+03, FES = 65100
Generation 660: Best Fitness = 1.700046e+03, FES = 66100
Generation 670: Best Fitness = 1.700045e+03, FES = 67100
Generation 680: Best Fitness = 1.700007e+03, FES = 68100
Generation 690: Best Fitness = 1.700006e+03, FES = 69100
Generation 700: Best Fitness = 1.700006e+03, FES = 70100
Generation 710: Best Fitness = 1.700006e+03, FES = 71100
Generation 720: Best Fitness = 1.700003e+03, FES = 72100
Generation 730: Best Fitness = 1.700002e+03, FES = 73100
Generation 740: Best Fitness = 1.700002e+03, FES = 74100
Generation 750: Best Fitness = 1.700001e+03, FES = 75100
Generation 760: Best Fitness = 1.700001e+03, FES = 76100
Generation 770: Best Fitness = 1.700000e+03, FES = 77100
Generation 780: Best Fitness = 1.700000e+03, FES = 78100
Generation 790: Best Fitness = 1.700000e+03, FES = 79100
Generation 800: Best Fitness = 1.700000e+03, FES = 80100
Generation 810: Best Fitness = 1.700000e+03, FES = 81100
Generation 820: Best Fitness = 1.700000e+03, FES = 82100
Generation 830: Best Fitness = 1.700000e+03, FES = 83100
Generation 840: Best Fitness = 1.700000e+03, FES = 84100
Generation 850: Best Fitness = 1.700000e+03, FES = 85100
Generation 860: Best Fitness = 1.700000e+03, FES = 86100
Generation 870: Best Fitness = 1.700000e+03, FES = 87100
Generation 880: Best Fitness = 1.700000e+03, FES = 88100
Generation 890: Best Fitness = 1.700000e+03, FES = 89100
Generation 900: Best Fitness = 1.700000e+03, FES = 90100
Generation 910: Best Fitness = 1.700000e+03, FES = 91100
Generation 920: Best Fitness = 1.700000e+03, FES = 92100
Generation 930: Best Fitness = 1.700000e+03, FES = 93100
Generation 940: Best Fitness = 1.700000e+03, FES = 94100
Generation 950: Best Fitness = 1.700000e+03, FES = 95100
Generation 960: Best Fitness = 1.700000e+03, FES = 96100
Generation 970: Best Fitness = 1.700000e+03, FES = 97100
Generation 980: Best Fitness = 1.700000e+03, FES = 98100
Generation 990: Best Fitness = 1.700000e+03, FES = 99100

Optimization completed!
Final Best Fitness = 1.700000e+03
Total Function Evaluations = 100000

======================================================================
```

```
==========
Results:
  Best Fitness           : 1.700000e+03
  Global Optimum         : 1.700000e+03
  Error (|best-optimum|) : 2.545740e-05
  Best Solution (5 dim)  : [-38.8413879   -7.16024381 -22.21762556
28.25934376 -53.05441624]
  D^(0) (init. diversity): 0.0000e+00
==========================================================================
==========
```

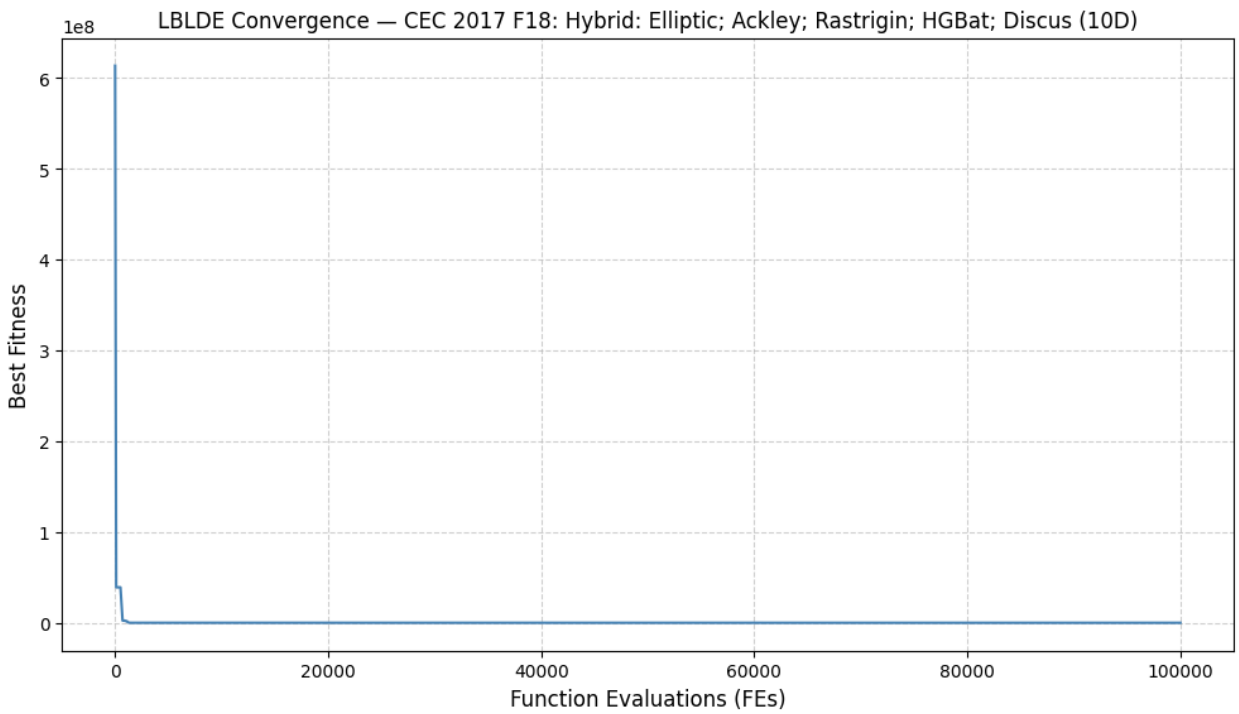



```
Graph saved as convergence_F17.png


# ============================================================
# F18 — Hybrid Function 8 (Elliptic; Ackley; Rastrigin; HGBat; Discus)
# ============================================================
run_and_plot('F18', make_wrapper(hybrid.f18), 1800.0, 'Hybrid:
Elliptic; Ackley; Rastrigin; HGBat; Discus')

==========================================================================
==========
LBLDE on CEC 2017 F18 — Single Run
==========================================================================
==========
Function       : Hybrid: Elliptic; Ackley; Rastrigin; HGBat; Discus
(10D)
```

```
Bounds         : [-100, 100]
Global optimum : 1800.0
Generation 0: Best Fitness = 6.135676e+08
Generation 10: Best Fitness = 1.834567e+06, FES = 1100
Generation 20: Best Fitness = 2.999261e+04, FES = 2100
Generation 30: Best Fitness = 3.710367e+03, FES = 3100
Generation 40: Best Fitness = 2.900791e+03, FES = 4100
Generation 50: Best Fitness = 2.153698e+03, FES = 5100
Generation 60: Best Fitness = 1.920486e+03, FES = 6100
Generation 70: Best Fitness = 1.871995e+03, FES = 7100
Generation 80: Best Fitness = 1.837634e+03, FES = 8100
Generation 90: Best Fitness = 1.830184e+03, FES = 9100
Generation 100: Best Fitness = 1.827821e+03, FES = 10100
Generation 110: Best Fitness = 1.824083e+03, FES = 11100
Generation 120: Best Fitness = 1.824083e+03, FES = 12100
Generation 130: Best Fitness = 1.824083e+03, FES = 13100
Generation 140: Best Fitness = 1.823686e+03, FES = 14100
Generation 150: Best Fitness = 1.814045e+03, FES = 15100
Generation 160: Best Fitness = 1.814045e+03, FES = 16100
Generation 170: Best Fitness = 1.813578e+03, FES = 17100
Generation 180: Best Fitness = 1.813578e+03, FES = 18100
Generation 190: Best Fitness = 1.812042e+03, FES = 19100
Generation 200: Best Fitness = 1.810253e+03, FES = 20100
Generation 210: Best Fitness = 1.807099e+03, FES = 21100
Generation 220: Best Fitness = 1.806893e+03, FES = 22100
Generation 230: Best Fitness = 1.805126e+03, FES = 23100
Generation 240: Best Fitness = 1.804848e+03, FES = 24100
Generation 250: Best Fitness = 1.803769e+03, FES = 25100
Generation 260: Best Fitness = 1.803188e+03, FES = 26100
Generation 270: Best Fitness = 1.802546e+03, FES = 27100
Generation 280: Best Fitness = 1.802527e+03, FES = 28100
Generation 290: Best Fitness = 1.802041e+03, FES = 29100
Generation 300: Best Fitness = 1.801075e+03, FES = 30100
Generation 310: Best Fitness = 1.801017e+03, FES = 31100
Generation 320: Best Fitness = 1.800903e+03, FES = 32100
Generation 330: Best Fitness = 1.800853e+03, FES = 33100
Generation 340: Best Fitness = 1.800623e+03, FES = 34100
Generation 350: Best Fitness = 1.800602e+03, FES = 35100
Generation 360: Best Fitness = 1.800471e+03, FES = 36100
Generation 370: Best Fitness = 1.800471e+03, FES = 37100
Generation 380: Best Fitness = 1.800467e+03, FES = 38100
Generation 390: Best Fitness = 1.800421e+03, FES = 39100
Generation 400: Best Fitness = 1.800393e+03, FES = 40100
Generation 410: Best Fitness = 1.800385e+03, FES = 41100
Generation 420: Best Fitness = 1.800384e+03, FES = 42100
Generation 430: Best Fitness = 1.800352e+03, FES = 43100
Generation 440: Best Fitness = 1.800335e+03, FES = 44100
Generation 450: Best Fitness = 1.800319e+03, FES = 45100
Generation 460: Best Fitness = 1.800286e+03, FES = 46100
```

```
Generation 470: Best Fitness = 1.800259e+03, FES = 47100
Generation 480: Best Fitness = 1.800242e+03, FES = 48100
Generation 490: Best Fitness = 1.800241e+03, FES = 49100
Generation 500: Best Fitness = 1.800232e+03, FES = 50100
Generation 510: Best Fitness = 1.800221e+03, FES = 51100
Generation 520: Best Fitness = 1.800205e+03, FES = 52100
Generation 530: Best Fitness = 1.800205e+03, FES = 53100
Generation 540: Best Fitness = 1.800203e+03, FES = 54100
Generation 550: Best Fitness = 1.800196e+03, FES = 55100
Generation 560: Best Fitness = 1.800195e+03, FES = 56100
Generation 570: Best Fitness = 1.800193e+03, FES = 57100
Generation 580: Best Fitness = 1.800190e+03, FES = 58100
Generation 590: Best Fitness = 1.800189e+03, FES = 59100
Generation 600: Best Fitness = 1.800188e+03, FES = 60100
Generation 610: Best Fitness = 1.800186e+03, FES = 61100
Generation 620: Best Fitness = 1.800186e+03, FES = 62100
Generation 630: Best Fitness = 1.800185e+03, FES = 63100
Generation 640: Best Fitness = 1.800185e+03, FES = 64100
Generation 650: Best Fitness = 1.800184e+03, FES = 65100
Generation 660: Best Fitness = 1.800184e+03, FES = 66100
Generation 670: Best Fitness = 1.800184e+03, FES = 67100
Generation 680: Best Fitness = 1.800184e+03, FES = 68100
Generation 690: Best Fitness = 1.800184e+03, FES = 69100
Generation 700: Best Fitness = 1.800183e+03, FES = 70100
Generation 710: Best Fitness = 1.800183e+03, FES = 71100
Generation 720: Best Fitness = 1.800183e+03, FES = 72100
Generation 730: Best Fitness = 1.800183e+03, FES = 73100
Generation 740: Best Fitness = 1.800183e+03, FES = 74100
Generation 750: Best Fitness = 1.800183e+03, FES = 75100
Generation 760: Best Fitness = 1.800183e+03, FES = 76100
Generation 770: Best Fitness = 1.800183e+03, FES = 77100
Generation 780: Best Fitness = 1.800183e+03, FES = 78100
Generation 790: Best Fitness = 1.800183e+03, FES = 79100
Generation 800: Best Fitness = 1.800183e+03, FES = 80100
Generation 810: Best Fitness = 1.800183e+03, FES = 81100
Generation 820: Best Fitness = 1.800183e+03, FES = 82100
Generation 830: Best Fitness = 1.800183e+03, FES = 83100
Generation 840: Best Fitness = 1.800183e+03, FES = 84100
Generation 850: Best Fitness = 1.800183e+03, FES = 85100
Generation 860: Best Fitness = 1.800183e+03, FES = 86100
Generation 870: Best Fitness = 1.800183e+03, FES = 87100
Generation 880: Best Fitness = 1.800183e+03, FES = 88100
Generation 890: Best Fitness = 1.800183e+03, FES = 89100
Generation 900: Best Fitness = 1.800183e+03, FES = 90100
Generation 910: Best Fitness = 1.800183e+03, FES = 91100
Generation 920: Best Fitness = 1.800183e+03, FES = 92100
Generation 930: Best Fitness = 1.800183e+03, FES = 93100
Generation 940: Best Fitness = 1.800183e+03, FES = 94100
Generation 950: Best Fitness = 1.800183e+03, FES = 95100
```

```
Generation 960: Best Fitness = 1.800183e+03, FES = 96100
Generation 970: Best Fitness = 1.800183e+03, FES = 97100
Generation 980: Best Fitness = 1.800183e+03, FES = 98100
Generation 990: Best Fitness = 1.800183e+03, FES = 99100

Optimization completed!
Final Best Fitness = 1.800183e+03
Total Function Evaluations = 100000

======================================================================
==========
Results:
  Best Fitness           : 1.800183e+03
  Global Optimum         : 1.800000e+03
  Error (|best-optimum|) : 1.829275e-01
  Best Solution (5 dim)  : [ 18.15449764   1.84791763  17.95157717 -
58.38565291 -11.41756804]
  D^(0) (init. diversity): 0.0000e+00
======================================================================
==========
```



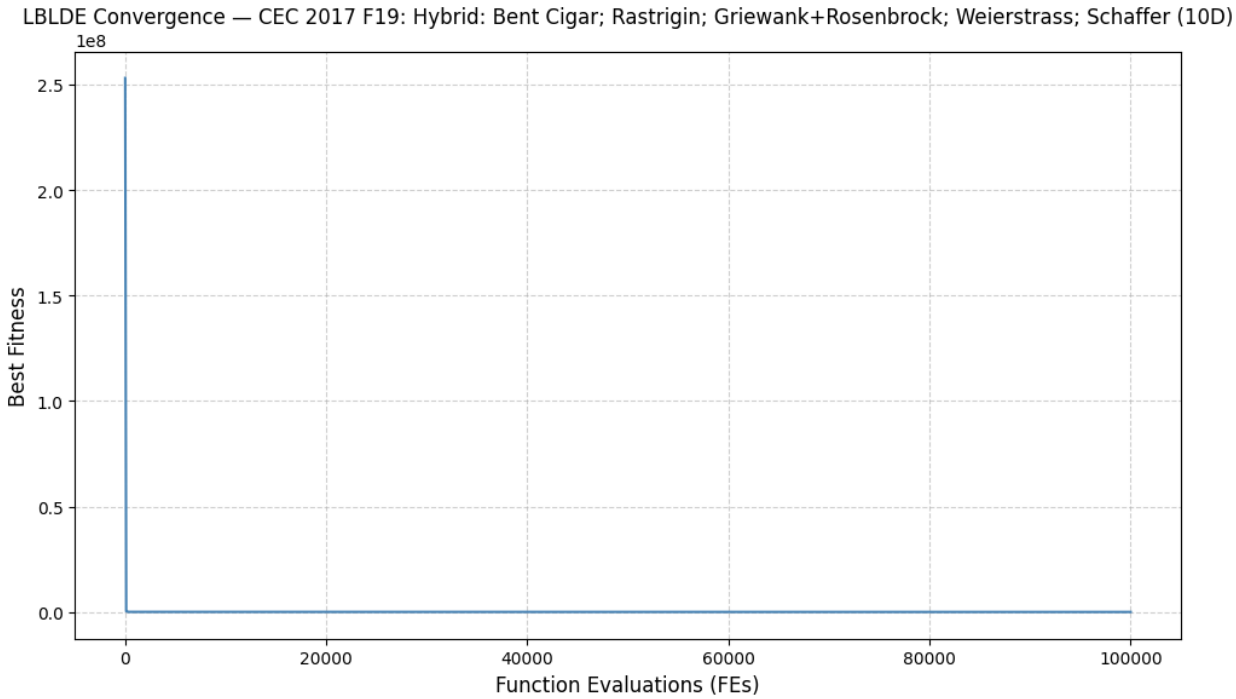

```
Graph saved as convergence_F18.png


# ============================================================
# F19 — Hybrid Function 9 (Bent Cigar; Rastrigin; Griewank+Rosenbrock;
Weierstrass; Schaffer)
```

```
# ============================================================
run_and_plot('F19', make_wrapper(hybrid.f19), 1900.0, 'Hybrid: Bent
Cigar; Rastrigin; Griewank+Rosenbrock; Weierstrass; Schaffer')
```

```
=====================================================================
==========
LBLDE on CEC 2017 F19 — Single Run
=====================================================================
==========
Function       : Hybrid: Bent Cigar; Rastrigin; Griewank+Rosenbrock;
Weierstrass; Schaffer (10D)
Bounds         : [-100, 100]
Global optimum : 1900.0
Generation 0: Best Fitness = 2.530389e+08
Generation 10: Best Fitness = 4.935248e+03, FES = 1100
Generation 20: Best Fitness = 2.636804e+03, FES = 2100
Generation 30: Best Fitness = 1.945360e+03, FES = 3100
Generation 40: Best Fitness = 1.919619e+03, FES = 4100
Generation 50: Best Fitness = 1.908322e+03, FES = 5100
Generation 60: Best Fitness = 1.908088e+03, FES = 6100
Generation 70: Best Fitness = 1.907805e+03, FES = 7100
Generation 80: Best Fitness = 1.903542e+03, FES = 8100
Generation 90: Best Fitness = 1.903542e+03, FES = 9100
Generation 100: Best Fitness = 1.903542e+03, FES = 10100
Generation 110: Best Fitness = 1.903077e+03, FES = 11100
Generation 120: Best Fitness = 1.902255e+03, FES = 12100
Generation 130: Best Fitness = 1.902255e+03, FES = 13100
Generation 140: Best Fitness = 1.902157e+03, FES = 14100
Generation 150: Best Fitness = 1.902096e+03, FES = 15100
Generation 160: Best Fitness = 1.901764e+03, FES = 16100
Generation 170: Best Fitness = 1.901568e+03, FES = 17100
Generation 180: Best Fitness = 1.901090e+03, FES = 18100
Generation 190: Best Fitness = 1.901028e+03, FES = 19100
Generation 200: Best Fitness = 1.901026e+03, FES = 20100
Generation 210: Best Fitness = 1.900977e+03, FES = 21100
Generation 220: Best Fitness = 1.900977e+03, FES = 22100
Generation 230: Best Fitness = 1.900608e+03, FES = 23100
Generation 240: Best Fitness = 1.900478e+03, FES = 24100
Generation 250: Best Fitness = 1.900478e+03, FES = 25100
Generation 260: Best Fitness = 1.900478e+03, FES = 26100
Generation 270: Best Fitness = 1.900399e+03, FES = 27100
Generation 280: Best Fitness = 1.900399e+03, FES = 28100
Generation 290: Best Fitness = 1.900281e+03, FES = 29100
Generation 300: Best Fitness = 1.900239e+03, FES = 30100
Generation 310: Best Fitness = 1.900186e+03, FES = 31100
Generation 320: Best Fitness = 1.900130e+03, FES = 32100
Generation 330: Best Fitness = 1.900130e+03, FES = 33100
Generation 340: Best Fitness = 1.900082e+03, FES = 34100
Generation 350: Best Fitness = 1.900036e+03, FES = 35100
Generation 360: Best Fitness = 1.900036e+03, FES = 36100
```

```
Generation 370: Best Fitness = 1.900036e+03, FES = 37100
Generation 380: Best Fitness = 1.900035e+03, FES = 38100
Generation 390: Best Fitness = 1.900029e+03, FES = 39100
Generation 400: Best Fitness = 1.900015e+03, FES = 40100
Generation 410: Best Fitness = 1.900015e+03, FES = 41100
Generation 420: Best Fitness = 1.900015e+03, FES = 42100
Generation 430: Best Fitness = 1.900011e+03, FES = 43100
Generation 440: Best Fitness = 1.900011e+03, FES = 44100
Generation 450: Best Fitness = 1.900011e+03, FES = 45100
Generation 460: Best Fitness = 1.900007e+03, FES = 46100
Generation 470: Best Fitness = 1.900007e+03, FES = 47100
Generation 480: Best Fitness = 1.900005e+03, FES = 48100
Generation 490: Best Fitness = 1.900001e+03, FES = 49100
Generation 500: Best Fitness = 1.900001e+03, FES = 50100
Generation 510: Best Fitness = 1.900001e+03, FES = 51100
Generation 520: Best Fitness = 1.900001e+03, FES = 52100
Generation 530: Best Fitness = 1.900001e+03, FES = 53100
Generation 540: Best Fitness = 1.900001e+03, FES = 54100
Generation 550: Best Fitness = 1.900001e+03, FES = 55100
Generation 560: Best Fitness = 1.900001e+03, FES = 56100
Generation 570: Best Fitness = 1.900000e+03, FES = 57100
Generation 580: Best Fitness = 1.900000e+03, FES = 58100
Generation 590: Best Fitness = 1.900000e+03, FES = 59100
Generation 600: Best Fitness = 1.900000e+03, FES = 60100
Generation 610: Best Fitness = 1.900000e+03, FES = 61100
Generation 620: Best Fitness = 1.900000e+03, FES = 62100
Generation 630: Best Fitness = 1.900000e+03, FES = 63100
Generation 640: Best Fitness = 1.900000e+03, FES = 64100
Generation 650: Best Fitness = 1.900000e+03, FES = 65100
Generation 660: Best Fitness = 1.900000e+03, FES = 66100
Generation 670: Best Fitness = 1.900000e+03, FES = 67100
Generation 680: Best Fitness = 1.900000e+03, FES = 68100
Generation 690: Best Fitness = 1.900000e+03, FES = 69100
Generation 700: Best Fitness = 1.900000e+03, FES = 70100
Generation 710: Best Fitness = 1.900000e+03, FES = 71100
Generation 720: Best Fitness = 1.900000e+03, FES = 72100
Generation 730: Best Fitness = 1.900000e+03, FES = 73100
Generation 740: Best Fitness = 1.900000e+03, FES = 74100
Generation 750: Best Fitness = 1.900000e+03, FES = 75100
Generation 760: Best Fitness = 1.900000e+03, FES = 76100
Generation 770: Best Fitness = 1.900000e+03, FES = 77100
Generation 780: Best Fitness = 1.900000e+03, FES = 78100
Generation 790: Best Fitness = 1.900000e+03, FES = 79100
Generation 800: Best Fitness = 1.900000e+03, FES = 80100
Generation 810: Best Fitness = 1.900000e+03, FES = 81100
Generation 820: Best Fitness = 1.900000e+03, FES = 82100
Generation 830: Best Fitness = 1.900000e+03, FES = 83100
Generation 840: Best Fitness = 1.900000e+03, FES = 84100
Generation 850: Best Fitness = 1.900000e+03, FES = 85100
```

```
Generation 860: Best Fitness = 1.900000e+03, FES = 86100
Generation 870: Best Fitness = 1.900000e+03, FES = 87100
Generation 880: Best Fitness = 1.900000e+03, FES = 88100
Generation 890: Best Fitness = 1.900000e+03, FES = 89100
Generation 900: Best Fitness = 1.900000e+03, FES = 90100
Generation 910: Best Fitness = 1.900000e+03, FES = 91100
Generation 920: Best Fitness = 1.900000e+03, FES = 92100
Generation 930: Best Fitness = 1.900000e+03, FES = 93100
Generation 940: Best Fitness = 1.900000e+03, FES = 94100
Generation 950: Best Fitness = 1.900000e+03, FES = 95100
Generation 960: Best Fitness = 1.900000e+03, FES = 96100
Generation 970: Best Fitness = 1.900000e+03, FES = 97100
Generation 980: Best Fitness = 1.900000e+03, FES = 98100
Generation 990: Best Fitness = 1.900000e+03, FES = 99100

Optimization completed!
Final Best Fitness = 1.900000e+03
Total Function Evaluations = 100000

==========================================================================
==========
Results:
  Best Fitness           : 1.900000e+03
  Global Optimum         : 1.900000e+03
  Error (|best-optimum|) : 0.000000e+00
  Best Solution (5 dim)  : [ 79.29118504   6.94740051 -35.04044538
35.23447602  25.83828371]
  D^(0) (init. diversity): 0.0000e+00
==========================================================================
==========
```

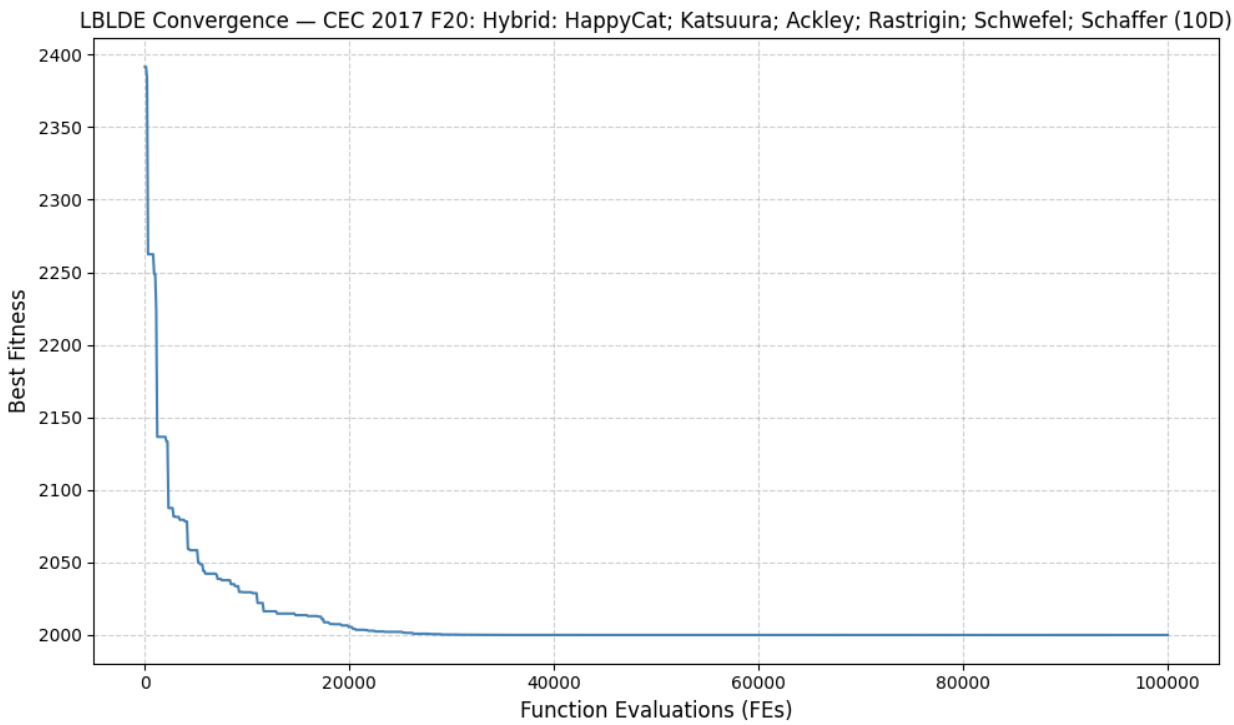



```
Graph saved as convergence_F19.png


# ============================================================
# F20 — Hybrid Function 10 (HappyCat; Katsuura; Ackley; Rastrigin;
Schwefel; Schaffer)
# ============================================================
run_and_plot('F20', make_wrapper(hybrid.f20), 2000.0, 'Hybrid:
HappyCat; Katsuura; Ackley; Rastrigin; Schwefel; Schaffer')

======================================================================
==========
LBLDE on CEC 2017 F20 — Single Run
======================================================================
==========
Function       : Hybrid: HappyCat; Katsuura; Ackley; Rastrigin;
Schwefel; Schaffer (10D)
Bounds         : [-100, 100]
Global optimum : 2000.0
Generation 0: Best Fitness = 2.391561e+03
Generation 10: Best Fitness = 2.248848e+03, FES = 1100
Generation 20: Best Fitness = 2.136551e+03, FES = 2100
Generation 30: Best Fitness = 2.081400e+03, FES = 3100
Generation 40: Best Fitness = 2.078281e+03, FES = 4100
Generation 50: Best Fitness = 2.058421e+03, FES = 5100
Generation 60: Best Fitness = 2.042295e+03, FES = 6100
Generation 70: Best Fitness = 2.041487e+03, FES = 7100
Generation 80: Best Fitness = 2.037758e+03, FES = 8100
```

```
Generation 90: Best Fitness = 2.033713e+03, FES = 9100
Generation 100: Best Fitness = 2.029487e+03, FES = 10100
Generation 110: Best Fitness = 2.022087e+03, FES = 11100
Generation 120: Best Fitness = 2.016360e+03, FES = 12100
Generation 130: Best Fitness = 2.014797e+03, FES = 13100
Generation 140: Best Fitness = 2.014797e+03, FES = 14100
Generation 150: Best Fitness = 2.013703e+03, FES = 15100
Generation 160: Best Fitness = 2.013077e+03, FES = 16100
Generation 170: Best Fitness = 2.012784e+03, FES = 17100
Generation 180: Best Fitness = 2.008168e+03, FES = 18100
Generation 190: Best Fitness = 2.007461e+03, FES = 19100
Generation 200: Best Fitness = 2.005525e+03, FES = 20100
Generation 210: Best Fitness = 2.003586e+03, FES = 21100
Generation 220: Best Fitness = 2.002965e+03, FES = 22100
Generation 230: Best Fitness = 2.002433e+03, FES = 23100
Generation 240: Best Fitness = 2.002199e+03, FES = 24100
Generation 250: Best Fitness = 2.002162e+03, FES = 25100
Generation 260: Best Fitness = 2.001478e+03, FES = 26100
Generation 270: Best Fitness = 2.000922e+03, FES = 27100
Generation 280: Best Fitness = 2.000637e+03, FES = 28100
Generation 290: Best Fitness = 2.000335e+03, FES = 29100
Generation 300: Best Fitness = 2.000332e+03, FES = 30100
Generation 310: Best Fitness = 2.000214e+03, FES = 31100
Generation 320: Best Fitness = 2.000127e+03, FES = 32100
Generation 330: Best Fitness = 2.000121e+03, FES = 33100
Generation 340: Best Fitness = 2.000102e+03, FES = 34100
Generation 350: Best Fitness = 2.000074e+03, FES = 35100
Generation 360: Best Fitness = 2.000042e+03, FES = 36100
Generation 370: Best Fitness = 2.000036e+03, FES = 37100
Generation 380: Best Fitness = 2.000016e+03, FES = 38100
Generation 390: Best Fitness = 2.000016e+03, FES = 39100
Generation 400: Best Fitness = 2.000010e+03, FES = 40100
Generation 410: Best Fitness = 2.000007e+03, FES = 41100
Generation 420: Best Fitness = 2.000006e+03, FES = 42100
Generation 430: Best Fitness = 2.000004e+03, FES = 43100
Generation 440: Best Fitness = 2.000003e+03, FES = 44100
Generation 450: Best Fitness = 2.000002e+03, FES = 45100
Generation 460: Best Fitness = 2.000001e+03, FES = 46100
Generation 470: Best Fitness = 2.000001e+03, FES = 47100
Generation 480: Best Fitness = 2.000001e+03, FES = 48100
Generation 490: Best Fitness = 2.000001e+03, FES = 49100
Generation 500: Best Fitness = 2.000000e+03, FES = 50100
Generation 510: Best Fitness = 2.000000e+03, FES = 51100
Generation 520: Best Fitness = 2.000000e+03, FES = 52100
Generation 530: Best Fitness = 2.000000e+03, FES = 53100
Generation 540: Best Fitness = 2.000000e+03, FES = 54100
Generation 550: Best Fitness = 2.000000e+03, FES = 55100
Generation 560: Best Fitness = 2.000000e+03, FES = 56100
Generation 570: Best Fitness = 2.000000e+03, FES = 57100
```

```
Generation 580: Best Fitness = 2.000000e+03, FES = 58100
Generation 590: Best Fitness = 2.000000e+03, FES = 59100
Generation 600: Best Fitness = 2.000000e+03, FES = 60100
Generation 610: Best Fitness = 2.000000e+03, FES = 61100
Generation 620: Best Fitness = 2.000000e+03, FES = 62100
Generation 630: Best Fitness = 2.000000e+03, FES = 63100
Generation 640: Best Fitness = 2.000000e+03, FES = 64100
Generation 650: Best Fitness = 2.000000e+03, FES = 65100
Generation 660: Best Fitness = 2.000000e+03, FES = 66100
Generation 670: Best Fitness = 2.000000e+03, FES = 67100
Generation 680: Best Fitness = 2.000000e+03, FES = 68100
Generation 690: Best Fitness = 2.000000e+03, FES = 69100
Generation 700: Best Fitness = 2.000000e+03, FES = 70100
Generation 710: Best Fitness = 2.000000e+03, FES = 71100
Generation 720: Best Fitness = 2.000000e+03, FES = 72100
Generation 730: Best Fitness = 2.000000e+03, FES = 73100
Generation 740: Best Fitness = 2.000000e+03, FES = 74100
Generation 750: Best Fitness = 2.000000e+03, FES = 75100
Generation 760: Best Fitness = 2.000000e+03, FES = 76100
Generation 770: Best Fitness = 2.000000e+03, FES = 77100
Generation 780: Best Fitness = 2.000000e+03, FES = 78100
Generation 790: Best Fitness = 2.000000e+03, FES = 79100
Generation 800: Best Fitness = 2.000000e+03, FES = 80100
Generation 810: Best Fitness = 2.000000e+03, FES = 81100
Generation 820: Best Fitness = 2.000000e+03, FES = 82100
Generation 830: Best Fitness = 2.000000e+03, FES = 83100
Generation 840: Best Fitness = 2.000000e+03, FES = 84100
Generation 850: Best Fitness = 2.000000e+03, FES = 85100
Generation 860: Best Fitness = 2.000000e+03, FES = 86100
Generation 870: Best Fitness = 2.000000e+03, FES = 87100
Generation 880: Best Fitness = 2.000000e+03, FES = 88100
Generation 890: Best Fitness = 2.000000e+03, FES = 89100
Generation 900: Best Fitness = 2.000000e+03, FES = 90100
Generation 910: Best Fitness = 2.000000e+03, FES = 91100
Generation 920: Best Fitness = 2.000000e+03, FES = 92100
Generation 930: Best Fitness = 2.000000e+03, FES = 93100
Generation 940: Best Fitness = 2.000000e+03, FES = 94100
Generation 950: Best Fitness = 2.000000e+03, FES = 95100
Generation 960: Best Fitness = 2.000000e+03, FES = 96100
Generation 970: Best Fitness = 2.000000e+03, FES = 97100
Generation 980: Best Fitness = 2.000000e+03, FES = 98100
Generation 990: Best Fitness = 2.000000e+03, FES = 99100

Optimization completed!
Final Best Fitness = 2.000000e+03
Total Function Evaluations = 100000

==========================================================================
==========
```

```
Results:
  Best Fitness           : 2.000000e+03
  Global Optimum         : 2.000000e+03
  Error (|best-optimum|) : 2.545513e-05
  Best Solution (5 dim)  : [-30.59057464 -55.78803892 -29.57451971
61.04539565  69.96466593]
  D^(0) (init. diversity): 0.0000e+00
==========================================================================
==========
```




```
Graph saved as convergence_F20.png
```

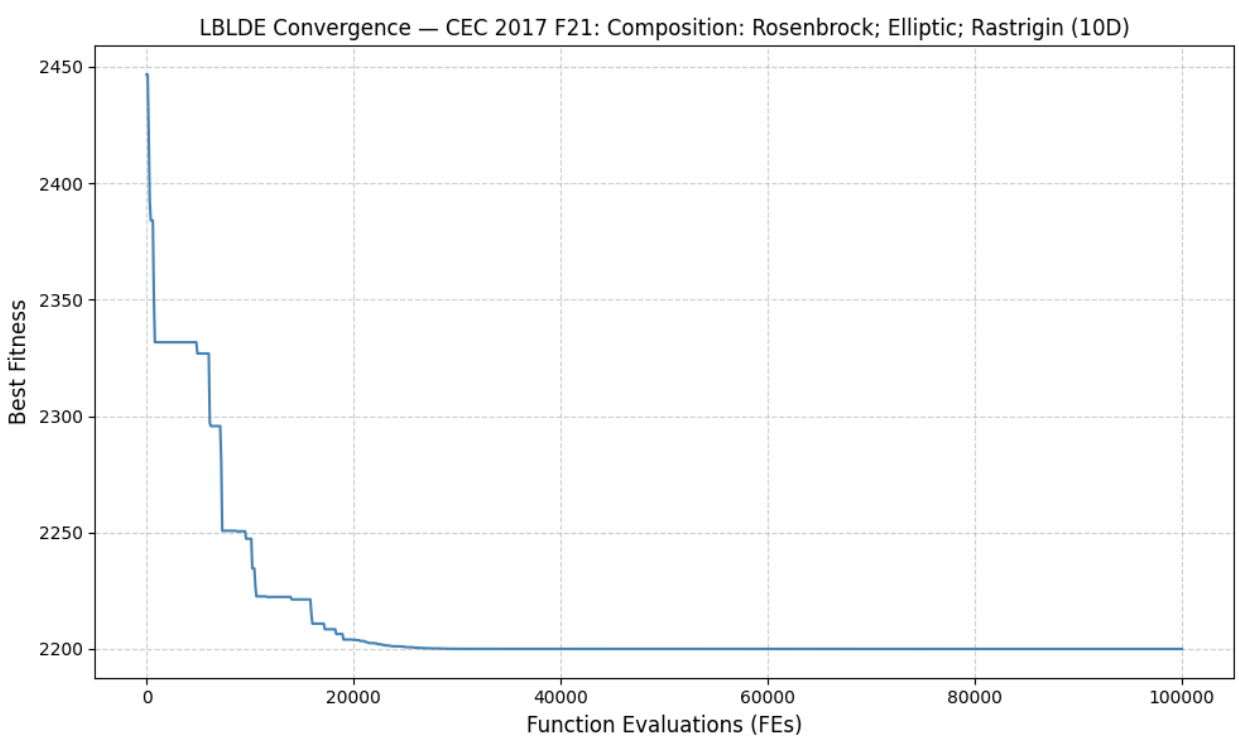
6.4 Composite Functions (F21-F30)

```
# ============================================================
# F21 — Composition Function 1 (Rosenbrock; Elliptic; Rastrigin)
# ============================================================
run_and_plot('F21', make_wrapper(composition.f21), 2100.0,
'Composition: Rosenbrock; Elliptic; Rastrigin')

==========================================================================
==========
LBLDE on CEC 2017 F21 — Single Run
==========================================================================
==========
```

```
Function       : Composition: Rosenbrock; Elliptic; Rastrigin (10D)
Bounds         : [-100, 100]
Global optimum : 2100.0
Generation 0: Best Fitness = 2.446690e+03
Generation 10: Best Fitness = 2.331717e+03, FES = 1100
Generation 20: Best Fitness = 2.331717e+03, FES = 2100
Generation 30: Best Fitness = 2.331717e+03, FES = 3100
Generation 40: Best Fitness = 2.331717e+03, FES = 4100
Generation 50: Best Fitness = 2.326901e+03, FES = 5100
Generation 60: Best Fitness = 2.326873e+03, FES = 6100
Generation 70: Best Fitness = 2.295721e+03, FES = 7100
Generation 80: Best Fitness = 2.250756e+03, FES = 8100
Generation 90: Best Fitness = 2.250545e+03, FES = 9100
Generation 100: Best Fitness = 2.247326e+03, FES = 10100
Generation 110: Best Fitness = 2.222573e+03, FES = 11100
Generation 120: Best Fitness = 2.222336e+03, FES = 12100
Generation 130: Best Fitness = 2.222336e+03, FES = 13100
Generation 140: Best Fitness = 2.221278e+03, FES = 14100
Generation 150: Best Fitness = 2.221278e+03, FES = 15100
Generation 160: Best Fitness = 2.210877e+03, FES = 16100
Generation 170: Best Fitness = 2.210877e+03, FES = 17100
Generation 180: Best Fitness = 2.208467e+03, FES = 18100
Generation 190: Best Fitness = 2.204038e+03, FES = 19100
Generation 200: Best Fitness = 2.203893e+03, FES = 20100
Generation 210: Best Fitness = 2.203430e+03, FES = 21100
Generation 220: Best Fitness = 2.202379e+03, FES = 22100
Generation 230: Best Fitness = 2.201486e+03, FES = 23100
Generation 240: Best Fitness = 2.201145e+03, FES = 24100
Generation 250: Best Fitness = 2.200766e+03, FES = 25100
Generation 260: Best Fitness = 2.200582e+03, FES = 26100
Generation 270: Best Fitness = 2.200362e+03, FES = 27100
Generation 280: Best Fitness = 2.200239e+03, FES = 28100
Generation 290: Best Fitness = 2.200124e+03, FES = 29100
Generation 300: Best Fitness = 2.200049e+03, FES = 30100
Generation 310: Best Fitness = 2.200032e+03, FES = 31100
Generation 320: Best Fitness = 2.200015e+03, FES = 32100
Generation 330: Best Fitness = 2.200005e+03, FES = 33100
Generation 340: Best Fitness = 2.200002e+03, FES = 34100
Generation 350: Best Fitness = 2.200001e+03, FES = 35100
Generation 360: Best Fitness = 2.200000e+03, FES = 36100
Generation 370: Best Fitness = 2.200000e+03, FES = 37100
Generation 380: Best Fitness = 2.200000e+03, FES = 38100
Generation 390: Best Fitness = 2.200000e+03, FES = 39100
Generation 400: Best Fitness = 2.200000e+03, FES = 40100
Generation 410: Best Fitness = 2.200000e+03, FES = 41100
Generation 420: Best Fitness = 2.200000e+03, FES = 42100
Generation 430: Best Fitness = 2.200000e+03, FES = 43100
Generation 440: Best Fitness = 2.200000e+03, FES = 44100
Generation 450: Best Fitness = 2.200000e+03, FES = 45100
```

```
Generation 460: Best Fitness = 2.200000e+03, FES = 46100
Generation 470: Best Fitness = 2.200000e+03, FES = 47100
Generation 480: Best Fitness = 2.200000e+03, FES = 48100
Generation 490: Best Fitness = 2.200000e+03, FES = 49100
Generation 500: Best Fitness = 2.200000e+03, FES = 50100
Generation 510: Best Fitness = 2.200000e+03, FES = 51100
Generation 520: Best Fitness = 2.200000e+03, FES = 52100
Generation 530: Best Fitness = 2.200000e+03, FES = 53100
Generation 540: Best Fitness = 2.200000e+03, FES = 54100
Generation 550: Best Fitness = 2.200000e+03, FES = 55100
Generation 560: Best Fitness = 2.200000e+03, FES = 56100
Generation 570: Best Fitness = 2.200000e+03, FES = 57100
Generation 580: Best Fitness = 2.200000e+03, FES = 58100
Generation 590: Best Fitness = 2.200000e+03, FES = 59100
Generation 600: Best Fitness = 2.200000e+03, FES = 60100
Generation 610: Best Fitness = 2.200000e+03, FES = 61100
Generation 620: Best Fitness = 2.200000e+03, FES = 62100
Generation 630: Best Fitness = 2.200000e+03, FES = 63100
Generation 640: Best Fitness = 2.200000e+03, FES = 64100
Generation 650: Best Fitness = 2.200000e+03, FES = 65100
Generation 660: Best Fitness = 2.200000e+03, FES = 66100
Generation 670: Best Fitness = 2.200000e+03, FES = 67100
Generation 680: Best Fitness = 2.200000e+03, FES = 68100
Generation 690: Best Fitness = 2.200000e+03, FES = 69100
Generation 700: Best Fitness = 2.200000e+03, FES = 70100
Generation 710: Best Fitness = 2.200000e+03, FES = 71100
Generation 720: Best Fitness = 2.200000e+03, FES = 72100
Generation 730: Best Fitness = 2.200000e+03, FES = 73100
Generation 740: Best Fitness = 2.200000e+03, FES = 74100
Generation 750: Best Fitness = 2.200000e+03, FES = 75100
Generation 760: Best Fitness = 2.200000e+03, FES = 76100
Generation 770: Best Fitness = 2.200000e+03, FES = 77100
Generation 780: Best Fitness = 2.200000e+03, FES = 78100
Generation 790: Best Fitness = 2.200000e+03, FES = 79100
Generation 800: Best Fitness = 2.200000e+03, FES = 80100
Generation 810: Best Fitness = 2.200000e+03, FES = 81100
Generation 820: Best Fitness = 2.200000e+03, FES = 82100
Generation 830: Best Fitness = 2.200000e+03, FES = 83100
Generation 840: Best Fitness = 2.200000e+03, FES = 84100
Generation 850: Best Fitness = 2.200000e+03, FES = 85100
Generation 860: Best Fitness = 2.200000e+03, FES = 86100
Generation 870: Best Fitness = 2.200000e+03, FES = 87100
Generation 880: Best Fitness = 2.200000e+03, FES = 88100
Generation 890: Best Fitness = 2.200000e+03, FES = 89100
Generation 900: Best Fitness = 2.200000e+03, FES = 90100
Generation 910: Best Fitness = 2.200000e+03, FES = 91100
Generation 920: Best Fitness = 2.200000e+03, FES = 92100
Generation 930: Best Fitness = 2.200000e+03, FES = 93100
Generation 940: Best Fitness = 2.200000e+03, FES = 94100
```

```
Generation 950: Best Fitness = 2.200000e+03, FES = 95100
Generation 960: Best Fitness = 2.200000e+03, FES = 96100
Generation 970: Best Fitness = 2.200000e+03, FES = 97100
Generation 980: Best Fitness = 2.200000e+03, FES = 98100
Generation 990: Best Fitness = 2.200000e+03, FES = 99100

Optimization completed!
Final Best Fitness = 2.200000e+03
Total Function Evaluations = 100000

================================================================================
Results:
  Best Fitness           : 2.200000e+03
  Global Optimum         : 2.100000e+03
  Error (|best-optimum|) : 1.000000e+02
  Best Solution (5 dim)  : [ 37.15806064 -63.08043142 -40.97860684 -
71.06301346 -23.17762719]
  D^(0) (init. diversity): 0.0000e+00
================================================================================
```



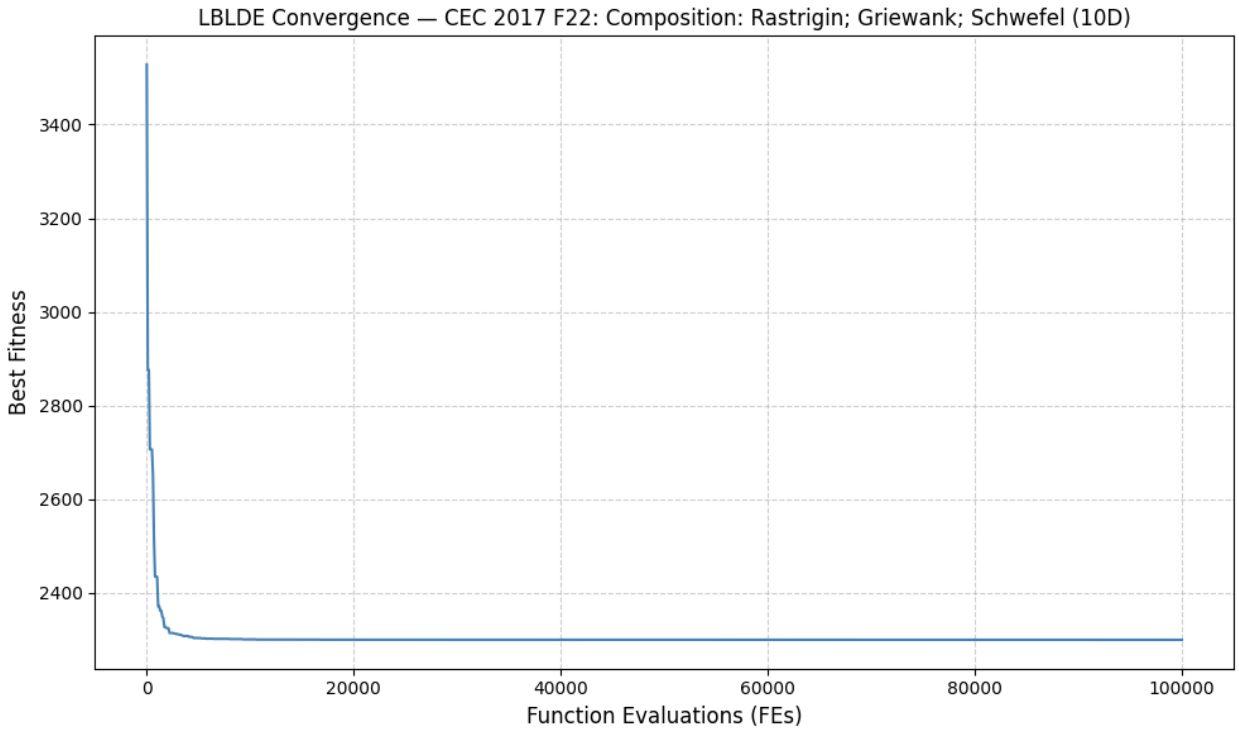

```
Graph saved as convergence_F21.png


# ============================================================
# F22 — Composition Function 2 (Rastrigin; Griewank; Schwefel)
```

```
# ============================================================
run_and_plot('F22', make_wrapper(composition.f22), 2200.0,
'Composition: Rastrigin; Griewank; Schwefel')
```

```
==================================================================
==========
LBLDE on CEC 2017 F22 — Single Run
==================================================================
==========
Function       : Composition: Rastrigin; Griewank; Schwefel (10D)
Bounds         : [-100, 100]
Global optimum : 2200.0
Generation 0: Best Fitness = 3.528003e+03
Generation 10: Best Fitness = 2.434846e+03, FES = 1100
Generation 20: Best Fitness = 2.324872e+03, FES = 2100
Generation 30: Best Fitness = 2.311597e+03, FES = 3100
Generation 40: Best Fitness = 2.308102e+03, FES = 4100
Generation 50: Best Fitness = 2.304007e+03, FES = 5100
Generation 60: Best Fitness = 2.302447e+03, FES = 6100
Generation 70: Best Fitness = 2.302083e+03, FES = 7100
Generation 80: Best Fitness = 2.301663e+03, FES = 8100
Generation 90: Best Fitness = 2.301525e+03, FES = 9100
Generation 100: Best Fitness = 2.300959e+03, FES = 10100
Generation 110: Best Fitness = 2.300524e+03, FES = 11100
Generation 120: Best Fitness = 2.300524e+03, FES = 12100
Generation 130: Best Fitness = 2.300524e+03, FES = 13100
Generation 140: Best Fitness = 2.300524e+03, FES = 14100
Generation 150: Best Fitness = 2.300524e+03, FES = 15100
Generation 160: Best Fitness = 2.300463e+03, FES = 16100
Generation 170: Best Fitness = 2.300212e+03, FES = 17100
Generation 180: Best Fitness = 2.300209e+03, FES = 18100
Generation 190: Best Fitness = 2.300209e+03, FES = 19100
Generation 200: Best Fitness = 2.300174e+03, FES = 20100
Generation 210: Best Fitness = 2.300116e+03, FES = 21100
Generation 220: Best Fitness = 2.300106e+03, FES = 22100
Generation 230: Best Fitness = 2.300053e+03, FES = 23100
Generation 240: Best Fitness = 2.300042e+03, FES = 24100
Generation 250: Best Fitness = 2.300022e+03, FES = 25100
Generation 260: Best Fitness = 2.300011e+03, FES = 26100
Generation 270: Best Fitness = 2.300008e+03, FES = 27100
Generation 280: Best Fitness = 2.300005e+03, FES = 28100
Generation 290: Best Fitness = 2.300002e+03, FES = 29100
Generation 300: Best Fitness = 2.300001e+03, FES = 30100
Generation 310: Best Fitness = 2.300000e+03, FES = 31100
Generation 320: Best Fitness = 2.300000e+03, FES = 32100
Generation 330: Best Fitness = 2.300000e+03, FES = 33100
Generation 340: Best Fitness = 2.300000e+03, FES = 34100
Generation 350: Best Fitness = 2.300000e+03, FES = 35100
Generation 360: Best Fitness = 2.300000e+03, FES = 36100
Generation 370: Best Fitness = 2.300000e+03, FES = 37100
```

```
Generation 380: Best Fitness = 2.300000e+03, FES = 38100
Generation 390: Best Fitness = 2.300000e+03, FES = 39100
Generation 400: Best Fitness = 2.300000e+03, FES = 40100
Generation 410: Best Fitness = 2.300000e+03, FES = 41100
Generation 420: Best Fitness = 2.300000e+03, FES = 42100
Generation 430: Best Fitness = 2.300000e+03, FES = 43100
Generation 440: Best Fitness = 2.300000e+03, FES = 44100
Generation 450: Best Fitness = 2.300000e+03, FES = 45100
Generation 460: Best Fitness = 2.300000e+03, FES = 46100
Generation 470: Best Fitness = 2.300000e+03, FES = 47100
Generation 480: Best Fitness = 2.300000e+03, FES = 48100
Generation 490: Best Fitness = 2.300000e+03, FES = 49100
Generation 500: Best Fitness = 2.300000e+03, FES = 50100
Generation 510: Best Fitness = 2.300000e+03, FES = 51100
Generation 520: Best Fitness = 2.300000e+03, FES = 52100
Generation 530: Best Fitness = 2.300000e+03, FES = 53100
Generation 540: Best Fitness = 2.300000e+03, FES = 54100
Generation 550: Best Fitness = 2.300000e+03, FES = 55100
Generation 560: Best Fitness = 2.300000e+03, FES = 56100
Generation 570: Best Fitness = 2.300000e+03, FES = 57100
Generation 580: Best Fitness = 2.300000e+03, FES = 58100
Generation 590: Best Fitness = 2.300000e+03, FES = 59100
Generation 600: Best Fitness = 2.300000e+03, FES = 60100
Generation 610: Best Fitness = 2.300000e+03, FES = 61100
Generation 620: Best Fitness = 2.300000e+03, FES = 62100
Generation 630: Best Fitness = 2.300000e+03, FES = 63100
Generation 640: Best Fitness = 2.300000e+03, FES = 64100
Generation 650: Best Fitness = 2.300000e+03, FES = 65100
Generation 660: Best Fitness = 2.300000e+03, FES = 66100
Generation 670: Best Fitness = 2.300000e+03, FES = 67100
Generation 680: Best Fitness = 2.300000e+03, FES = 68100
Generation 690: Best Fitness = 2.300000e+03, FES = 69100
Generation 700: Best Fitness = 2.300000e+03, FES = 70100
Generation 710: Best Fitness = 2.300000e+03, FES = 71100
Generation 720: Best Fitness = 2.300000e+03, FES = 72100
Generation 730: Best Fitness = 2.300000e+03, FES = 73100
Generation 740: Best Fitness = 2.300000e+03, FES = 74100
Generation 750: Best Fitness = 2.300000e+03, FES = 75100
Generation 760: Best Fitness = 2.300000e+03, FES = 76100
Generation 770: Best Fitness = 2.300000e+03, FES = 77100
Generation 780: Best Fitness = 2.300000e+03, FES = 78100
Generation 790: Best Fitness = 2.300000e+03, FES = 79100
Generation 800: Best Fitness = 2.300000e+03, FES = 80100
Generation 810: Best Fitness = 2.300000e+03, FES = 81100
Generation 820: Best Fitness = 2.300000e+03, FES = 82100
Generation 830: Best Fitness = 2.300000e+03, FES = 83100
Generation 840: Best Fitness = 2.300000e+03, FES = 84100
Generation 850: Best Fitness = 2.300000e+03, FES = 85100
Generation 860: Best Fitness = 2.300000e+03, FES = 86100
```

```
Generation 870: Best Fitness = 2.300000e+03, FES = 87100
Generation 880: Best Fitness = 2.300000e+03, FES = 88100
Generation 890: Best Fitness = 2.300000e+03, FES = 89100
Generation 900: Best Fitness = 2.300000e+03, FES = 90100
Generation 910: Best Fitness = 2.300000e+03, FES = 91100
Generation 920: Best Fitness = 2.300000e+03, FES = 92100
Generation 930: Best Fitness = 2.300000e+03, FES = 93100
Generation 940: Best Fitness = 2.300000e+03, FES = 94100
Generation 950: Best Fitness = 2.300000e+03, FES = 95100
Generation 960: Best Fitness = 2.300000e+03, FES = 96100
Generation 970: Best Fitness = 2.300000e+03, FES = 97100
Generation 980: Best Fitness = 2.300000e+03, FES = 98100
Generation 990: Best Fitness = 2.300000e+03, FES = 99100

Optimization completed!
Final Best Fitness = 2.300000e+03
Total Function Evaluations = 100000

=====================================================================
==========
Results:
  Best Fitness           : 2.300000e+03
  Global Optimum         : 2.200000e+03
  Error (|best-optimum|) : 1.000000e+02
  Best Solution (5 dim)  : [-52.41266063  11.16831479  23.05456506
35.08347775  30.48935237]
  D^(0) (init. diversity): 0.0000e+00
=====================================================================
==========
```

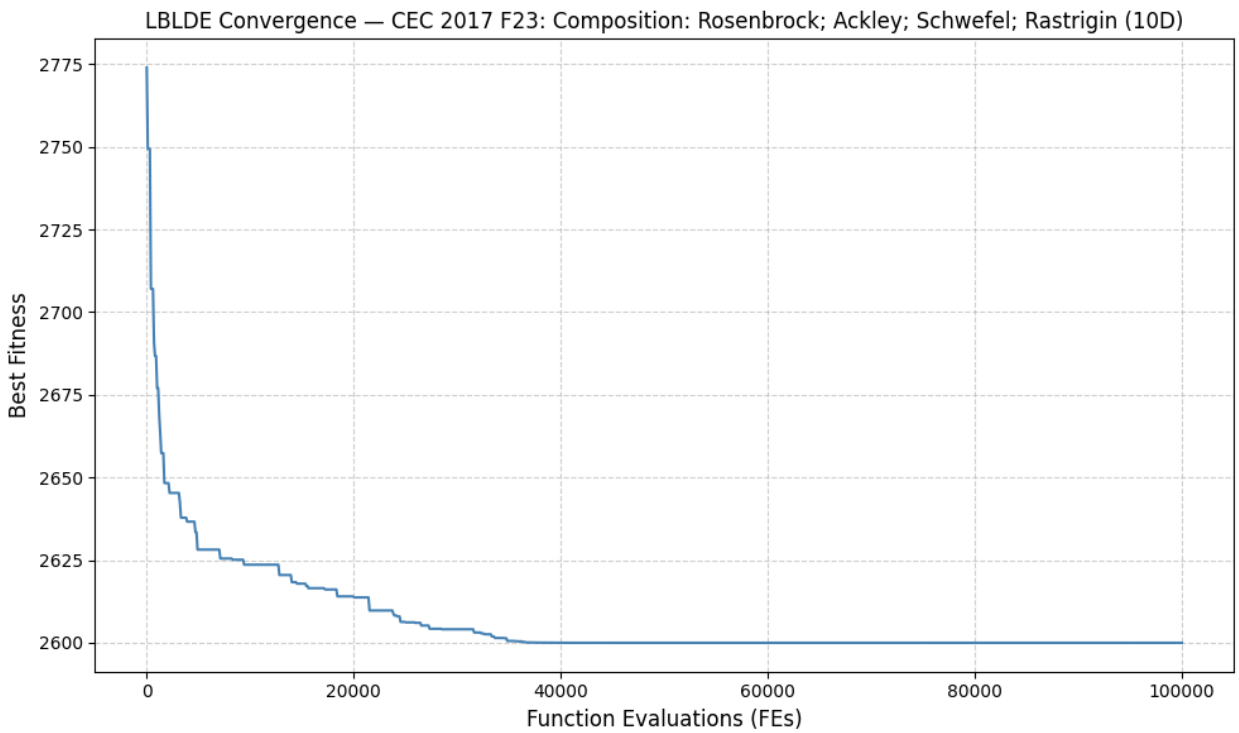



```
Graph saved as convergence_F22.png


# ============================================================
# F23 — Composition Function 3 (Rosenbrock; Ackley; Schwefel;
Rastrigin)
# ============================================================
run_and_plot('F23', make_wrapper(composition.f23), 2300.0,
'Composition: Rosenbrock; Ackley; Schwefel; Rastrigin')

======================================================================
==========
LBLDE on CEC 2017 F23 — Single Run
======================================================================
==========
Function       : Composition: Rosenbrock; Ackley; Schwefel; Rastrigin
(10D)
Bounds         : [-100, 100]
Global optimum : 2300.0
Generation 0: Best Fitness = 2.773921e+03
Generation 10: Best Fitness = 2.677021e+03, FES = 1100
Generation 20: Best Fitness = 2.648303e+03, FES = 2100
Generation 30: Best Fitness = 2.645320e+03, FES = 3100
Generation 40: Best Fitness = 2.636632e+03, FES = 4100
Generation 50: Best Fitness = 2.628176e+03, FES = 5100
Generation 60: Best Fitness = 2.628176e+03, FES = 6100
Generation 70: Best Fitness = 2.628176e+03, FES = 7100
```

```
Generation 80: Best Fitness = 2.625478e+03, FES = 8100
Generation 90: Best Fitness = 2.625126e+03, FES = 9100
Generation 100: Best Fitness = 2.623611e+03, FES = 10100
Generation 110: Best Fitness = 2.623611e+03, FES = 11100
Generation 120: Best Fitness = 2.623611e+03, FES = 12100
Generation 130: Best Fitness = 2.620504e+03, FES = 13100
Generation 140: Best Fitness = 2.618329e+03, FES = 14100
Generation 150: Best Fitness = 2.617892e+03, FES = 15100
Generation 160: Best Fitness = 2.616526e+03, FES = 16100
Generation 170: Best Fitness = 2.616526e+03, FES = 17100
Generation 180: Best Fitness = 2.616132e+03, FES = 18100
Generation 190: Best Fitness = 2.614041e+03, FES = 19100
Generation 200: Best Fitness = 2.613725e+03, FES = 20100
Generation 210: Best Fitness = 2.613725e+03, FES = 21100
Generation 220: Best Fitness = 2.609773e+03, FES = 22100
Generation 230: Best Fitness = 2.609773e+03, FES = 23100
Generation 240: Best Fitness = 2.608251e+03, FES = 24100
Generation 250: Best Fitness = 2.606187e+03, FES = 25100
Generation 260: Best Fitness = 2.606045e+03, FES = 26100
Generation 270: Best Fitness = 2.605216e+03, FES = 27100
Generation 280: Best Fitness = 2.604242e+03, FES = 28100
Generation 290: Best Fitness = 2.604096e+03, FES = 29100
Generation 300: Best Fitness = 2.604096e+03, FES = 30100
Generation 310: Best Fitness = 2.604096e+03, FES = 31100
Generation 320: Best Fitness = 2.603117e+03, FES = 32100
Generation 330: Best Fitness = 2.602585e+03, FES = 33100
Generation 340: Best Fitness = 2.601457e+03, FES = 34100
Generation 350: Best Fitness = 2.600558e+03, FES = 35100
Generation 360: Best Fitness = 2.600437e+03, FES = 36100
Generation 370: Best Fitness = 2.600122e+03, FES = 37100
Generation 380: Best Fitness = 2.600063e+03, FES = 38100
Generation 390: Best Fitness = 2.600035e+03, FES = 39100
Generation 400: Best Fitness = 2.600010e+03, FES = 40100
Generation 410: Best Fitness = 2.600004e+03, FES = 41100
Generation 420: Best Fitness = 2.600002e+03, FES = 42100
Generation 430: Best Fitness = 2.600001e+03, FES = 43100
Generation 440: Best Fitness = 2.600000e+03, FES = 44100
Generation 450: Best Fitness = 2.600000e+03, FES = 45100
Generation 460: Best Fitness = 2.600000e+03, FES = 46100
Generation 470: Best Fitness = 2.600000e+03, FES = 47100
Generation 480: Best Fitness = 2.600000e+03, FES = 48100
Generation 490: Best Fitness = 2.600000e+03, FES = 49100
Generation 500: Best Fitness = 2.600000e+03, FES = 50100
Generation 510: Best Fitness = 2.600000e+03, FES = 51100
Generation 520: Best Fitness = 2.600000e+03, FES = 52100
Generation 530: Best Fitness = 2.600000e+03, FES = 53100
Generation 540: Best Fitness = 2.600000e+03, FES = 54100
Generation 550: Best Fitness = 2.600000e+03, FES = 55100
Generation 560: Best Fitness = 2.600000e+03, FES = 56100
```

```
Generation 570: Best Fitness = 2.600000e+03, FES = 57100
Generation 580: Best Fitness = 2.600000e+03, FES = 58100
Generation 590: Best Fitness = 2.600000e+03, FES = 59100
Generation 600: Best Fitness = 2.600000e+03, FES = 60100
Generation 610: Best Fitness = 2.600000e+03, FES = 61100
Generation 620: Best Fitness = 2.600000e+03, FES = 62100
Generation 630: Best Fitness = 2.600000e+03, FES = 63100
Generation 640: Best Fitness = 2.600000e+03, FES = 64100
Generation 650: Best Fitness = 2.600000e+03, FES = 65100
Generation 660: Best Fitness = 2.600000e+03, FES = 66100
Generation 670: Best Fitness = 2.600000e+03, FES = 67100
Generation 680: Best Fitness = 2.600000e+03, FES = 68100
Generation 690: Best Fitness = 2.600000e+03, FES = 69100
Generation 700: Best Fitness = 2.600000e+03, FES = 70100
Generation 710: Best Fitness = 2.600000e+03, FES = 71100
Generation 720: Best Fitness = 2.600000e+03, FES = 72100
Generation 730: Best Fitness = 2.600000e+03, FES = 73100
Generation 740: Best Fitness = 2.600000e+03, FES = 74100
Generation 750: Best Fitness = 2.600000e+03, FES = 75100
Generation 760: Best Fitness = 2.600000e+03, FES = 76100
Generation 770: Best Fitness = 2.600000e+03, FES = 77100
Generation 780: Best Fitness = 2.600000e+03, FES = 78100
Generation 790: Best Fitness = 2.600000e+03, FES = 79100
Generation 800: Best Fitness = 2.600000e+03, FES = 80100
Generation 810: Best Fitness = 2.600000e+03, FES = 81100
Generation 820: Best Fitness = 2.600000e+03, FES = 82100
Generation 830: Best Fitness = 2.600000e+03, FES = 83100
Generation 840: Best Fitness = 2.600000e+03, FES = 84100
Generation 850: Best Fitness = 2.600000e+03, FES = 85100
Generation 860: Best Fitness = 2.600000e+03, FES = 86100
Generation 870: Best Fitness = 2.600000e+03, FES = 87100
Generation 880: Best Fitness = 2.600000e+03, FES = 88100
Generation 890: Best Fitness = 2.600000e+03, FES = 89100
Generation 900: Best Fitness = 2.600000e+03, FES = 90100
Generation 910: Best Fitness = 2.600000e+03, FES = 91100
Generation 920: Best Fitness = 2.600000e+03, FES = 92100
Generation 930: Best Fitness = 2.600000e+03, FES = 93100
Generation 940: Best Fitness = 2.600000e+03, FES = 94100
Generation 950: Best Fitness = 2.600000e+03, FES = 95100
Generation 960: Best Fitness = 2.600000e+03, FES = 96100
Generation 970: Best Fitness = 2.600000e+03, FES = 97100
Generation 980: Best Fitness = 2.600000e+03, FES = 98100
Generation 990: Best Fitness = 2.600000e+03, FES = 99100

Optimization completed!
Final Best Fitness = 2.600000e+03
Total Function Evaluations = 100000

======================================================================
```

```
==========
Results:
  Best Fitness           : 2.600000e+03
  Global Optimum         : 2.300000e+03
  Error (|best-optimum|) : 3.000000e+02
  Best Solution (5 dim)  : [-77.31782769 -77.20035565 -53.51102581
69.61738777 -57.02810436]
  D^(0) (init. diversity): 0.0000e+00
==========================================================================
==========
```

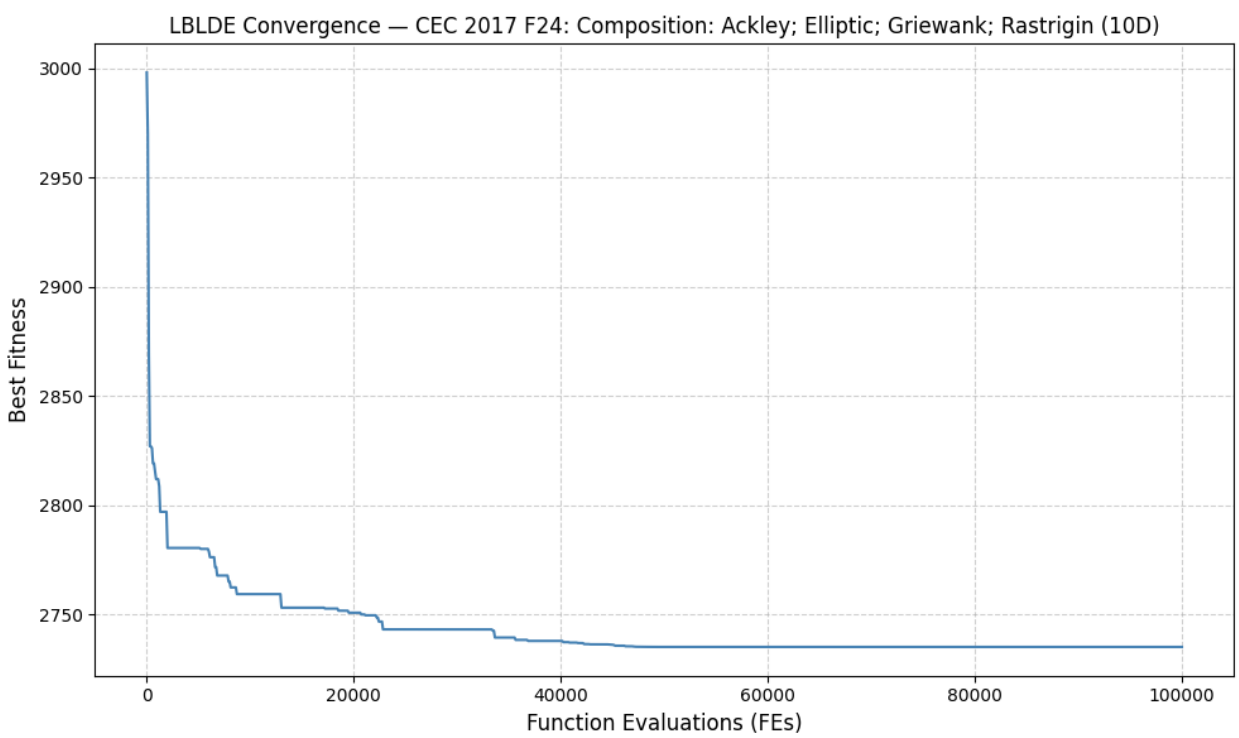



```
Graph saved as convergence_F23.png


# ============================================================
# F24 — Composition Function 4 (Ackley; Elliptic; Griewank; Rastrigin)
# ============================================================
run_and_plot('F24', make_wrapper(composition.f24), 2400.0,
'Composition: Ackley; Elliptic; Griewank; Rastrigin')

==========================================================================
==========
LBLDE on CEC 2017 F24 — Single Run
==========================================================================
==========
Function       : Composition: Ackley; Elliptic; Griewank; Rastrigin
(10D)
```

```
Bounds         : [-100, 100]
Global optimum : 2400.0
Generation 0: Best Fitness = 2.997998e+03
Generation 10: Best Fitness = 2.812031e+03, FES = 1100
Generation 20: Best Fitness = 2.780551e+03, FES = 2100
Generation 30: Best Fitness = 2.780551e+03, FES = 3100
Generation 40: Best Fitness = 2.780551e+03, FES = 4100
Generation 50: Best Fitness = 2.780551e+03, FES = 5100
Generation 60: Best Fitness = 2.778877e+03, FES = 6100
Generation 70: Best Fitness = 2.767934e+03, FES = 7100
Generation 80: Best Fitness = 2.765154e+03, FES = 8100
Generation 90: Best Fitness = 2.759445e+03, FES = 9100
Generation 100: Best Fitness = 2.759445e+03, FES = 10100
Generation 110: Best Fitness = 2.759445e+03, FES = 11100
Generation 120: Best Fitness = 2.759445e+03, FES = 12100
Generation 130: Best Fitness = 2.753237e+03, FES = 13100
Generation 140: Best Fitness = 2.753237e+03, FES = 14100
Generation 150: Best Fitness = 2.753237e+03, FES = 15100
Generation 160: Best Fitness = 2.753237e+03, FES = 16100
Generation 170: Best Fitness = 2.753237e+03, FES = 17100
Generation 180: Best Fitness = 2.752836e+03, FES = 18100
Generation 190: Best Fitness = 2.751866e+03, FES = 19100
Generation 200: Best Fitness = 2.750937e+03, FES = 20100
Generation 210: Best Fitness = 2.750136e+03, FES = 21100
Generation 220: Best Fitness = 2.749755e+03, FES = 22100
Generation 230: Best Fitness = 2.743276e+03, FES = 23100
Generation 240: Best Fitness = 2.743276e+03, FES = 24100
Generation 250: Best Fitness = 2.743276e+03, FES = 25100
Generation 260: Best Fitness = 2.743276e+03, FES = 26100
Generation 270: Best Fitness = 2.743276e+03, FES = 27100
Generation 280: Best Fitness = 2.743276e+03, FES = 28100
Generation 290: Best Fitness = 2.743276e+03, FES = 29100
Generation 300: Best Fitness = 2.743276e+03, FES = 30100
Generation 310: Best Fitness = 2.743276e+03, FES = 31100
Generation 320: Best Fitness = 2.743276e+03, FES = 32100
Generation 330: Best Fitness = 2.743276e+03, FES = 33100
Generation 340: Best Fitness = 2.739577e+03, FES = 34100
Generation 350: Best Fitness = 2.739577e+03, FES = 35100
Generation 360: Best Fitness = 2.738499e+03, FES = 36100
Generation 370: Best Fitness = 2.738026e+03, FES = 37100
Generation 380: Best Fitness = 2.738026e+03, FES = 38100
Generation 390: Best Fitness = 2.738026e+03, FES = 39100
Generation 400: Best Fitness = 2.738026e+03, FES = 40100
Generation 410: Best Fitness = 2.737265e+03, FES = 41100
Generation 420: Best Fitness = 2.737026e+03, FES = 42100
Generation 430: Best Fitness = 2.736473e+03, FES = 43100
Generation 440: Best Fitness = 2.736442e+03, FES = 44100
Generation 450: Best Fitness = 2.736249e+03, FES = 45100
Generation 460: Best Fitness = 2.735825e+03, FES = 46100
```

```
Generation 470: Best Fitness = 2.735435e+03, FES = 47100
Generation 480: Best Fitness = 2.735306e+03, FES = 48100
Generation 490: Best Fitness = 2.735281e+03, FES = 49100
Generation 500: Best Fitness = 2.735274e+03, FES = 50100
Generation 510: Best Fitness = 2.735273e+03, FES = 51100
Generation 520: Best Fitness = 2.735272e+03, FES = 52100
Generation 530: Best Fitness = 2.735272e+03, FES = 53100
Generation 540: Best Fitness = 2.735272e+03, FES = 54100
Generation 550: Best Fitness = 2.735272e+03, FES = 55100
Generation 560: Best Fitness = 2.735272e+03, FES = 56100
Generation 570: Best Fitness = 2.735272e+03, FES = 57100
Generation 580: Best Fitness = 2.735272e+03, FES = 58100
Generation 590: Best Fitness = 2.735272e+03, FES = 59100
Generation 600: Best Fitness = 2.735272e+03, FES = 60100
Generation 610: Best Fitness = 2.735272e+03, FES = 61100
Generation 620: Best Fitness = 2.735272e+03, FES = 62100
Generation 630: Best Fitness = 2.735272e+03, FES = 63100
Generation 640: Best Fitness = 2.735272e+03, FES = 64100
Generation 650: Best Fitness = 2.735272e+03, FES = 65100
Generation 660: Best Fitness = 2.735272e+03, FES = 66100
Generation 670: Best Fitness = 2.735272e+03, FES = 67100
Generation 680: Best Fitness = 2.735272e+03, FES = 68100
Generation 690: Best Fitness = 2.735272e+03, FES = 69100
Generation 700: Best Fitness = 2.735272e+03, FES = 70100
Generation 710: Best Fitness = 2.735272e+03, FES = 71100
Generation 720: Best Fitness = 2.735272e+03, FES = 72100
Generation 730: Best Fitness = 2.735272e+03, FES = 73100
Generation 740: Best Fitness = 2.735272e+03, FES = 74100
Generation 750: Best Fitness = 2.735272e+03, FES = 75100
Generation 760: Best Fitness = 2.735272e+03, FES = 76100
Generation 770: Best Fitness = 2.735272e+03, FES = 77100
Generation 780: Best Fitness = 2.735272e+03, FES = 78100
Generation 790: Best Fitness = 2.735272e+03, FES = 79100
Generation 800: Best Fitness = 2.735272e+03, FES = 80100
Generation 810: Best Fitness = 2.735272e+03, FES = 81100
Generation 820: Best Fitness = 2.735272e+03, FES = 82100
Generation 830: Best Fitness = 2.735272e+03, FES = 83100
Generation 840: Best Fitness = 2.735272e+03, FES = 84100
Generation 850: Best Fitness = 2.735272e+03, FES = 85100
Generation 860: Best Fitness = 2.735272e+03, FES = 86100
Generation 870: Best Fitness = 2.735272e+03, FES = 87100
Generation 880: Best Fitness = 2.735272e+03, FES = 88100
Generation 890: Best Fitness = 2.735272e+03, FES = 89100
Generation 900: Best Fitness = 2.735272e+03, FES = 90100
Generation 910: Best Fitness = 2.735272e+03, FES = 91100
Generation 920: Best Fitness = 2.735272e+03, FES = 92100
Generation 930: Best Fitness = 2.735272e+03, FES = 93100
Generation 940: Best Fitness = 2.735272e+03, FES = 94100
Generation 950: Best Fitness = 2.735272e+03, FES = 95100
```

```
Generation 960: Best Fitness = 2.735272e+03, FES = 96100
Generation 970: Best Fitness = 2.735272e+03, FES = 97100
Generation 980: Best Fitness = 2.735272e+03, FES = 98100
Generation 990: Best Fitness = 2.735272e+03, FES = 99100

Optimization completed!
Final Best Fitness = 2.735272e+03
Total Function Evaluations = 100000

======================================================================
==========
Results:
  Best Fitness           : 2.735272e+03
  Global Optimum         : 2.400000e+03
  Error (|best-optimum|) : 3.352723e+02
  Best Solution (5 dim)  : [ 46.5857532   77.80113249 -64.57208147 -
84.47125237  83.12726671]
  D^(0) (init. diversity): 0.0000e+00
======================================================================
==========
```



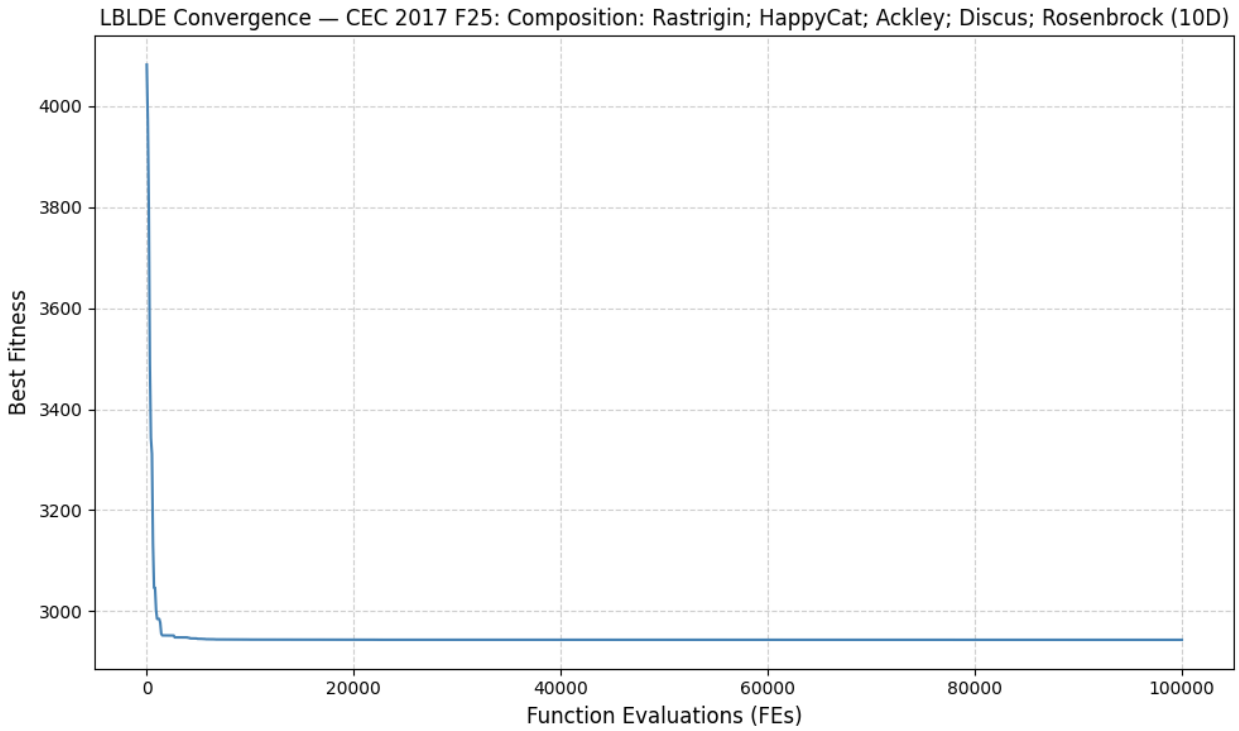

```
Graph saved as convergence_F24.png


# ============================================================
# F25 — Composition Function 5 (Rastrigin; HappyCat; Ackley; Discus;
Rosenbrock)
```

```
# ============================================================
run_and_plot('F25', make_wrapper(composition.f25), 2500.0,
'Composition: Rastrigin; HappyCat; Ackley; Discus; Rosenbrock')
```

```
======================================================================
==========
LBLDE on CEC 2017 F25 — Single Run
======================================================================
==========
Function       : Composition: Rastrigin; HappyCat; Ackley; Discus;
Rosenbrock (10D)
Bounds         : [-100, 100]
Global optimum : 2500.0
Generation 0: Best Fitness = 4.081798e+03
Generation 10: Best Fitness = 2.984791e+03, FES = 1100
Generation 20: Best Fitness = 2.952102e+03, FES = 2100
Generation 30: Best Fitness = 2.948213e+03, FES = 3100
Generation 40: Best Fitness = 2.947295e+03, FES = 4100
Generation 50: Best Fitness = 2.945402e+03, FES = 5100
Generation 60: Best Fitness = 2.944558e+03, FES = 6100
Generation 70: Best Fitness = 2.944150e+03, FES = 7100
Generation 80: Best Fitness = 2.943945e+03, FES = 8100
Generation 90: Best Fitness = 2.943945e+03, FES = 9100
Generation 100: Best Fitness = 2.943825e+03, FES = 10100
Generation 110: Best Fitness = 2.943825e+03, FES = 11100
Generation 120: Best Fitness = 2.943805e+03, FES = 12100
Generation 130: Best Fitness = 2.943707e+03, FES = 13100
Generation 140: Best Fitness = 2.943653e+03, FES = 14100
Generation 150: Best Fitness = 2.943642e+03, FES = 15100
Generation 160: Best Fitness = 2.943642e+03, FES = 16100
Generation 170: Best Fitness = 2.943637e+03, FES = 17100
Generation 180: Best Fitness = 2.943623e+03, FES = 18100
Generation 190: Best Fitness = 2.943586e+03, FES = 19100
Generation 200: Best Fitness = 2.943577e+03, FES = 20100
Generation 210: Best Fitness = 2.943558e+03, FES = 21100
Generation 220: Best Fitness = 2.943517e+03, FES = 22100
Generation 230: Best Fitness = 2.943465e+03, FES = 23100
Generation 240: Best Fitness = 2.943463e+03, FES = 24100
Generation 250: Best Fitness = 2.943463e+03, FES = 25100
Generation 260: Best Fitness = 2.943413e+03, FES = 26100
Generation 270: Best Fitness = 2.943413e+03, FES = 27100
Generation 280: Best Fitness = 2.943413e+03, FES = 28100
Generation 290: Best Fitness = 2.943392e+03, FES = 29100
Generation 300: Best Fitness = 2.943392e+03, FES = 30100
Generation 310: Best Fitness = 2.943379e+03, FES = 31100
Generation 320: Best Fitness = 2.943377e+03, FES = 32100
Generation 330: Best Fitness = 2.943377e+03, FES = 33100
Generation 340: Best Fitness = 2.943377e+03, FES = 34100
Generation 350: Best Fitness = 2.943377e+03, FES = 35100
Generation 360: Best Fitness = 2.943376e+03, FES = 36100
```

```
Generation 370: Best Fitness = 2.943368e+03, FES = 37100
Generation 380: Best Fitness = 2.943365e+03, FES = 38100
Generation 390: Best Fitness = 2.943359e+03, FES = 39100
Generation 400: Best Fitness = 2.943357e+03, FES = 40100
Generation 410: Best Fitness = 2.943357e+03, FES = 41100
Generation 420: Best Fitness = 2.943357e+03, FES = 42100
Generation 430: Best Fitness = 2.943357e+03, FES = 43100
Generation 440: Best Fitness = 2.943357e+03, FES = 44100
Generation 450: Best Fitness = 2.943357e+03, FES = 45100
Generation 460: Best Fitness = 2.943357e+03, FES = 46100
Generation 470: Best Fitness = 2.943357e+03, FES = 47100
Generation 480: Best Fitness = 2.943357e+03, FES = 48100
Generation 490: Best Fitness = 2.943357e+03, FES = 49100
Generation 500: Best Fitness = 2.943357e+03, FES = 50100
Generation 510: Best Fitness = 2.943357e+03, FES = 51100
Generation 520: Best Fitness = 2.943357e+03, FES = 52100
Generation 530: Best Fitness = 2.943357e+03, FES = 53100
Generation 540: Best Fitness = 2.943357e+03, FES = 54100
Generation 550: Best Fitness = 2.943357e+03, FES = 55100
Generation 560: Best Fitness = 2.943357e+03, FES = 56100
Generation 570: Best Fitness = 2.943357e+03, FES = 57100
Generation 580: Best Fitness = 2.943357e+03, FES = 58100
Generation 590: Best Fitness = 2.943357e+03, FES = 59100
Generation 600: Best Fitness = 2.943357e+03, FES = 60100
Generation 610: Best Fitness = 2.943357e+03, FES = 61100
Generation 620: Best Fitness = 2.943357e+03, FES = 62100
Generation 630: Best Fitness = 2.943357e+03, FES = 63100
Generation 640: Best Fitness = 2.943357e+03, FES = 64100
Generation 650: Best Fitness = 2.943357e+03, FES = 65100
Generation 660: Best Fitness = 2.943357e+03, FES = 66100
Generation 670: Best Fitness = 2.943357e+03, FES = 67100
Generation 680: Best Fitness = 2.943357e+03, FES = 68100
Generation 690: Best Fitness = 2.943357e+03, FES = 69100
Generation 700: Best Fitness = 2.943357e+03, FES = 70100
Generation 710: Best Fitness = 2.943357e+03, FES = 71100
Generation 720: Best Fitness = 2.943357e+03, FES = 72100
Generation 730: Best Fitness = 2.943357e+03, FES = 73100
Generation 740: Best Fitness = 2.943357e+03, FES = 74100
Generation 750: Best Fitness = 2.943357e+03, FES = 75100
Generation 760: Best Fitness = 2.943357e+03, FES = 76100
Generation 770: Best Fitness = 2.943357e+03, FES = 77100
Generation 780: Best Fitness = 2.943357e+03, FES = 78100
Generation 790: Best Fitness = 2.943357e+03, FES = 79100
Generation 800: Best Fitness = 2.943357e+03, FES = 80100
Generation 810: Best Fitness = 2.943357e+03, FES = 81100
Generation 820: Best Fitness = 2.943357e+03, FES = 82100
Generation 830: Best Fitness = 2.943357e+03, FES = 83100
Generation 840: Best Fitness = 2.943357e+03, FES = 84100
Generation 850: Best Fitness = 2.943357e+03, FES = 85100
```

```
Generation 860: Best Fitness = 2.943357e+03, FES = 86100
Generation 870: Best Fitness = 2.943357e+03, FES = 87100
Generation 880: Best Fitness = 2.943357e+03, FES = 88100
Generation 890: Best Fitness = 2.943357e+03, FES = 89100
Generation 900: Best Fitness = 2.943357e+03, FES = 90100
Generation 910: Best Fitness = 2.943357e+03, FES = 91100
Generation 920: Best Fitness = 2.943357e+03, FES = 92100
Generation 930: Best Fitness = 2.943357e+03, FES = 93100
Generation 940: Best Fitness = 2.943357e+03, FES = 94100
Generation 950: Best Fitness = 2.943357e+03, FES = 95100
Generation 960: Best Fitness = 2.943357e+03, FES = 96100
Generation 970: Best Fitness = 2.943357e+03, FES = 97100
Generation 980: Best Fitness = 2.943357e+03, FES = 98100
Generation 990: Best Fitness = 2.943357e+03, FES = 99100

Optimization completed!
Final Best Fitness = 2.943357e+03
Total Function Evaluations = 100000

==========================================================================
==========
Results:
  Best Fitness           : 2.943357e+03
  Global Optimum         : 2.500000e+03
  Error (|best-optimum|) : 4.433567e+02
  Best Solution (5 dim)  : [ 14.78992897  70.76517876 -47.28636762 -
17.07127599  44.95935138]
  D^(0) (init. diversity): 0.0000e+00
==========================================================================
==========
```

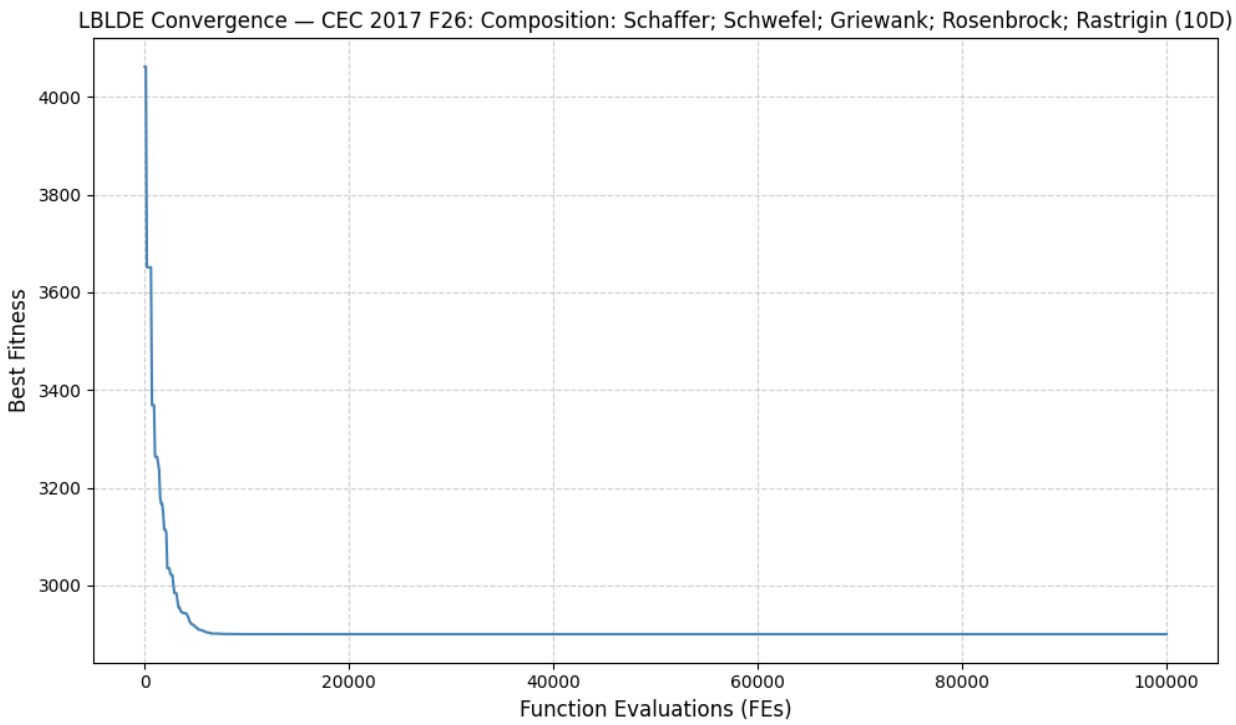



```
Graph saved as convergence_F25.png


# ============================================================
# F26 — Composition Function 6 (Schaffer; Schwefel; Griewank;
Rosenbrock; Rastrigin)
# ============================================================
run_and_plot('F26', make_wrapper(composition.f26), 2600.0,
'Composition: Schaffer; Schwefel; Griewank; Rosenbrock; Rastrigin')

======================================================================
==========
LBLDE on CEC 2017 F26 — Single Run
======================================================================
==========
Function       : Composition: Schaffer; Schwefel; Griewank;
Rosenbrock; Rastrigin (10D)
Bounds         : [-100, 100]
Global optimum : 2600.0
Generation 0: Best Fitness = 4.061684e+03
Generation 10: Best Fitness = 3.263167e+03, FES = 1100
Generation 20: Best Fitness = 3.114227e+03, FES = 2100
Generation 30: Best Fitness = 2.983701e+03, FES = 3100
Generation 40: Best Fitness = 2.942940e+03, FES = 4100
Generation 50: Best Fitness = 2.914062e+03, FES = 5100
Generation 60: Best Fitness = 2.904567e+03, FES = 6100
Generation 70: Best Fitness = 2.901151e+03, FES = 7100
```

```
Generation 80: Best Fitness = 2.900334e+03, FES = 8100
Generation 90: Best Fitness = 2.900128e+03, FES = 9100
Generation 100: Best Fitness = 2.900048e+03, FES = 10100
Generation 110: Best Fitness = 2.900016e+03, FES = 11100
Generation 120: Best Fitness = 2.900006e+03, FES = 12100
Generation 130: Best Fitness = 2.900002e+03, FES = 13100
Generation 140: Best Fitness = 2.900001e+03, FES = 14100
Generation 150: Best Fitness = 2.900000e+03, FES = 15100
Generation 160: Best Fitness = 2.900000e+03, FES = 16100
Generation 170: Best Fitness = 2.900000e+03, FES = 17100
Generation 180: Best Fitness = 2.900000e+03, FES = 18100
Generation 190: Best Fitness = 2.900000e+03, FES = 19100
Generation 200: Best Fitness = 2.900000e+03, FES = 20100
Generation 210: Best Fitness = 2.900000e+03, FES = 21100
Generation 220: Best Fitness = 2.900000e+03, FES = 22100
Generation 230: Best Fitness = 2.900000e+03, FES = 23100
Generation 240: Best Fitness = 2.900000e+03, FES = 24100
Generation 250: Best Fitness = 2.900000e+03, FES = 25100
Generation 260: Best Fitness = 2.900000e+03, FES = 26100
Generation 270: Best Fitness = 2.900000e+03, FES = 27100
Generation 280: Best Fitness = 2.900000e+03, FES = 28100
Generation 290: Best Fitness = 2.900000e+03, FES = 29100
Generation 300: Best Fitness = 2.900000e+03, FES = 30100
Generation 310: Best Fitness = 2.900000e+03, FES = 31100
Generation 320: Best Fitness = 2.900000e+03, FES = 32100
Generation 330: Best Fitness = 2.900000e+03, FES = 33100
Generation 340: Best Fitness = 2.900000e+03, FES = 34100
Generation 350: Best Fitness = 2.900000e+03, FES = 35100
Generation 360: Best Fitness = 2.900000e+03, FES = 36100
Generation 370: Best Fitness = 2.900000e+03, FES = 37100
Generation 380: Best Fitness = 2.900000e+03, FES = 38100
Generation 390: Best Fitness = 2.900000e+03, FES = 39100
Generation 400: Best Fitness = 2.900000e+03, FES = 40100
Generation 410: Best Fitness = 2.900000e+03, FES = 41100
Generation 420: Best Fitness = 2.900000e+03, FES = 42100
Generation 430: Best Fitness = 2.900000e+03, FES = 43100
Generation 440: Best Fitness = 2.900000e+03, FES = 44100
Generation 450: Best Fitness = 2.900000e+03, FES = 45100
Generation 460: Best Fitness = 2.900000e+03, FES = 46100
Generation 470: Best Fitness = 2.900000e+03, FES = 47100
Generation 480: Best Fitness = 2.900000e+03, FES = 48100
Generation 490: Best Fitness = 2.900000e+03, FES = 49100
Generation 500: Best Fitness = 2.900000e+03, FES = 50100
Generation 510: Best Fitness = 2.900000e+03, FES = 51100
Generation 520: Best Fitness = 2.900000e+03, FES = 52100
Generation 530: Best Fitness = 2.900000e+03, FES = 53100
Generation 540: Best Fitness = 2.900000e+03, FES = 54100
Generation 550: Best Fitness = 2.900000e+03, FES = 55100
Generation 560: Best Fitness = 2.900000e+03, FES = 56100
```

```
Generation 570: Best Fitness = 2.900000e+03, FES = 57100
Generation 580: Best Fitness = 2.900000e+03, FES = 58100
Generation 590: Best Fitness = 2.900000e+03, FES = 59100
Generation 600: Best Fitness = 2.900000e+03, FES = 60100
Generation 610: Best Fitness = 2.900000e+03, FES = 61100
Generation 620: Best Fitness = 2.900000e+03, FES = 62100
Generation 630: Best Fitness = 2.900000e+03, FES = 63100
Generation 640: Best Fitness = 2.900000e+03, FES = 64100
Generation 650: Best Fitness = 2.900000e+03, FES = 65100
Generation 660: Best Fitness = 2.900000e+03, FES = 66100
Generation 670: Best Fitness = 2.900000e+03, FES = 67100
Generation 680: Best Fitness = 2.900000e+03, FES = 68100
Generation 690: Best Fitness = 2.900000e+03, FES = 69100
Generation 700: Best Fitness = 2.900000e+03, FES = 70100
Generation 710: Best Fitness = 2.900000e+03, FES = 71100
Generation 720: Best Fitness = 2.900000e+03, FES = 72100
Generation 730: Best Fitness = 2.900000e+03, FES = 73100
Generation 740: Best Fitness = 2.900000e+03, FES = 74100
Generation 750: Best Fitness = 2.900000e+03, FES = 75100
Generation 760: Best Fitness = 2.900000e+03, FES = 76100
Generation 770: Best Fitness = 2.900000e+03, FES = 77100
Generation 780: Best Fitness = 2.900000e+03, FES = 78100
Generation 790: Best Fitness = 2.900000e+03, FES = 79100
Generation 800: Best Fitness = 2.900000e+03, FES = 80100
Generation 810: Best Fitness = 2.900000e+03, FES = 81100
Generation 820: Best Fitness = 2.900000e+03, FES = 82100
Generation 830: Best Fitness = 2.900000e+03, FES = 83100
Generation 840: Best Fitness = 2.900000e+03, FES = 84100
Generation 850: Best Fitness = 2.900000e+03, FES = 85100
Generation 860: Best Fitness = 2.900000e+03, FES = 86100
Generation 870: Best Fitness = 2.900000e+03, FES = 87100
Generation 880: Best Fitness = 2.900000e+03, FES = 88100
Generation 890: Best Fitness = 2.900000e+03, FES = 89100
Generation 900: Best Fitness = 2.900000e+03, FES = 90100
Generation 910: Best Fitness = 2.900000e+03, FES = 91100
Generation 920: Best Fitness = 2.900000e+03, FES = 92100
Generation 930: Best Fitness = 2.900000e+03, FES = 93100
Generation 940: Best Fitness = 2.900000e+03, FES = 94100
Generation 950: Best Fitness = 2.900000e+03, FES = 95100
Generation 960: Best Fitness = 2.900000e+03, FES = 96100
Generation 970: Best Fitness = 2.900000e+03, FES = 97100
Generation 980: Best Fitness = 2.900000e+03, FES = 98100
Generation 990: Best Fitness = 2.900000e+03, FES = 99100

Optimization completed!
Final Best Fitness = 2.900000e+03
Total Function Evaluations = 100000

======================================================================
```

```
==========
Results:
  Best Fitness           : 2.900000e+03
  Global Optimum         : 2.600000e+03
  Error (|best-optimum|) : 3.000000e+02
  Best Solution (5 dim)  : [ 24.746414   -25.74629564  74.04615394 -
62.35027147  35.80221832]
  D^(0) (init. diversity): 0.0000e+00
==========================================================================
==========
```

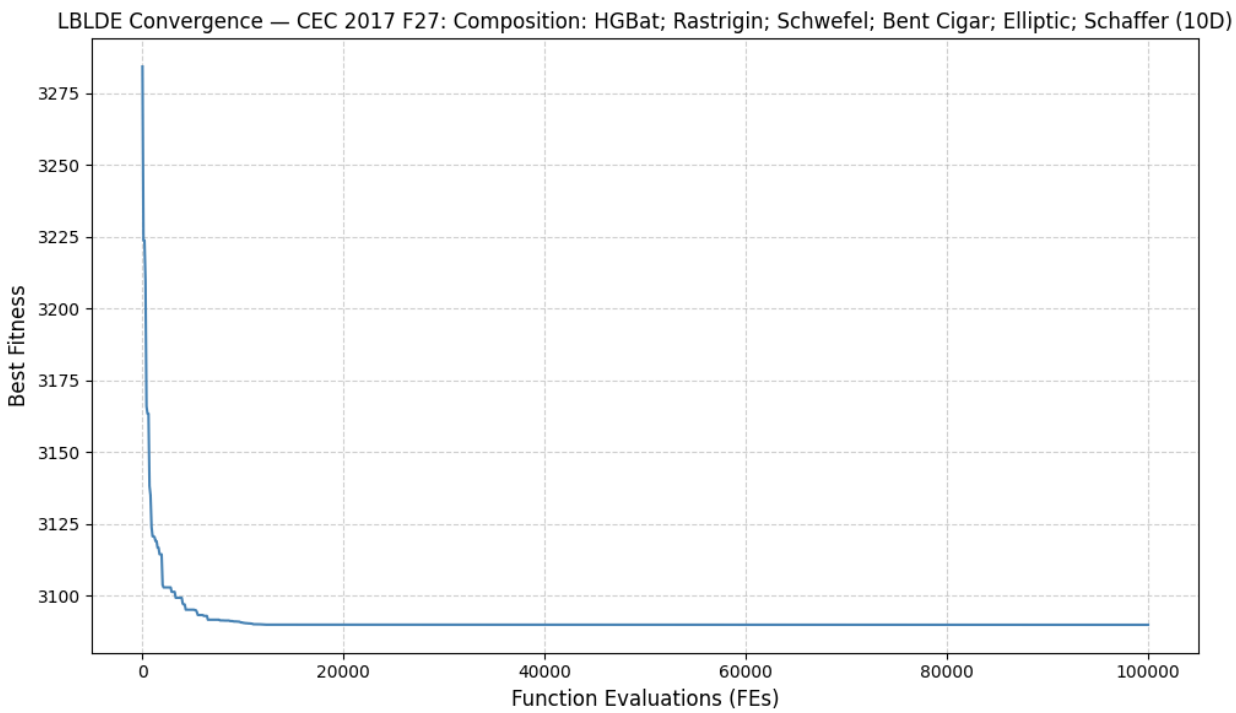



```
Graph saved as convergence_F26.png


# ============================================================
# F27 — Composition Function 7 (HGBat; Rastrigin; Schwefel; Bent
Cigar; Elliptic; Schaffer)
# ============================================================
run_and_plot('F27', make_wrapper(composition.f27), 2700.0,
'Composition: HGBat; Rastrigin; Schwefel; Bent Cigar; Elliptic;
Schaffer')

==========================================================================
==========
LBLDE on CEC 2017 F27 — Single Run
==========================================================================
==========
```

```
Function       : Composition: HGBat; Rastrigin; Schwefel; Bent Cigar;
Elliptic; Schaffer (10D)
Bounds         : [-100, 100]
Global optimum : 2700.0
Generation 0: Best Fitness = 3.284356e+03
Generation 10: Best Fitness = 3.120771e+03, FES = 1100
Generation 20: Best Fitness = 3.103839e+03, FES = 2100
Generation 30: Best Fitness = 3.101423e+03, FES = 3100
Generation 40: Best Fitness = 3.097315e+03, FES = 4100
Generation 50: Best Fitness = 3.095211e+03, FES = 5100
Generation 60: Best Fitness = 3.093337e+03, FES = 6100
Generation 70: Best Fitness = 3.091726e+03, FES = 7100
Generation 80: Best Fitness = 3.091473e+03, FES = 8100
Generation 90: Best Fitness = 3.091176e+03, FES = 9100
Generation 100: Best Fitness = 3.090648e+03, FES = 10100
Generation 110: Best Fitness = 3.090164e+03, FES = 11100
Generation 120: Best Fitness = 3.090044e+03, FES = 12100
Generation 130: Best Fitness = 3.089982e+03, FES = 13100
Generation 140: Best Fitness = 3.089965e+03, FES = 14100
Generation 150: Best Fitness = 3.089957e+03, FES = 15100
Generation 160: Best Fitness = 3.089956e+03, FES = 16100
Generation 170: Best Fitness = 3.089955e+03, FES = 17100
Generation 180: Best Fitness = 3.089955e+03, FES = 18100
Generation 190: Best Fitness = 3.089955e+03, FES = 19100
Generation 200: Best Fitness = 3.089955e+03, FES = 20100
Generation 210: Best Fitness = 3.089955e+03, FES = 21100
Generation 220: Best Fitness = 3.089955e+03, FES = 22100
Generation 230: Best Fitness = 3.089955e+03, FES = 23100
Generation 240: Best Fitness = 3.089955e+03, FES = 24100
Generation 250: Best Fitness = 3.089955e+03, FES = 25100
Generation 260: Best Fitness = 3.089955e+03, FES = 26100
Generation 270: Best Fitness = 3.089955e+03, FES = 27100
Generation 280: Best Fitness = 3.089955e+03, FES = 28100
Generation 290: Best Fitness = 3.089955e+03, FES = 29100
Generation 300: Best Fitness = 3.089955e+03, FES = 30100
Generation 310: Best Fitness = 3.089955e+03, FES = 31100
Generation 320: Best Fitness = 3.089955e+03, FES = 32100
Generation 330: Best Fitness = 3.089955e+03, FES = 33100
Generation 340: Best Fitness = 3.089955e+03, FES = 34100
Generation 350: Best Fitness = 3.089955e+03, FES = 35100
Generation 360: Best Fitness = 3.089955e+03, FES = 36100
Generation 370: Best Fitness = 3.089955e+03, FES = 37100
Generation 380: Best Fitness = 3.089955e+03, FES = 38100
Generation 390: Best Fitness = 3.089955e+03, FES = 39100
Generation 400: Best Fitness = 3.089955e+03, FES = 40100
Generation 410: Best Fitness = 3.089955e+03, FES = 41100
Generation 420: Best Fitness = 3.089955e+03, FES = 42100
Generation 430: Best Fitness = 3.089955e+03, FES = 43100
Generation 440: Best Fitness = 3.089955e+03, FES = 44100
```

```
Generation 450: Best Fitness = 3.089955e+03, FES = 45100
Generation 460: Best Fitness = 3.089955e+03, FES = 46100
Generation 470: Best Fitness = 3.089955e+03, FES = 47100
Generation 480: Best Fitness = 3.089955e+03, FES = 48100
Generation 490: Best Fitness = 3.089955e+03, FES = 49100
Generation 500: Best Fitness = 3.089955e+03, FES = 50100
Generation 510: Best Fitness = 3.089955e+03, FES = 51100
Generation 520: Best Fitness = 3.089955e+03, FES = 52100
Generation 530: Best Fitness = 3.089955e+03, FES = 53100
Generation 540: Best Fitness = 3.089955e+03, FES = 54100
Generation 550: Best Fitness = 3.089955e+03, FES = 55100
Generation 560: Best Fitness = 3.089955e+03, FES = 56100
Generation 570: Best Fitness = 3.089955e+03, FES = 57100
Generation 580: Best Fitness = 3.089955e+03, FES = 58100
Generation 590: Best Fitness = 3.089955e+03, FES = 59100
Generation 600: Best Fitness = 3.089955e+03, FES = 60100
Generation 610: Best Fitness = 3.089955e+03, FES = 61100
Generation 620: Best Fitness = 3.089955e+03, FES = 62100
Generation 630: Best Fitness = 3.089955e+03, FES = 63100
Generation 640: Best Fitness = 3.089955e+03, FES = 64100
Generation 650: Best Fitness = 3.089955e+03, FES = 65100
Generation 660: Best Fitness = 3.089955e+03, FES = 66100
Generation 670: Best Fitness = 3.089955e+03, FES = 67100
Generation 680: Best Fitness = 3.089955e+03, FES = 68100
Generation 690: Best Fitness = 3.089955e+03, FES = 69100
Generation 700: Best Fitness = 3.089955e+03, FES = 70100
Generation 710: Best Fitness = 3.089955e+03, FES = 71100
Generation 720: Best Fitness = 3.089955e+03, FES = 72100
Generation 730: Best Fitness = 3.089955e+03, FES = 73100
Generation 740: Best Fitness = 3.089955e+03, FES = 74100
Generation 750: Best Fitness = 3.089955e+03, FES = 75100
Generation 760: Best Fitness = 3.089955e+03, FES = 76100
Generation 770: Best Fitness = 3.089955e+03, FES = 77100
Generation 780: Best Fitness = 3.089955e+03, FES = 78100
Generation 790: Best Fitness = 3.089955e+03, FES = 79100
Generation 800: Best Fitness = 3.089955e+03, FES = 80100
Generation 810: Best Fitness = 3.089955e+03, FES = 81100
Generation 820: Best Fitness = 3.089955e+03, FES = 82100
Generation 830: Best Fitness = 3.089955e+03, FES = 83100
Generation 840: Best Fitness = 3.089955e+03, FES = 84100
Generation 850: Best Fitness = 3.089955e+03, FES = 85100
Generation 860: Best Fitness = 3.089955e+03, FES = 86100
Generation 870: Best Fitness = 3.089955e+03, FES = 87100
Generation 880: Best Fitness = 3.089955e+03, FES = 88100
Generation 890: Best Fitness = 3.089955e+03, FES = 89100
Generation 900: Best Fitness = 3.089955e+03, FES = 90100
Generation 910: Best Fitness = 3.089955e+03, FES = 91100
Generation 920: Best Fitness = 3.089955e+03, FES = 92100
Generation 930: Best Fitness = 3.089955e+03, FES = 93100
```

```
Generation 940: Best Fitness = 3.089955e+03, FES = 94100
Generation 950: Best Fitness = 3.089955e+03, FES = 95100
Generation 960: Best Fitness = 3.089955e+03, FES = 96100
Generation 970: Best Fitness = 3.089955e+03, FES = 97100
Generation 980: Best Fitness = 3.089955e+03, FES = 98100
Generation 990: Best Fitness = 3.089955e+03, FES = 99100

Optimization completed!
Final Best Fitness = 3.089955e+03
Total Function Evaluations = 100000

==========================================================================
==========
Results:
  Best Fitness           : 3.089955e+03
  Global Optimum         : 2.700000e+03
  Error (|best-optimum|) : 3.899548e+02
  Best Solution (5 dim)  : [-76.25122425 -89.12870101 -93.19989096 -
88.83891536 100.        ]
  D^(0) (init. diversity): 0.0000e+00
==========================================================================
==========
```



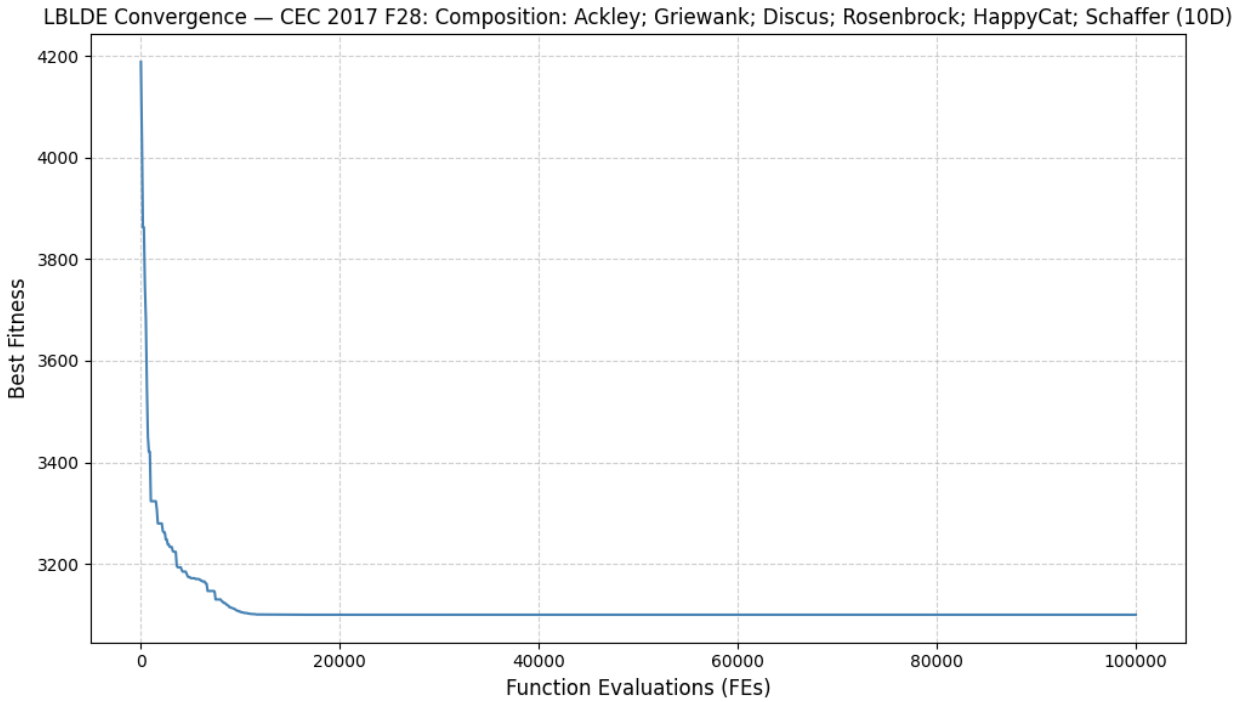

```
Graph saved as convergence_F27.png
```

```
# ==============================================================
# F28 — Composition Function 8 (Ackley; Griewank; Discus; Rosenbrock;
HappyCat; Schaffer)
# ==============================================================
run_and_plot('F28', make_wrapper(composition.f28), 2800.0,
'Composition: Ackley; Griewank; Discus; Rosenbrock; HappyCat;
Schaffer')
```

```
==========================================================================
==========
LBLDE on CEC 2017 F28 — Single Run
==========================================================================
==========
Function       : Composition: Ackley; Griewank; Discus; Rosenbrock;
HappyCat; Schaffer (10D)
Bounds         : [-100, 100]
Global optimum : 2800.0
Generation 0: Best Fitness = 4.188047e+03
Generation 10: Best Fitness = 3.323462e+03, FES = 1100
Generation 20: Best Fitness = 3.279650e+03, FES = 2100
Generation 30: Best Fitness = 3.233419e+03, FES = 3100
Generation 40: Best Fitness = 3.193409e+03, FES = 4100
Generation 50: Best Fitness = 3.172102e+03, FES = 5100
Generation 60: Best Fitness = 3.167359e+03, FES = 6100
Generation 70: Best Fitness = 3.146758e+03, FES = 7100
Generation 80: Best Fitness = 3.130118e+03, FES = 8100
Generation 90: Best Fitness = 3.114676e+03, FES = 9100
Generation 100: Best Fitness = 3.106106e+03, FES = 10100
Generation 110: Best Fitness = 3.101670e+03, FES = 11100
Generation 120: Best Fitness = 3.100569e+03, FES = 12100
Generation 130: Best Fitness = 3.100198e+03, FES = 13100
Generation 140: Best Fitness = 3.100082e+03, FES = 14100
Generation 150: Best Fitness = 3.100029e+03, FES = 15100
Generation 160: Best Fitness = 3.100011e+03, FES = 16100
Generation 170: Best Fitness = 3.100004e+03, FES = 17100
Generation 180: Best Fitness = 3.100002e+03, FES = 18100
Generation 190: Best Fitness = 3.100001e+03, FES = 19100
Generation 200: Best Fitness = 3.100000e+03, FES = 20100
Generation 210: Best Fitness = 3.100000e+03, FES = 21100
Generation 220: Best Fitness = 3.100000e+03, FES = 22100
Generation 230: Best Fitness = 3.100000e+03, FES = 23100
Generation 240: Best Fitness = 3.100000e+03, FES = 24100
Generation 250: Best Fitness = 3.100000e+03, FES = 25100
Generation 260: Best Fitness = 3.100000e+03, FES = 26100
Generation 270: Best Fitness = 3.100000e+03, FES = 27100
Generation 280: Best Fitness = 3.100000e+03, FES = 28100
Generation 290: Best Fitness = 3.100000e+03, FES = 29100
Generation 300: Best Fitness = 3.100000e+03, FES = 30100
Generation 310: Best Fitness = 3.100000e+03, FES = 31100
Generation 320: Best Fitness = 3.100000e+03, FES = 32100
```

```
Generation 330: Best Fitness = 3.100000e+03, FES = 33100
Generation 340: Best Fitness = 3.100000e+03, FES = 34100
Generation 350: Best Fitness = 3.100000e+03, FES = 35100
Generation 360: Best Fitness = 3.100000e+03, FES = 36100
Generation 370: Best Fitness = 3.100000e+03, FES = 37100
Generation 380: Best Fitness = 3.100000e+03, FES = 38100
Generation 390: Best Fitness = 3.100000e+03, FES = 39100
Generation 400: Best Fitness = 3.100000e+03, FES = 40100
Generation 410: Best Fitness = 3.100000e+03, FES = 41100
Generation 420: Best Fitness = 3.100000e+03, FES = 42100
Generation 430: Best Fitness = 3.100000e+03, FES = 43100
Generation 440: Best Fitness = 3.100000e+03, FES = 44100
Generation 450: Best Fitness = 3.100000e+03, FES = 45100
Generation 460: Best Fitness = 3.100000e+03, FES = 46100
Generation 470: Best Fitness = 3.100000e+03, FES = 47100
Generation 480: Best Fitness = 3.100000e+03, FES = 48100
Generation 490: Best Fitness = 3.100000e+03, FES = 49100
Generation 500: Best Fitness = 3.100000e+03, FES = 50100
Generation 510: Best Fitness = 3.100000e+03, FES = 51100
Generation 520: Best Fitness = 3.100000e+03, FES = 52100
Generation 530: Best Fitness = 3.100000e+03, FES = 53100
Generation 540: Best Fitness = 3.100000e+03, FES = 54100
Generation 550: Best Fitness = 3.100000e+03, FES = 55100
Generation 560: Best Fitness = 3.100000e+03, FES = 56100
Generation 570: Best Fitness = 3.100000e+03, FES = 57100
Generation 580: Best Fitness = 3.100000e+03, FES = 58100
Generation 590: Best Fitness = 3.100000e+03, FES = 59100
Generation 600: Best Fitness = 3.100000e+03, FES = 60100
Generation 610: Best Fitness = 3.100000e+03, FES = 61100
Generation 620: Best Fitness = 3.100000e+03, FES = 62100
Generation 630: Best Fitness = 3.100000e+03, FES = 63100
Generation 640: Best Fitness = 3.100000e+03, FES = 64100
Generation 650: Best Fitness = 3.100000e+03, FES = 65100
Generation 660: Best Fitness = 3.100000e+03, FES = 66100
Generation 670: Best Fitness = 3.100000e+03, FES = 67100
Generation 680: Best Fitness = 3.100000e+03, FES = 68100
Generation 690: Best Fitness = 3.100000e+03, FES = 69100
Generation 700: Best Fitness = 3.100000e+03, FES = 70100
Generation 710: Best Fitness = 3.100000e+03, FES = 71100
Generation 720: Best Fitness = 3.100000e+03, FES = 72100
Generation 730: Best Fitness = 3.100000e+03, FES = 73100
Generation 740: Best Fitness = 3.100000e+03, FES = 74100
Generation 750: Best Fitness = 3.100000e+03, FES = 75100
Generation 760: Best Fitness = 3.100000e+03, FES = 76100
Generation 770: Best Fitness = 3.100000e+03, FES = 77100
Generation 780: Best Fitness = 3.100000e+03, FES = 78100
Generation 790: Best Fitness = 3.100000e+03, FES = 79100
Generation 800: Best Fitness = 3.100000e+03, FES = 80100
Generation 810: Best Fitness = 3.100000e+03, FES = 81100
```

```
Generation 820: Best Fitness = 3.100000e+03, FES = 82100
Generation 830: Best Fitness = 3.100000e+03, FES = 83100
Generation 840: Best Fitness = 3.100000e+03, FES = 84100
Generation 850: Best Fitness = 3.100000e+03, FES = 85100
Generation 860: Best Fitness = 3.100000e+03, FES = 86100
Generation 870: Best Fitness = 3.100000e+03, FES = 87100
Generation 880: Best Fitness = 3.100000e+03, FES = 88100
Generation 890: Best Fitness = 3.100000e+03, FES = 89100
Generation 900: Best Fitness = 3.100000e+03, FES = 90100
Generation 910: Best Fitness = 3.100000e+03, FES = 91100
Generation 920: Best Fitness = 3.100000e+03, FES = 92100
Generation 930: Best Fitness = 3.100000e+03, FES = 93100
Generation 940: Best Fitness = 3.100000e+03, FES = 94100
Generation 950: Best Fitness = 3.100000e+03, FES = 95100
Generation 960: Best Fitness = 3.100000e+03, FES = 96100
Generation 970: Best Fitness = 3.100000e+03, FES = 97100
Generation 980: Best Fitness = 3.100000e+03, FES = 98100
Generation 990: Best Fitness = 3.100000e+03, FES = 99100

Optimization completed!
Final Best Fitness = 3.100000e+03
Total Function Evaluations = 100000

=====================================================================
==========
Results:
  Best Fitness           : 3.100000e+03
  Global Optimum         : 2.800000e+03
  Error (|best-optimum|) : 3.000000e+02
  Best Solution (5 dim)  : [-26.37193452 -34.00844064 -49.92961782
47.6166024  -66.54538272]
  D^(0) (init. diversity): 0.0000e+00
=====================================================================
==========
```

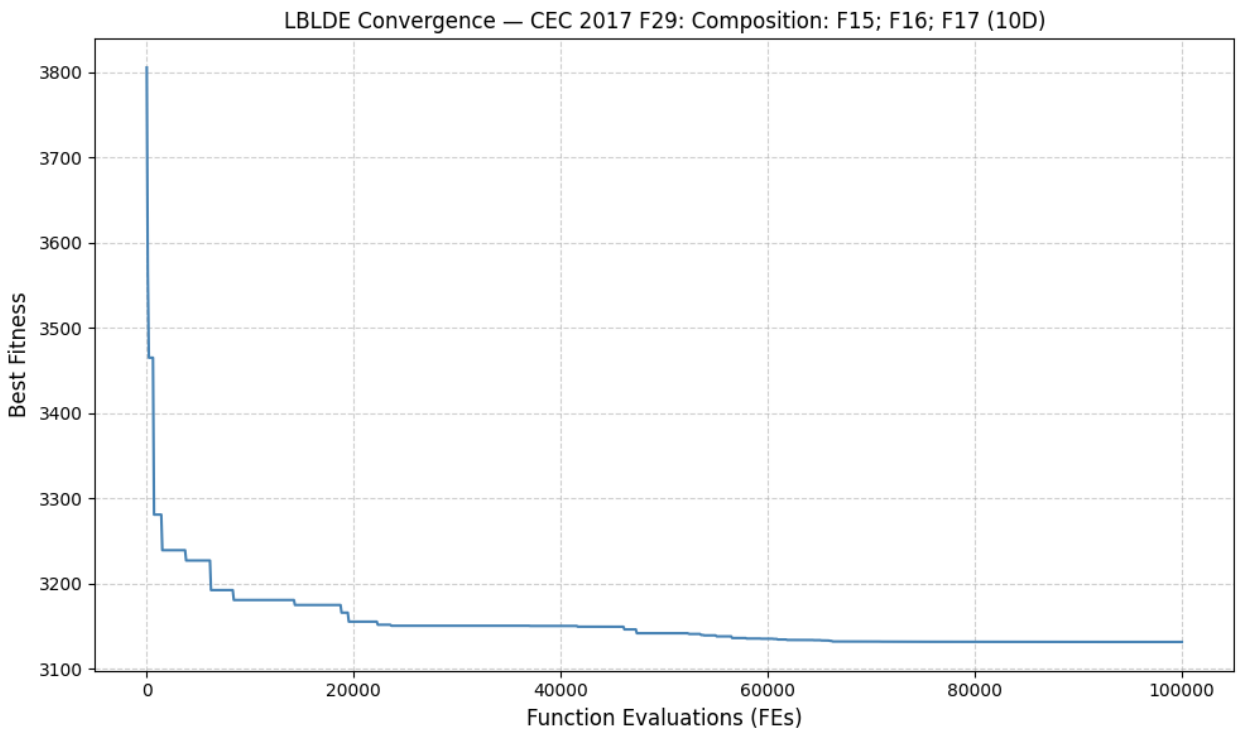



```
Graph saved as convergence_F28.png


# ============================================================
# F29 — Composition Function 9 (F15; F16; F17)
# ============================================================
run_and_plot('F29', make_wrapper(composition.f29), 2900.0,
'Composition: F15; F16; F17')

======================================================================
==========
LBLDE on CEC 2017 F29 — Single Run
======================================================================
==========
Function       : Composition: F15; F16; F17 (10D)
Bounds         : [-100, 100]
Global optimum : 2900.0
Generation 0: Best Fitness = 3.805371e+03
Generation 10: Best Fitness = 3.281103e+03, FES = 1100
Generation 20: Best Fitness = 3.239361e+03, FES = 2100
Generation 30: Best Fitness = 3.239361e+03, FES = 3100
Generation 40: Best Fitness = 3.227271e+03, FES = 4100
Generation 50: Best Fitness = 3.227271e+03, FES = 5100
Generation 60: Best Fitness = 3.227271e+03, FES = 6100
Generation 70: Best Fitness = 3.192570e+03, FES = 7100
Generation 80: Best Fitness = 3.192570e+03, FES = 8100
Generation 90: Best Fitness = 3.180958e+03, FES = 9100
Generation 100: Best Fitness = 3.180958e+03, FES = 10100
```

```
Generation 110: Best Fitness = 3.180958e+03, FES = 11100
Generation 120: Best Fitness = 3.180958e+03, FES = 12100
Generation 130: Best Fitness = 3.180958e+03, FES = 13100
Generation 140: Best Fitness = 3.180958e+03, FES = 14100
Generation 150: Best Fitness = 3.175115e+03, FES = 15100
Generation 160: Best Fitness = 3.175115e+03, FES = 16100
Generation 170: Best Fitness = 3.175115e+03, FES = 17100
Generation 180: Best Fitness = 3.175115e+03, FES = 18100
Generation 190: Best Fitness = 3.166116e+03, FES = 19100
Generation 200: Best Fitness = 3.155571e+03, FES = 20100
Generation 210: Best Fitness = 3.155571e+03, FES = 21100
Generation 220: Best Fitness = 3.155571e+03, FES = 22100
Generation 230: Best Fitness = 3.151953e+03, FES = 23100
Generation 240: Best Fitness = 3.150754e+03, FES = 24100
Generation 250: Best Fitness = 3.150754e+03, FES = 25100
Generation 260: Best Fitness = 3.150754e+03, FES = 26100
Generation 270: Best Fitness = 3.150754e+03, FES = 27100
Generation 280: Best Fitness = 3.150754e+03, FES = 28100
Generation 290: Best Fitness = 3.150754e+03, FES = 29100
Generation 300: Best Fitness = 3.150754e+03, FES = 30100
Generation 310: Best Fitness = 3.150754e+03, FES = 31100
Generation 320: Best Fitness = 3.150754e+03, FES = 32100
Generation 330: Best Fitness = 3.150754e+03, FES = 33100
Generation 340: Best Fitness = 3.150754e+03, FES = 34100
Generation 350: Best Fitness = 3.150754e+03, FES = 35100
Generation 360: Best Fitness = 3.150754e+03, FES = 36100
Generation 370: Best Fitness = 3.150538e+03, FES = 37100
Generation 380: Best Fitness = 3.150538e+03, FES = 38100
Generation 390: Best Fitness = 3.150538e+03, FES = 39100
Generation 400: Best Fitness = 3.150538e+03, FES = 40100
Generation 410: Best Fitness = 3.150538e+03, FES = 41100
Generation 420: Best Fitness = 3.149596e+03, FES = 42100
Generation 430: Best Fitness = 3.149596e+03, FES = 43100
Generation 440: Best Fitness = 3.149596e+03, FES = 44100
Generation 450: Best Fitness = 3.149596e+03, FES = 45100
Generation 460: Best Fitness = 3.149596e+03, FES = 46100
Generation 470: Best Fitness = 3.146492e+03, FES = 47100
Generation 480: Best Fitness = 3.141982e+03, FES = 48100
Generation 490: Best Fitness = 3.141982e+03, FES = 49100
Generation 500: Best Fitness = 3.141982e+03, FES = 50100
Generation 510: Best Fitness = 3.141982e+03, FES = 51100
Generation 520: Best Fitness = 3.141982e+03, FES = 52100
Generation 530: Best Fitness = 3.141134e+03, FES = 53100
Generation 540: Best Fitness = 3.139479e+03, FES = 54100
Generation 550: Best Fitness = 3.138314e+03, FES = 55100
Generation 560: Best Fitness = 3.138314e+03, FES = 56100
Generation 570: Best Fitness = 3.136367e+03, FES = 57100
Generation 580: Best Fitness = 3.135768e+03, FES = 58100
Generation 590: Best Fitness = 3.135768e+03, FES = 59100
```

```
Generation 600: Best Fitness = 3.135557e+03, FES = 60100
Generation 610: Best Fitness = 3.134665e+03, FES = 61100
Generation 620: Best Fitness = 3.134123e+03, FES = 62100
Generation 630: Best Fitness = 3.134123e+03, FES = 63100
Generation 640: Best Fitness = 3.134021e+03, FES = 64100
Generation 650: Best Fitness = 3.133851e+03, FES = 65100
Generation 660: Best Fitness = 3.132973e+03, FES = 66100
Generation 670: Best Fitness = 3.132321e+03, FES = 67100
Generation 680: Best Fitness = 3.132321e+03, FES = 68100
Generation 690: Best Fitness = 3.132321e+03, FES = 69100
Generation 700: Best Fitness = 3.132237e+03, FES = 70100
Generation 710: Best Fitness = 3.132136e+03, FES = 71100
Generation 720: Best Fitness = 3.132032e+03, FES = 72100
Generation 730: Best Fitness = 3.132032e+03, FES = 73100
Generation 740: Best Fitness = 3.132015e+03, FES = 74100
Generation 750: Best Fitness = 3.131973e+03, FES = 75100
Generation 760: Best Fitness = 3.131971e+03, FES = 76100
Generation 770: Best Fitness = 3.131971e+03, FES = 77100
Generation 780: Best Fitness = 3.131902e+03, FES = 78100
Generation 790: Best Fitness = 3.131902e+03, FES = 79100
Generation 800: Best Fitness = 3.131855e+03, FES = 80100
Generation 810: Best Fitness = 3.131855e+03, FES = 81100
Generation 820: Best Fitness = 3.131855e+03, FES = 82100
Generation 830: Best Fitness = 3.131847e+03, FES = 83100
Generation 840: Best Fitness = 3.131818e+03, FES = 84100
Generation 850: Best Fitness = 3.131814e+03, FES = 85100
Generation 860: Best Fitness = 3.131804e+03, FES = 86100
Generation 870: Best Fitness = 3.131804e+03, FES = 87100
Generation 880: Best Fitness = 3.131794e+03, FES = 88100
Generation 890: Best Fitness = 3.131789e+03, FES = 89100
Generation 900: Best Fitness = 3.131786e+03, FES = 90100
Generation 910: Best Fitness = 3.131778e+03, FES = 91100
Generation 920: Best Fitness = 3.131778e+03, FES = 92100
Generation 930: Best Fitness = 3.131776e+03, FES = 93100
Generation 940: Best Fitness = 3.131773e+03, FES = 94100
Generation 950: Best Fitness = 3.131772e+03, FES = 95100
Generation 960: Best Fitness = 3.131771e+03, FES = 96100
Generation 970: Best Fitness = 3.131770e+03, FES = 97100
Generation 980: Best Fitness = 3.131770e+03, FES = 98100
Generation 990: Best Fitness = 3.131769e+03, FES = 99100

Optimization completed!
Final Best Fitness = 3.131769e+03
Total Function Evaluations = 100000

=====================================================================
==========
Results:
  Best Fitness            : 3.131769e+03
```

```
  Global Optimum         : 2.900000e+03
  Error (|best-optimum|) : 2.317689e+02
  Best Solution (5 dim)  : [-99.99944172 -87.28553289 -66.20563655
97.69742987 -41.62877128]
  D^(0) (init. diversity): 0.0000e+00
======================================================================
==========
```

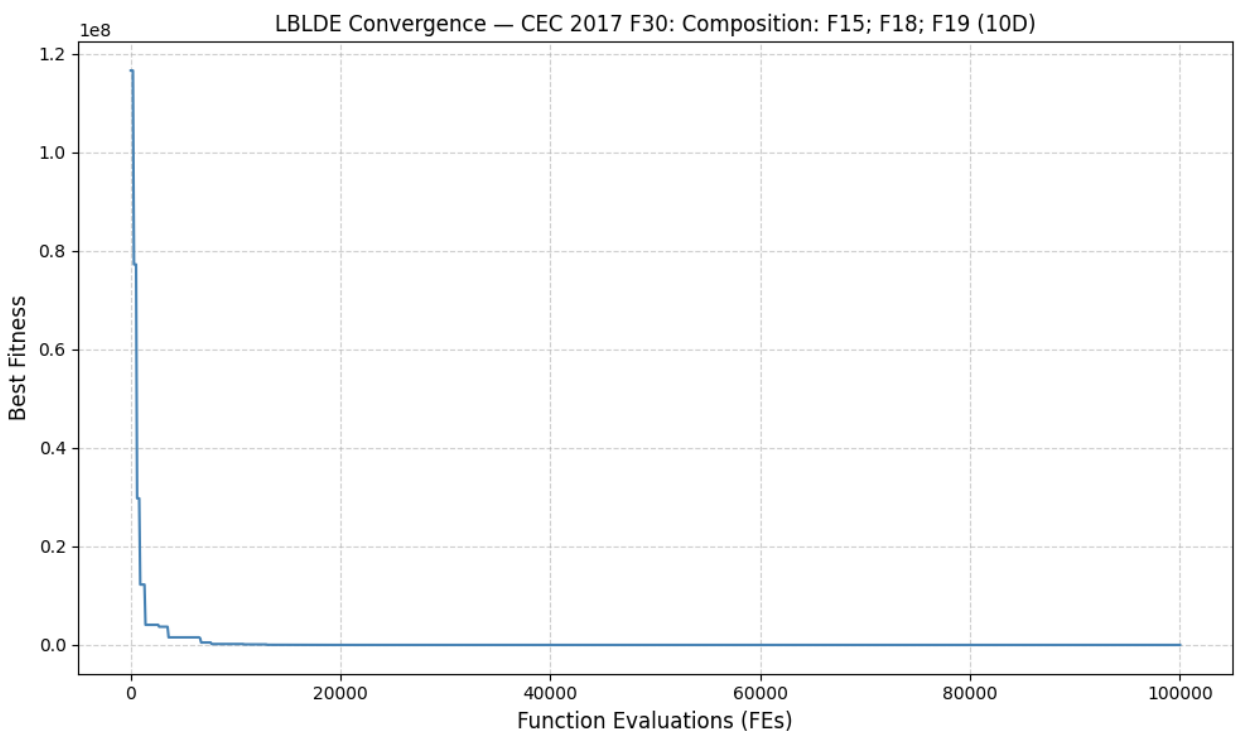



```
Graph saved as convergence_F29.png


# ============================================================
# F30 — Composition Function 10 (F15; F18; F19)
# ============================================================
run_and_plot('F30', make_wrapper(composition.f30), 3000.0,
'Composition: F15; F18; F19')

======================================================================
==========
LBLDE on CEC 2017 F30 — Single Run
======================================================================
==========
Function       : Composition: F15; F18; F19 (10D)
Bounds         : [-100, 100]
Global optimum : 3000.0
Generation 0: Best Fitness = 1.165390e+08
Generation 10: Best Fitness = 1.225792e+07, FES = 1100
```

```
Generation 20: Best Fitness = 4.089954e+06, FES = 2100
Generation 30: Best Fitness = 3.706738e+06, FES = 3100
Generation 40: Best Fitness = 1.546986e+06, FES = 4100
Generation 50: Best Fitness = 1.546986e+06, FES = 5100
Generation 60: Best Fitness = 1.546986e+06, FES = 6100
Generation 70: Best Fitness = 5.022993e+05, FES = 7100
Generation 80: Best Fitness = 1.933948e+05, FES = 8100
Generation 90: Best Fitness = 1.933948e+05, FES = 9100
Generation 100: Best Fitness = 1.933948e+05, FES = 10100
Generation 110: Best Fitness = 1.189030e+05, FES = 11100
Generation 120: Best Fitness = 1.189030e+05, FES = 12100
Generation 130: Best Fitness = 3.441522e+04, FES = 13100
Generation 140: Best Fitness = 2.491497e+04, FES = 14100
Generation 150: Best Fitness = 2.491497e+04, FES = 15100
Generation 160: Best Fitness = 1.693385e+04, FES = 16100
Generation 170: Best Fitness = 1.327341e+04, FES = 17100
Generation 180: Best Fitness = 8.737295e+03, FES = 18100
Generation 190: Best Fitness = 6.331806e+03, FES = 19100
Generation 200: Best Fitness = 4.782709e+03, FES = 20100
Generation 210: Best Fitness = 4.782709e+03, FES = 21100
Generation 220: Best Fitness = 4.264770e+03, FES = 22100
Generation 230: Best Fitness = 4.122119e+03, FES = 23100
Generation 240: Best Fitness = 3.855997e+03, FES = 24100
Generation 250: Best Fitness = 3.783709e+03, FES = 25100
Generation 260: Best Fitness = 3.744260e+03, FES = 26100
Generation 270: Best Fitness = 3.731315e+03, FES = 27100
Generation 280: Best Fitness = 3.689803e+03, FES = 28100
Generation 290: Best Fitness = 3.686934e+03, FES = 29100
Generation 300: Best Fitness = 3.662527e+03, FES = 30100
Generation 310: Best Fitness = 3.652532e+03, FES = 31100
Generation 320: Best Fitness = 3.643073e+03, FES = 32100
Generation 330: Best Fitness = 3.610860e+03, FES = 33100
Generation 340: Best Fitness = 3.610860e+03, FES = 34100
Generation 350: Best Fitness = 3.607020e+03, FES = 35100
Generation 360: Best Fitness = 3.605016e+03, FES = 36100
Generation 370: Best Fitness = 3.591756e+03, FES = 37100
Generation 380: Best Fitness = 3.591756e+03, FES = 38100
Generation 390: Best Fitness = 3.586952e+03, FES = 39100
Generation 400: Best Fitness = 3.583918e+03, FES = 40100
Generation 410: Best Fitness = 3.574917e+03, FES = 41100
Generation 420: Best Fitness = 3.568264e+03, FES = 42100
Generation 430: Best Fitness = 3.560087e+03, FES = 43100
Generation 440: Best Fitness = 3.555619e+03, FES = 44100
Generation 450: Best Fitness = 3.555619e+03, FES = 45100
Generation 460: Best Fitness = 3.554600e+03, FES = 46100
Generation 470: Best Fitness = 3.550593e+03, FES = 47100
Generation 480: Best Fitness = 3.544579e+03, FES = 48100
Generation 490: Best Fitness = 3.540874e+03, FES = 49100
Generation 500: Best Fitness = 3.540091e+03, FES = 50100
```

```
Generation 510: Best Fitness = 3.538322e+03, FES = 51100
Generation 520: Best Fitness = 3.537815e+03, FES = 52100
Generation 530: Best Fitness = 3.523685e+03, FES = 53100
Generation 540: Best Fitness = 3.523685e+03, FES = 54100
Generation 550: Best Fitness = 3.519806e+03, FES = 55100
Generation 560: Best Fitness = 3.515854e+03, FES = 56100
Generation 570: Best Fitness = 3.514146e+03, FES = 57100
Generation 580: Best Fitness = 3.507939e+03, FES = 58100
Generation 590: Best Fitness = 3.507058e+03, FES = 59100
Generation 600: Best Fitness = 3.496171e+03, FES = 60100
Generation 610: Best Fitness = 3.495498e+03, FES = 61100
Generation 620: Best Fitness = 3.492287e+03, FES = 62100
Generation 630: Best Fitness = 3.480146e+03, FES = 63100
Generation 640: Best Fitness = 3.479751e+03, FES = 64100
Generation 650: Best Fitness = 3.474739e+03, FES = 65100
Generation 660: Best Fitness = 3.469700e+03, FES = 66100
Generation 670: Best Fitness = 3.460258e+03, FES = 67100
Generation 680: Best Fitness = 3.460258e+03, FES = 68100
Generation 690: Best Fitness = 3.460258e+03, FES = 69100
Generation 700: Best Fitness = 3.452243e+03, FES = 70100
Generation 710: Best Fitness = 3.451321e+03, FES = 71100
Generation 720: Best Fitness = 3.450500e+03, FES = 72100
Generation 730: Best Fitness = 3.444856e+03, FES = 73100
Generation 740: Best Fitness = 3.444010e+03, FES = 74100
Generation 750: Best Fitness = 3.441431e+03, FES = 75100
Generation 760: Best Fitness = 3.437894e+03, FES = 76100
Generation 770: Best Fitness = 3.437894e+03, FES = 77100
Generation 780: Best Fitness = 3.433624e+03, FES = 78100
Generation 790: Best Fitness = 3.433624e+03, FES = 79100
Generation 800: Best Fitness = 3.433624e+03, FES = 80100
Generation 810: Best Fitness = 3.433624e+03, FES = 81100
Generation 820: Best Fitness = 3.432711e+03, FES = 82100
Generation 830: Best Fitness = 3.430436e+03, FES = 83100
Generation 840: Best Fitness = 3.428062e+03, FES = 84100
Generation 850: Best Fitness = 3.424453e+03, FES = 85100
Generation 860: Best Fitness = 3.423123e+03, FES = 86100
Generation 870: Best Fitness = 3.419015e+03, FES = 87100
Generation 880: Best Fitness = 3.418767e+03, FES = 88100
Generation 890: Best Fitness = 3.415879e+03, FES = 89100
Generation 900: Best Fitness = 3.414756e+03, FES = 90100
Generation 910: Best Fitness = 3.414469e+03, FES = 91100
Generation 920: Best Fitness = 3.411118e+03, FES = 92100
Generation 930: Best Fitness = 3.410816e+03, FES = 93100
Generation 940: Best Fitness = 3.409534e+03, FES = 94100
Generation 950: Best Fitness = 3.407715e+03, FES = 95100
Generation 960: Best Fitness = 3.407381e+03, FES = 96100
Generation 970: Best Fitness = 3.405873e+03, FES = 97100
Generation 980: Best Fitness = 3.404452e+03, FES = 98100
Generation 990: Best Fitness = 3.403852e+03, FES = 99100
```

```
Optimization completed!
Final Best Fitness = 3.403028e+03
Total Function Evaluations = 100000

=========================================================================
==========
Results:
  Best Fitness           : 3.403028e+03
  Global Optimum         : 3.000000e+03
  Error (|best-optimum|) : 4.030284e+02
  Best Solution (5 dim)  : [-44.06388849  78.76523452  99.04231263
97.64446644 -99.83669662]
  D^(0) (init. diversity): 0.0000e+00
=========================================================================
==========
```




```
Graph saved as convergence_F30.png
```

Section 7. Compiled Results

Print out a summary of all results in a tabular format

```
print('=' * 80)
print('SUMMARY — LBLDE on CEC 2017 (10D), Single Run, Seed=42')
print('=' * 80)
print(f'{"Function":<10} {"Best Fitness":>18} {"Global Optimum":>18}
{"Error |best-opt|":>18}')
print('-' * 70)
for fn in [f'F{i}' for i in range(1, 31)]:
    if fn not in all_results:
        continue
    r = all_results[fn]
    if r['best_fit'] is None:
        print(f'{fn:<10} {"N/A (deprecated)":>18}
{r["f_global"]:>18.2f} {"N/A":>18}')
    else:
        print(f'{fn:<10} {r["best_fit"]:>18.6e} {r["f_global"]:>18.2f}
{r["error"]:>18.6e}')
print('=' * 80)
```

```
================================================================================
SUMMARY — LBLDE on CEC 2017 (10D), Single Run, Seed=42
================================================================================
Function         Best Fitness     Global Optimum   Error |best-opt|
----------------------------------------------------------------------
F1               1.000000e+02             100.00       0.000000e+00
F2               2.000000e+02             200.00       0.000000e+00
F3               3.000000e+02             300.00       0.000000e+00
F4               4.004217e+02             400.00       4.217135e-01
F5               5.029849e+02             500.00       2.984877e+00
F6               6.000000e+02             600.00       0.000000e+00
F7               7.110264e+02             700.00       1.102645e+01
F8               8.019899e+02             800.00       1.989918e+00
F9               9.000000e+02             900.00       0.000000e+00
F10              1.015182e+03            1000.00       1.518216e+01
F11              1.100000e+03            1100.00       0.000000e+00
F12              1.211381e+03            1200.00       1.138114e+01
F13              1.305951e+03            1300.00       5.951113e+00
F14              1.400000e+03            1400.00       0.000000e+00
F15              1.500001e+03            1500.00       7.984507e-04
F16              1.600235e+03            1600.00       2.353018e-01
F17              1.700000e+03            1700.00       2.545740e-05
F18              1.800183e+03            1800.00       1.829275e-01
F19              1.900000e+03            1900.00       0.000000e+00
F20              2.000000e+03            2000.00       2.545513e-05
F21              2.200000e+03            2100.00       1.000000e+02
F22              2.300000e+03            2200.00       1.000000e+02
F23              2.600000e+03            2300.00       3.000000e+02
F24              2.735272e+03            2400.00       3.352723e+02
F25              2.943357e+03            2500.00       4.433567e+02
```

```
F26                2.900000e+03             2600.00       3.000000e+02
F27                3.089955e+03             2700.00       3.899548e+02
F28                3.100000e+03             2800.00       3.000000e+02
F29                3.131769e+03             2900.00       2.317689e+02
F30                3.403028e+03             3000.00       4.030284e+02
=====================================================================
==========
```